

Personality

The Psychodynamic Perspective

Robert Bornstein

Originating in the work of Sigmund Freud, the psychodynamic perspective emphasizes unconscious psychological processes (for example, wishes and fears of which we're not fully aware), and contends that childhood experiences are crucial in shaping adult personality. The psychodynamic perspective has evolved considerably since Freud's time, and now includes innovative new approaches such as object relations theory and neuropsychanalysis. Some psychodynamic concepts have held up well to empirical scrutiny while others have not, and aspects of the theory remain controversial, but the psychodynamic perspective continues to influence many different areas of contemporary psychology.

Learning Objectives

- Describe the major models of personality within the psychodynamic perspective.
- Define the concept of ego defense, and give examples of commonly used ego defenses.
- Identify psychodynamic concepts that have been supported by empirical research.
- Discuss current trends in psychodynamic theory.

Introduction

Have you ever done something that didn't make sense? Perhaps you waited until the last minute to begin studying for an exam, even though you knew that delaying so long would ensure that you got a poor grade. Or maybe you spotted a person you liked across the room—someone about whom you had romantic feelings—but instead of approaching that person

you headed the other way (and felt ashamed about it afterward). If you've ever done something that didn't seem to make sense—and who among us hasn't—the psychodynamic perspective on personality might be useful for you. It can help you understand why you chose not to study for that test, or why you ran the other way when the person of your dreams entered the room.

Psychodynamic theory (sometimes called *psychoanalytic theory*) explains personality in terms of unconscious psychological processes (for example, wishes and fears of which we're not fully aware), and contends that childhood experiences are crucial in shaping adult personality. Psychodynamic theory is most closely associated with the work of Sigmund Freud, and with psychoanalysis, a type of psychotherapy that attempts to explore the patient's unconscious thoughts and emotions so that the person is better able to understand him- or herself.



According to psychodynamic theory, a lot of our behaviors and preferences of adulthood are shaped by the experiences in our childhood. [Image: Rifqi Dahlgren, <https://goo.gl/hx4Oeb>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

Freud's work has been extremely influential, its impact extending far beyond psychology (several years ago *Time* magazine selected Freud as one of the most important thinkers of the 20th century). Freud's work has been not only influential, but quite controversial as well. As you might imagine, when Freud suggested in 1900 that much of our behavior is determined by psychological forces of which we're largely unaware—that we literally don't know what's going on in our own minds—people were (to put it mildly) displeased (Freud, 1900/1953a). When he suggested in 1905 that we humans have strong sexual feelings from a very early age, and that some of these sexual feelings are directed toward our parents, people were more than displeased—they were outraged (Freud, 1905/1953b). Few theories in psychology have evoked such strong reactions from other professionals and members of the public.

Controversy notwithstanding, no competent psychologist, or student of psychology, can ignore psychodynamic theory. It is simply too important for psychological science and practice, and continues to play an important role in a wide variety of disciplines within and outside psychology (for example, developmental psychology, social psychology, sociology, and

neuroscience; see Bornstein, 2005, 2006; Solms & Turnbull, 2011). This module reviews the psychodynamic perspective on personality. We begin with a brief discussion of the core assumptions of psychodynamic theory, followed by an overview of the evolution of the theory from Freud's time to today. We then discuss the place of psychodynamic theory within contemporary psychology, and look toward the future as well.

Core Assumptions of the Psychodynamic Perspective

The core assumptions of psychodynamic theory are surprisingly simple. Moreover, these assumptions are unique to the psychodynamic framework: No other theories of personality accept these three ideas in their purest form.

Assumption 1: Primacy of the Unconscious

Psychodynamic theorists contend that the majority of psychological processes take place outside conscious awareness. In psychoanalytic terms, the activities of the mind (or *psyche*) are presumed to be largely unconscious. Research confirms this basic premise of psychoanalysis: Many of our mental activities—memories, motives, feelings, and the like—are largely inaccessible to consciousness (Bargh & Morsella, 2008; Bornstein, 2010; Wilson, 2009).

Assumption 2: Critical Importance of Early Experiences

Psychodynamic theory is not alone in positing that early childhood events play a role in shaping personality, but the theory is unique in the degree to which it emphasizes these events as determinants of personality development and dynamics. According to the psychodynamic model, early experiences—including those occurring during the first weeks or months of life—set in motion personality processes that affect us years, even decades, later (Blatt & Levy, 2003; McWilliams, 2009). This is especially true of experiences that are outside the normal range (for example, losing a parent or sibling at a very early age).

Assumption 3: Psychic Causality

The third core assumption of psychodynamic theory is that nothing in mental life happens by chance—that there is no such thing as a random thought, feeling, motive, or behavior. This has come to be known as the principle of **psychic causality**, and though few psychologists accept the principle of psychic causality precisely as psychoanalysts conceive it, most theorists



Our every thought and behavior—even something as seemingly random as which seat you choose on the bus—results from biological or psychological influences. [Image: ryuu ji 竜次, <https://goo.gl/NrofGI>, CC BY 2.0, <https://goo.gl/BRvSA7>]

and researchers agree that thoughts, motives, emotional responses, and expressed behaviors do not arise randomly, but always stem from some combination of identifiable biological and psychological processes (Elliott, 2002; Robinson & Gordon, 2011).

The Evolution of Psychodynamic Theory

Given Freud's background in neurology, it is not surprising that the first incarnation of psychoanalytic theory was primarily biological: Freud set out to explain

psychological phenomena in terms that could be linked to neurological functioning as it was understood in his day. Because Freud's work in this area evolved over more than 50 years (he began in 1885, and continued until he died in 1939), there were numerous revisions along the way. Thus, it is most accurate to think of psychodynamic theory as a set of interrelated models that complement and build upon each other. Three are particularly important: the topographic model, the psychosexual stage model, and the structural model.

The Topographic Model

In his 1900 book, *The Interpretation of Dreams*, Freud introduced his topographic model of the mind, which contended that the mind could be divided into three regions: conscious, preconscious, and unconscious. The conscious part of the mind holds information that you're focusing on at this moment—what you're thinking and feeling right now. The preconscious contains material that is capable of becoming conscious but is not conscious at the moment because your attention is not being directed toward it. You can move material from the preconscious into consciousness simply by focusing your attention on it. Consider, for example, what you had for dinner last night. A moment ago that information was preconscious; now it's conscious, because you "pulled it up" into consciousness. (Not to worry, in a few moments it will be preconscious again, and you can move on to more important things.)

The unconscious—the most controversial part of the topographic model—contains anxiety-producing material (for example, sexual impulses, aggressive urges) that are deliberately

repressed (held outside of conscious awareness as a form of self-protection because they make you uncomfortable). The terms *conscious*, *preconscious*, and *unconscious* continue to be used today in psychology, and research has provided considerable support for Freud's thinking regarding conscious and preconscious processing (Erdelyi, 1985, 2004). The existence of the unconscious remains controversial, with some researchers arguing that evidence for it is compelling and others contending that "unconscious" processing can be accounted for without positing the existence of a Freudian repository of repressed wishes and troubling urges and impulses (Eagle, 2011; Luborsky & Barrett, 2006).

The Psychosexual Stage Model

Freud remained devoted to the topographic model, but by 1905 he had outlined the key elements of his **psychosexual stage model**, which argued that early in life we progress through a sequence of developmental stages, each with its own unique challenge and its own mode of sexual gratification. Freud's psychosexual stages—oral, anal, Oedipal, latency, and genital—are well-known even to non-analytic psychologists. Frustration or overgratification during a particular stage was hypothesized to result in "fixation" at that stage, and to the development of an oral, anal, or Oedipal personality style (Bornstein, 2005, 2006).

Table 1 illustrates the basic organization of Freud's (1905/1953b) psychosexual stage model, and the three personality styles that result. Note that—consistent with the developmental challenges that the child confronts during each stage—oral fixation is hypothesized to result in a dependent personality, whereas anal fixation results in a lifelong preoccupation with control. Oedipal fixation leads to an aggressive, competitive personality orientation.

The Structural Model

Ultimately, Freud recognized that the topographic model was helpful in understanding how people process and store information, but not all that useful in explaining other important psychological phenomena (for example, why certain people develop psychological disorders and others do not). To extend his theory, Freud developed a complementary framework to account for normal and abnormal personality development—the **structural model**—which posits the existence of three interacting mental structures called the *id*, *ego*, and *superego*. The *id* is the seat of drives and instincts, whereas the *ego* represents the logical, reality-oriented part of the mind, and the *superego* is basically your conscience—the moral guidelines, rules, and prohibitions that guide your behavior. (You acquire these through your family and through the culture in which you were raised.)

According to the structural model, our personality reflects the interplay of these three psychic structures, which differ across individuals in relative power and influence. When the id predominates and instincts rule, the result is an impulsive personality style. When the superego is strongest, moral prohibitions reign supreme, and a restrained, overcontrolled personality ensues. When the ego is dominant, a more balanced set of personality traits develop (Eagle, 2011; McWilliams, 2009).

The Ego and Its Defenses

In addition to being the logical, rational, reality-oriented part of the mind, the ego serves another important function: It helps us manage anxiety through the use of ego defenses. Ego defenses are basically mental strategies that we use automatically and unconsciously when we feel threatened (Cramer, 2000, 2006). They help us navigate upsetting events, but there's a cost as well: All ego defenses involve some distortion of reality. For example, repression (the most basic ego defense, according to Freud) involves removing from consciousness upsetting thoughts and feelings, and moving those thoughts and feelings to the unconscious. When you read about a person who “blocked out” upsetting memories of child abuse, that's an example of repression.

Another ego defense is denial. In denial (unlike repression), we are aware that a particular event occurred, but we don't allow ourselves to see the implications of that event. When you hear a person with a substance abuse problem say “I'm fine—even though people complain about my drinking I never miss a day of work,” that person is using denial. Table 2 lists some common ego defenses in psychodynamic theory, along with a definition and example of each.

Psychodynamic Theories: Where Are We Now?

The topographic model, psychosexual stage model, and structural model continue to influence contemporary psychology, but it is important to keep in mind that psychodynamic theory is never static, ever changing and evolving in response to new ideas and findings. In the following sections we discuss four current trends in the psychodynamic perspective: object relations theory, the empirical testing of psychodynamic concepts, psychoanalysis and culture, and the opportunities and challenges of neuroscience.

Object Relations Theory and the Growth of the Psychodynamic Perspective

In recent years a number of new psychodynamic frameworks have emerged to explain personality development and dynamics. The most important of these is **object relations theory**. (In psychoanalytic language, the term “object” refers to a person, so object relations theory is really something more like “interpersonal relations theory.”)

Object relations theory contends that personality can be understood as reflecting the mental images of significant figures (especially the parents) that we form early in life in response to interactions taking place within the family (Kernberg, 2004; Wachtel, 1997). These mental images (sometimes called *introjects*) serve as templates for later interpersonal relationships—almost like relationship blueprints or “scripts.” So if you internalized positive introjects early in life (for example, a mental image of mom or dad as warm and accepting), that’s what you expect to occur in later relationships as well. If you internalized a mental image of mom or dad as harsh and judgmental, you might instead become a self-critical person, and feel that you can never live up to other people’s standards . . . or your own (Luyten & Blatt, 2013).

Object relations theory has increased many psychologists’ interest in studying psychodynamic ideas and concepts, in part because it represents a natural bridge between the psychodynamic perspective and research in other areas of psychology. For example, developmental and social psychologists also believe that mental representations of significant people play an important role in shaping our behavior. In developmental psychology you might read about this in the context of *attachment theory* (which argues that attachments—or bonds—to significant people are key to understanding human behavior; Fraley, 2002). In social psychology, mental representations of significant figures play an important role in *social cognition* (thoughts and feelings regarding other people; Bargh & Morsella, 2008; Robinson & Gordon, 2011).

Empirical Research on Psychodynamic Theories

Empirical research assessing psychodynamic concepts has produced mixed results, with some concepts receiving good empirical support, and others not faring as well. For example, the notion that we express strong sexual feelings from a very early age, as the psychosexual stage model suggests, has not held up to empirical scrutiny. On the other hand, the idea that there are dependent, control-oriented, and competitive personality types—an idea also derived from the psychosexual stage model—does seem useful.

Many ideas from the psychodynamic perspective have been studied empirically. Luborsky and Barrett (2006) reviewed much of this research; other useful reviews are provided by Bornstein (2005), Gerber (2007), and Huprich (2009). For now, let’s look at three psychodynamic hypotheses that have received strong empirical support.

- *Unconscious processes influence our behavior as the psychodynamic perspective predicts.* We perceive and process much more information than we realize, and much of our behavior is shaped by feelings and motives of which we are, at best, only partially aware (Bornstein, 2009, 2010). Evidence for the importance of unconscious influences is so compelling that it has become a central element of contemporary cognitive and social psychology (Robinson & Gordon, 2011).
- *We all use ego defenses and they help determine our psychological adjustment and physical health.* People really do differ in the degree that they rely on different ego defenses—so much so that researchers now study each person’s “defense style” (the unique constellation of defenses that we use). It turns out that certain defenses are more adaptive than others: Rationalization and sublimation are healthier (psychologically speaking) than repression and reaction formation (Cramer, 2006). Denial is, quite literally, bad for your health, because people who use denial tend to ignore symptoms of illness until it’s too late (Bond, 2004).
- *Mental representations of self and others do indeed serve as blueprints for later relationships.* Dozens of studies have shown that mental images of our parents, and other significant figures, really do shape our expectations for later friendships and romantic relationships. The idea that you choose a romantic partner who resembles mom or dad is a myth, but it’s true that you expect to be treated by others as you were treated by your parents early in life (Silverstein, 2007; Wachtel, 1997).

Psychoanalysis and Culture

One of Freud’s lifelong goals was to use psychoanalytic principles to understand culture and improve intergroup relations (he actually exchanged several letters with Albert Einstein prior to World War II, in which they discussed this issue). During the past several decades, as society has become increasingly multicultural, this effort has taken on new importance; psychoanalysts have been active in incorporating ideas and findings regarding cultural influences into their research and clinical work. For example, studies have shown that individuals raised in individualistic, independence-focused cultures (for example, the United States, Great Britain) tend to define themselves primarily in terms of personal attributes (like attitudes and interests), whereas individuals raised in more sociocentric, interdependent cultures (for example, Japan, India) are more likely to describe themselves in terms of interpersonal relations and connections with others (Oyserman, Coon, & Kemmelmeier, 2002). Our self-representations are, quite literally, a product of our cultural milieu (Markus & Kitayama, 2010).

The Opportunities and Challenges of Neuroscience

Fifteen years ago, Nobel Laureate Eric Kandel (1998) articulated a vision for an empirically oriented psychodynamic perspective firmly embedded within the principles and findings of neuroscience. Kandel's vision ultimately led to the development of **neuropsychanalysis**, an integration of psychodynamic and neuropsychological concepts that has enhanced researchers' understanding of numerous aspects of human behavior and mental functioning (Solms & Turnbull, 2011). Some of the first efforts to integrate psychodynamic principles with findings from neuroscience involved sleep and dreams, and contemporary models of dream formation now incorporate principles from both domains (Levin & Nielsen, 2007). Neuroimaging techniques such as functional magnetic resonance imagery (fMRI) have begun to play an increasingly central role in this ongoing psychoanalysis–neuroscience integration as well (Gerber, 2007; Slipp, 2000).

Looking Ahead: Psychodynamic Theory in the 21st Century (and Beyond)

Despite being surrounded by controversy, the psychodynamic perspective on personality has survived for more than a century, reinventing itself in response to new empirical findings, theoretical shifts, and changing social forces. The psychodynamic perspective evolved considerably during the 20th century and will continue to evolve throughout the 21st century as well. Psychodynamic theory may be the closest thing we have to an overarching, all-encompassing theory in psychology. It deals with a broad range of issues—normal and pathological functioning, motivation and emotion, childhood and adulthood, individual and culture—and the psychodynamic perspective continues to have tremendous potential for integrating ideas and findings across the many domains of contemporary psychology.

Outside Resources

Institution: Institute for Psychoanalytic Training and Research (IPTAR) - A branch of the International Psychoanalytic Association, IPTAR plays an active role in supporting empirical research on psychoanalytic theory and therapy.

<http://www.iptar.org/>

Institution: The American Psychoanalytic Association - The American Psychoanalytic Association supports psychodynamic training and research, and sponsors a number of workshops (as well as two annual meetings) each year.

<http://www.apsa.org/>

Institution: The American Psychological Association Division of Psychoanalysis - Division 39 of the American Psychological Association is the “psychological home” of psychodynamic theory and research.

<http://www.apadivisions.org/division-39/>

Web: Library of Congress Exhibit – Freud: Conflict and Culture. This is a terrific website full of photos, original manuscripts, and links to various Freud artifacts. Toward the end of Section Three (From the Individual to Society) there is a link to Freud’s 1938 BBC radio address; play it and you’ll get to hear Freud’s voice.

<http://www.loc.gov/exhibits/freud/>

Discussion Questions

1. What is psychic causality?
2. What are the main differences between the preconscious and the unconscious in Freud’s topographic model?
3. What are the three key structures in the structural model of the mind—and what does each structure do?
4. Which ego defense do you think is more adaptive: reaction formation or sublimation? Why?
5. How do people raised in individualistic societies differ from those raised in more sociocentric societies with respect to their self-concept—how do they perceive and describe themselves?
6. According to object relations theory, how do early relationships with our parents and other

significant figures affect later friendships and romantic relationships?

7. Which field has the potential to benefit more from the emerging new discipline of neuropsychanalysis: neuroscience, or psychoanalysis? Why?

Vocabulary

Ego defenses

Mental strategies, rooted in the ego, that we use to manage anxiety when we feel threatened (some examples include repression, denial, sublimation, and reaction formation).

Neuropsychanalysis

An integrative, interdisciplinary domain of inquiry seeking to integrate psychoanalytic and neuropsychological ideas and findings to enhance both areas of inquiry (you can learn more by visiting the webpage of the International Neuropsychanalysis Society at <http://www.neuropsa.org.uk/>).

Object relations theory

A modern offshoot of the psychodynamic perspective, this theory contends that personality can be understood as reflecting mental images of significant figures (especially the parents) that we form early in life in response to interactions taking place within the family; these mental images serve as templates (or “scripts”) for later interpersonal relationships.

Primacy of the Unconscious

The hypothesis—supported by contemporary empirical research—that the vast majority of mental activity takes place outside conscious awareness.

Psychic causality

The assumption that nothing in mental life happens by chance—that there is no such thing as a “random” thought or feeling.

Psychosexual stage model

Probably the most controversial aspect of psychodynamic theory, the psychosexual stage model contends that early in life we progress through a sequence of developmental stages (oral, anal, Oedipal, latency, and genital), each with its own unique mode of sexual gratification.

Structural model

Developed to complement and extend the topographic model, the structural model of the mind posits the existence of three interacting mental structures called the id, ego, and superego.

Topographic model

Freud’s first model of the mind, which contended that the mind could be divided into three

regions: conscious, preconscious, and unconscious. (The “topographic” comes from the fact that topography is the study of maps.)

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Personality Traits

Edward Diener & Richard E. Lucas

Personality traits reflect people's characteristic patterns of thoughts, feelings, and behaviors. Personality traits imply consistency and stability—someone who scores high on a specific trait like Extraversion is expected to be sociable in different situations and over time. Thus, trait psychology rests on the idea that people differ from one another in terms of where they stand on a set of basic trait dimensions that persist over time and across situations. The most widely used system of traits is called the Five-Factor Model. This system includes five broad traits that can be remembered with the acronym OCEAN: Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. Each of the major traits from the Big Five can be divided into facets to give a more fine-grained analysis of someone's personality. In addition, some trait theorists argue that there are other traits that cannot be completely captured by the Five-Factor Model. Critics of the trait concept argue that people do not act consistently from one situation to the next and that people are very influenced by situational forces. Thus, one major debate in the field concerns the relative power of people's traits versus the situations in which they find themselves as predictors of their behavior.

Learning Objectives

- List and describe the “Big Five” (“OCEAN”) personality traits that comprise the Five-Factor Model of personality.
- Describe how the facet approach extends broad personality traits.
- Explain a critique of the personality-trait concept.
- Describe in what ways personality traits may be manifested in everyday behavior.
- Describe each of the Big Five personality traits, and the low and high end of the dimension.
- Give examples of each of the Big Five personality traits, including both a low and high example.

- Describe how traits and social learning combine to predict your social activities.
- Describe your theory of how personality traits get refined by social learning.

Introduction

When we observe people around us, one of the first things that strikes us is how different people are from one another. Some people are very talkative while others are very quiet. Some are active whereas others are couch potatoes. Some worry a lot, others almost never seem anxious. Each time we use one of these words, words like “talkative,” “quiet,” “active,” or “anxious,” to describe those around us, we are talking about a person’s **personality**—the characteristic ways that people differ from one another. Personality psychologists try to describe and understand these differences.

Although there are many ways to think about the personalities that people have, Gordon Allport and other “personologists” claimed that we can best understand the differences between individuals by understanding their personality traits. **Personality traits** reflect basic dimensions on which people differ (Matthews, Deary, & Whiteman, 2003). According to trait psychologists, there are a limited number of these dimensions (dimensions like Extraversion, Conscientiousness, or Agreeableness), and each individual falls somewhere on each dimension, meaning that they could be low, medium, or high on any specific trait.

An important feature of personality traits is that they reflect **continuous distributions** rather than distinct personality types. This means that when personality psychologists talk about Introverts and Extraverts, they are not really talking about two distinct types of people who are completely and qualitatively different from one another. Instead, they are talking about people who score relatively low or relatively high along a



“Are you an introvert?” In popular culture it’s common to talk about people being introverts or extroverts as if these were precise descriptions that meant the same thing for everyone. But research shows that these traits and others are quite variable within individuals. [Image: Nguyen Hung Vu, <https://goo.gl/qKJUAC>, CC BY 2.0, <https://goo.gl/BRvSA7>]

continuous distribution. In fact, when personality psychologists measure traits like **Extraversion**, they typically find that most people score somewhere in the middle, with smaller numbers showing more extreme levels. The figure below shows the distribution of Extraversion scores from a survey of thousands of people. As you can see, most people report being moderately, but not extremely, extraverted, with fewer people reporting very high or very low scores.

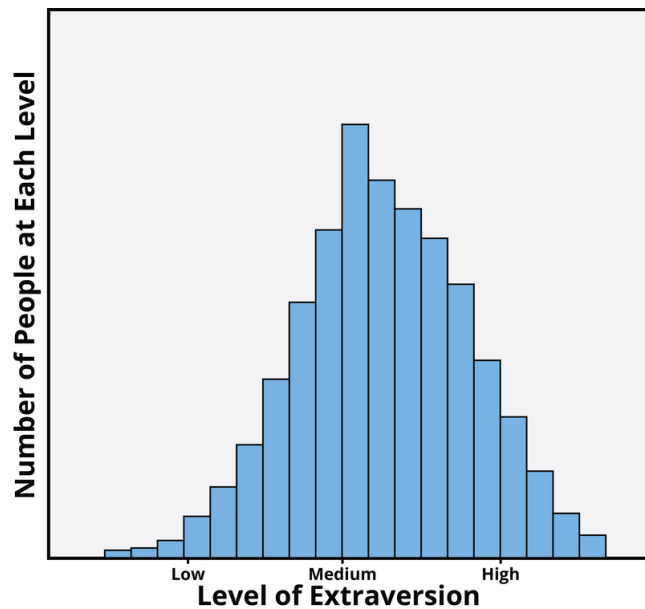


Figure 1. Distribution of Extraversion Scores in a Sample Higher bars mean that more people have scores of that level. This figure shows that most people score towards the middle of the extraversion scale, with fewer people who are highly extraverted or highly introverted.

There are three criteria that are characterize personality traits: (1) consistency, (2) stability, and (3) individual differences.

1. To have a personality trait, individuals must be somewhat consistent across situations in their behaviors related to the trait. For example, if they are talkative at home, they tend also to be talkative at work.

2. Individuals with a trait are also somewhat stable over time in behaviors related to the trait. If they are talkative, for example, at age 30, they will also tend to be talkative at age 40.

3. People differ from one another on behaviors related to the trait. Using speech is not a personality trait and neither is walking on two feet—virtually

all individuals do these activities, and there are almost no individual differences. But people differ on how frequently they talk and how active they are, and thus personality traits such as Talkativeness and Activity Level do exist.

A challenge of the trait approach was to discover the major traits on which all people differ. Scientists for many decades generated hundreds of new traits, so that it was soon difficult to keep track and make sense of them. For instance, one psychologist might focus on individual differences in “friendliness,” whereas another might focus on the highly related concept of “sociability.” Scientists began seeking ways to reduce the number of traits in some systematic way and to discover the basic traits that describe most of the differences between people.

The way that Gordon Allport and his colleague Henry Odbert approached this was to search

the dictionary for all descriptors of personality (Allport & Odbert, 1936). Their approach was guided by the **lexical hypothesis**, which states that all important personality characteristics should be reflected in the language that we use to describe other people. Therefore, if we want to understand the fundamental ways in which people differ from one another, we can turn to the words that people use to describe one another. So if we want to know what words people use to describe one another, where should we look? Allport and Odbert looked in the most obvious place—the dictionary. Specifically, they took all the personality descriptors that they could find in the dictionary (they started with almost 18,000 words but quickly reduced that list to a more manageable number) and then used statistical techniques to determine which words “went together.” In other words, if everyone who said that they were “friendly” also said that they were “sociable,” then this might mean that personality psychologists would only need a single trait to capture individual differences in these characteristics. Statistical techniques were used to determine whether a small number of dimensions might underlie all of the thousands of words we use to describe people.

The Five-Factor Model of Personality

Research that used the lexical approach showed that many of the personality descriptors found in the dictionary do indeed overlap. In other words, many of the words that we use to describe people are synonyms. Thus, if we want to know what a person is like, we do not necessarily need to ask how sociable they are, how friendly they are, and how gregarious they are. Instead, because sociable people tend to be friendly and gregarious, we can summarize this personality dimension with a single term. Someone who is sociable, friendly, and gregarious would typically be described as an “Extravert.” Once we know she is an extravert, we can assume that she is sociable, friendly, and gregarious.

Statistical methods (specifically, a technique called **factor analysis**) helped to determine whether a small number of dimensions underlie the diversity of words that people like Allport and Odbert identified. The most widely accepted system to emerge from this approach was “The Big Five” or “**Five-Factor Model**” (Goldberg, 1990; McCrae & John, 1992; McCrae & Costa, 1987). The Big Five comprises five major traits shown in the Figure 2 below. A way to remember these five is with the acronym OCEAN (O is for **Openness**; C is for **Conscientiousness**; E is for **Extraversion**; A is for **Agreeableness**; N is for **Neuroticism**). Figure 3 provides descriptions of people who would score high and low on each of these traits.

Scores on the Big Five traits are mostly independent. That means that a person’s standing on one trait tells very little about their standing on the other traits of the Big Five. For example, a person can be extremely high in Extraversion and be either high or low on Neuroticism.

Big 5 Trait	Definition
Openness	The tendency to appreciate new art, ideas, values, feelings, and behaviors.
Conscientiousness	The tendency to be careful, on-time for appointments, to follow rules, and to be hardworking.
Extraversion	The tendency to be talkative, sociable, and to enjoy others; the tendency to have a dominant style.
Agreeableness	The tendency to agree and go along with others rather than to assert one's own opinions and choices.
Neuroticism	The tendency to frequently experience negative emotions such as anger, worry, and sadness, as well as being interpersonally sensitive.

Figure 2. Descriptions of the Big Five Personality Traits

Similarly, a person can be low in Agreeableness and be either high or low in Conscientiousness. Thus, in the Five-Factor Model, you need five scores to describe most of an individual's personality.

In the Appendix to this module, we present a short scale to assess the Five-Factor Model of personality (Donnellan, Oswald, Baird, & Lucas, 2006). You can take this test to see where you stand in terms of your Big Five scores. John Johnson has also created a helpful website that has personality scales that can be used and taken by the general public:

<http://www.personal.psu.edu/j5j/IPIP/ipipneo120.htm>

After seeing your scores, you can judge for yourself whether you think such tests are valid.

Traits are important and interesting because they describe stable patterns of behavior that persist for long periods of time (Caspi, Roberts, & Shiner, 2005). Importantly, these stable patterns can have broad-ranging consequences for many areas of our life (Roberts, Kuncel, Shiner, Caspi, & Goldberg, 2007). For instance, think about the factors that determine success in college. If you were asked to guess what factors predict good grades in college, you might guess something like intelligence. This guess would be correct, but we know much more about who is likely to do well. Specifically, personality researchers have also found the personality traits like Conscientiousness play an important role in college and beyond, probably because highly conscientious individuals study hard, get their work done on time, and are less distracted by nonessential activities that take time away from school work. In addition, highly conscientious people are often healthier than people low in conscientiousness because they are more likely to maintain healthy diets, to exercise, and to follow basic safety procedures like wearing seat belts or bicycle helmets. Over the long term, this consistent pattern of

Big 5 Trait	Example Behavior for LOW Scorers	Example Behavior for HIGH Scorers
<i>Openness</i>	Prefers not to be exposed to alternative moral systems; narrow interests; inartistic; not analytical; down-to-earth	Enjoys seeing people with new types of haircuts and body piercing; curious; imaginative; untraditional
<i>Conscientiousness</i>	Prefers spur-of-the-moment action to planning; unreliable; hedonistic; careless; lax	Never late for a date; organized; hardworking; neat; persevering; punctual; self-disciplined
<i>Extraversion</i>	Preferring a quiet evening reading to a loud party; sober; aloof; unenthusiastic	Being the life of the party; active; optimistic; fun-loving; affectionate
<i>Agreeableness</i>	Quickly and confidently asserts own rights; irritable; manipulative; uncooperative; rude	Agrees with others about political opinions; good-natured; forgiving; gullible; helpful; forgiving
<i>Neuroticism</i>	Not getting irritated by small annoyances; calm, unemotional; hardy; secure; self-satisfied	Constantly worrying about little things; insecure; hypochondriacal; feeling inadequate

Figure 3. Example behaviors for those scoring low and high for the big 5 traits

behaviors can add up to meaningful differences in health and longevity. Thus, personality traits are not just a useful way to describe people you know; they actually help psychologists predict how good a worker someone will be, how long he or she will live, and the types of jobs and activities the person will enjoy. Thus, there is growing interest in personality psychology among psychologists who work in applied settings, such as health psychology or organizational psychology.

Facets of Traits (Subtraits)

So how does it feel to be told that your entire personality can be summarized with scores on just five personality traits? Do you think these five scores capture the complexity of your own and others' characteristic patterns of thoughts, feelings, and behaviors? Most people would probably say no, pointing to some exception in their behavior that goes against the general pattern that others might see. For instance, you may know people who are warm and friendly and find it easy to talk with strangers at a party yet are terrified if they have to perform in front of others or speak to large groups of people. The fact that there are different ways of being extraverted or conscientious shows that there is value in considering lower-level units of personality that are more specific than the Big Five traits. These more specific, lower-level units of personality are often called **facets**.

To give you a sense of what these narrow units are like, Figure 4 shows facets for each of the

Trait	Facets of Trait
Openness	• Fantasy prone • Open to feelings • Open to diverse behaviors • Open to new and different ideas • Open to various values and beliefs
Conscientiousness	• Competent • Orderly • Dutiful • Achievement oriented • Self-disciplined • Deliberate
Extraversion	• Gregarious (sociable) • Warm • Assertive • Active • Excitement-seeking • Positive emotionality
Agreeableness	• Trusting • Straightforward • Altruistic • Compliant • Modest • Tender-minded
Neuroticism	• Anxious • Angry • Depressed • Self-consciousness • Impulsive • Vulnerable

Figure 4. Facets of Traits

Big Five traits. It is important to note that although personality researchers generally agree about the value of the Big Five traits as a way to summarize one's personality, there is no widely accepted list of facets that should be studied. The list seen here, based on work by researchers Paul Costa and Jeff McCrae, thus reflects just one possible list among many. It should, however, give you an idea of some of the facets making up each of the Five-Factor Model.

Facets can be useful because they provide more specific descriptions of what a person is like. For instance, if we take our friend who loves parties but hates public speaking, we might say that this person scores high on the "gregariousness" and "warmth" facets of extraversion, while scoring lower on facets such as "assertiveness" or "excitement-seeking." This precise profile of facet scores not only provides a better description, it might also allow us to better predict how this friend will do in a variety of different jobs (for example, jobs that require public speaking versus jobs that involve one-on-one interactions with customers; Paunonen & Ashton, 2001). Because different facets within a broad, global trait like extraversion

tend to go together (those who are gregarious are often but not always assertive), the broad trait often provides a useful summary of what a person is like. But when we really want to know a person, facet scores add to our knowledge in important ways.

Other Traits Beyond the Five-Factor Model

Despite the popularity of the Five-Factor Model, it is certainly not the only model that exists. Some suggest that there are more than five major traits, or perhaps even fewer. For example, in one of the first comprehensive models to be proposed, Hans Eysenck suggested that

Extraversion and Neuroticism are most important. Eysenck believed that by combining people's standing on these two major traits, we could account for many of the differences in personality that we see in people (Eysenck, 1981). So for instance, a neurotic introvert would be shy and nervous, while a stable introvert might avoid social situations and prefer solitary activities, but he may do so with a calm, steady attitude and little anxiety or emotion. Interestingly, Eysenck attempted to link these two major dimensions to underlying differences in people's biology. For instance, he suggested that introverts experienced too much sensory stimulation and arousal, which made them want to seek out quiet settings and less stimulating environments. More recently, Jeffrey Gray suggested that these two broad traits are related to fundamental reward and avoidance systems in the brain—extraverts might be motivated to seek reward and thus exhibit assertive, reward-seeking behavior, whereas people high in neuroticism might be motivated to avoid punishment and thus may experience anxiety as a result of their heightened awareness of the threats in the world around them (Gray, 1981. This model has since been updated; see Gray & McNaughton, 2000). These early theories have led to a burgeoning interest in identifying the physiological underpinnings of the individual differences that we observe.

Another revision of the Big Five is the **HEXACO model** of traits (Ashton & Lee, 2007). This model is similar to the Big Five, but it posits slightly different versions of some of the traits, and its proponents argue that one important class of individual differences was omitted from the Five-Factor Model. The HEXACO adds Honesty-Humility as a sixth dimension of personality. People high in this trait are sincere, fair, and modest, whereas those low in the trait are manipulative, narcissistic, and self-centered. Thus, trait theorists are agreed that personality traits are important in understanding behavior, but there are still debates on the exact number and composition of the traits that are most important.

There are other important traits that are not included in comprehensive models like the Big Five. Although the five factors capture much that is important about personality, researchers have suggested other traits that capture interesting aspects of our behavior. In Figure 5 below we present just a few, out of hundreds, of the other traits that have been studied by personologists.

Not all of the above traits are currently popular with scientists, yet each of them has experienced popularity in the past. Although the Five-Factor Model has been the target of more rigorous research than some of the traits above, these additional personality characteristics give a good idea of the wide range of behaviors and attitudes that traits can cover.

The Person-Situation Debate and Alternatives to the Trait Perspective

Personality Trait	Description
<i>Machiavellianism</i>	Named after the famous political philosopher, Niccolo Machiavelli, this trait refers to individuals who manipulate the behavior of others, often through duplicity. Machiavellians are often interested in money and power, and pragmatically use others in this quest.
<i>Need for Achievement</i>	Those high in need for achievement want to accomplish a lot and set high standards of excellence for themselves. They are able to work persistently and hard for distant goals. David McClelland argued that economic growth depends in part on citizens with high need for achievement.
<i>Need for Cognition</i>	People high in need for cognition find it rewarding to understand things, and are willing to use considerable cognitive effort in this quest. Such individuals enjoy learning, and the process of trying to understand new things.
<i>Authoritarianism</i>	Authoritarians believe in strict social hierarchies, in which they are totally obedient to those above them, and expect complete obedience from their subordinates. Rigid in adherence to rules, the authoritarian personality is very uncomfortable with uncertainty.
<i>Narcissism</i>	The narcissistic personality has self-love that is so strong that it results in high levels of vanity, conceit, and selfishness. The narcissistic individual often has problems feeling empathetic toward others and grateful to others.
<i>Self-esteem</i>	The tendency to evaluate oneself positively. Self-esteem does not imply that one believes that he or she is better than others, only that he or she is a person of worth.
<i>Optimism</i>	The tendency to expect positive outcomes in the future. People who are optimistic expect good things to happen, and indeed they often have more positive outcomes, perhaps because they work harder to achieve them.
<i>Alexithymia</i>	The inability to recognize and label emotions in oneself. The individual also has a difficult time recognizing emotions in others, and often has difficulties in relationships.

Figure 5. Other Traits Beyond Those Included in the Big Five

The ideas described in this module should probably seem familiar, if not obvious to you. When asked to think about what our friends, enemies, family members, and colleagues are like, some of the first things that come to mind are their personality characteristics. We might think about how warm and helpful our first teacher was, how irresponsible and careless our brother is, or how demanding and insulting our first boss was. Each of these descriptors reflects a personality trait, and most of us generally think that the descriptions that we use for individuals accurately reflect their “characteristic pattern of thoughts, feelings, and behaviors,” or in other

words, their personality.

But what if this idea were wrong? What if our belief in personality traits were an illusion and people are not consistent from one situation to the next? This was a possibility that shook the foundation of personality psychology in the late 1960s when Walter Mischel published a book called *Personality and Assessment* (1968). In this book, Mischel suggested that if one looks closely at people's behavior across many different situations, the consistency is really not that impressive. In other words, children who cheat on tests at school may steadfastly follow all rules when playing games and may never tell a lie to their parents. In other words, he suggested, there may not be any general trait of honesty that links these seemingly related behaviors. Furthermore, Mischel

suggested that observers may believe that broad personality traits like honesty exist, when in fact, this belief is an illusion. The debate that followed the publication of Mischel's book was called the person-situation debate because it pitted the power of personality against the power of situational factors as determinants of the behavior that people exhibit.

Because of the findings that Mischel emphasized, many psychologists focused on an alternative to the trait perspective. Instead of studying broad, context-free descriptions, like the trait terms we've described so far, Mischel thought that psychologists should focus on people's distinctive reactions to specific situations. For instance, although there may not be a broad and general trait of honesty, some children may be especially likely to cheat on a test when the risk of being caught is low and the rewards for cheating are high. Others might be motivated by the sense of risk involved in cheating and may do so even when the rewards are not very high. Thus, the behavior itself results from the child's unique evaluation of the risks and rewards present at that moment, along with her evaluation of her abilities and values. Because of this, the same child might act very differently in different situations. Thus, Mischel thought that specific behaviors were driven by the interaction between very specific, psychologically meaningful features of the situation in which people found themselves, the person's unique way of perceiving that situation, and his or her abilities for dealing with it.



The way people behave is only in part a product of their natural personality. Situations also influence how a person behaves. Are you for instance a “different person” as a student in a classroom compared to when you’re a member of a close-knit social group? [Image: UO Education, <https://goo.gl/yIgv9T>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

Mischel and others argued that it was these social-cognitive processes that underlie people's reactions to specific situations that provide some consistency when situational features are the same. If so, then studying these broad traits might be more fruitful than cataloging and measuring narrow, context-free traits like Extraversion or Neuroticism.

In the years after the publication of Mischel's (1968) book, debates raged about whether personality truly exists, and if so, how it should be studied. And, as is often the case, it turns out that a more moderate middle ground than what the situationists proposed could be reached. It is certainly true, as Mischel pointed out, that a person's behavior in one specific situation is not a good guide to how that person will behave in a very different specific situation. Someone who is extremely talkative at one specific party may sometimes be reticent to speak up during class and may even act like a wallflower at a different party. But this does not mean that personality does not exist, nor does it mean that people's behavior is completely determined by situational factors. Indeed, research conducted after the person-situation debate shows that on average, the effect of the "situation" is about as large as that of personality traits. However, it is also true that if psychologists assess a broad range of behaviors across many different situations, there are general tendencies that emerge. Personality traits give an indication about how people will act on average, but frequently they are not so good at predicting how a person will act in a specific situation at a certain moment in time. Thus, to best capture broad traits, one must assess *aggregate* behaviors, averaged over time and across many different types of situations. Most modern personality researchers agree that there is a place for broad personality traits and for the narrower units such as those studied by Walter Mischel.

Appendix

The Mini-IPIP Scale

(Donnellan, Oswald, Baird, & Lucas, 2006)

Instructions: Below are phrases describing people's behaviors. Please use the rating scale below to describe how accurately each statement describes you. Describe yourself as you generally are now, not as you wish to be in the future. Describe yourself as you honestly see yourself, in relation to other people you know of the same sex as you are, and roughly your same age. Please read each statement carefully, and put a number from 1 to 5 next to it to describe how accurately the statement describes you.

1 = Very inaccurate

2 = Moderately inaccurate

3 = Neither inaccurate nor accurate

4 = Moderately accurate

5 = Very accurate

1. _____ Am the life of the party (E)
2. _____ Sympathize with others' feelings (A)
3. _____ Get chores done right away (C)
4. _____ Have frequent mood swings (N)
5. _____ Have a vivid imagination (O)
6. _____ Don't talk a lot (E)
7. _____ Am not interested in other people's problems (A)
8. _____ Often forget to put things back in their proper place (C)
9. _____ Am relaxed most of the time (N)
10. _____ Am not interested in abstract ideas (O)
11. _____ Talk to a lot of different people at parties (E)
12. _____ Feel others' emotions (A)
13. _____ Like order (C)
14. _____ Get upset easily (N)
15. _____ Have difficulty understanding abstract ideas (O)
16. _____ Keep in the background (E)
17. _____ Am not really interested in others (A)
18. _____ Make a mess of things (C)
19. _____ Seldom feel blue (N)
20. _____ Do not have a good imagination (O)

Scoring: The first thing you must do is to reverse the items that are worded in the opposite direction. In order to do this, subtract the number you put for that item from 6. So if you put a 4, for instance, it will become a 2. Cross out the score you put when you took the scale, and

put the new number in representing your score subtracted from the number 6.

Items to be reversed in this way: 6, 7, 8, 9, 10, 15, 16, 17, 18, 19, 20

Next, you need to add up the scores for each of the five OCEAN scales (including the reversed numbers where relevant). Each OCEAN score will be the sum of four items. Place the sum next to each scale below.

_____ Openness: Add items 5, 10, 15, 20

_____ Conscientiousness: Add items 3, 8, 13, 18

_____ Extraversion: Add items 1, 6, 11, 16

_____ Agreeableness: Add items 2, 7, 12, 17

_____ Neuroticism: Add items 4, 9, 14, 19

Compare your scores to the norms below to see where you stand on each scale. If you are low on a trait, it means you are the opposite of the trait label. For example, low on Extraversion is Introversion, low on Openness is Conventional, and low on Agreeableness is Assertive.

19–20 Extremely High, 17–18 Very High, 14–16 High,

11–13 Neither high nor low; in the middle, 8–10 Low, 6–7 Very low, 4–5 Extremely low

Outside Resources

Video 1: Gabriela Cintron's – 5 Factors of Personality (OCEAN Song). This is a student-made video which cleverly describes, through song, common behavioral characteristics of the Big 5 personality traits. It was one of the winning entries in the 2016-17 Noba + Psi Chi Student Video Award.

https://www.youtube.com/watch?feature=youtu.be&v=Rk8CDXMb8_U&app=desktop

Video 2: Michael Harris' – Personality Traits: The Big 5 and More. This is a student-made video that looks at characteristics of the OCEAN traits through a series of funny vignettes. It also presents on the Person vs Situation Debate. It was one of the winning entries in the 2016-17 Noba + Psi Chi Student Video Award.

<https://vimeo.com/218245492>

Video 3: David M. Cole's – Grouchy with a Chance of Stomping. This is a student-made video that makes a very important point about the relationship between personality traits and behavior using a handy weather analogy. It was one of the winning entries in the 2016-17 Noba + Psi Chi Student Video Award.

<https://www.youtube.com/watch?v=GnaFMjaItlY>

Web: International Personality Item Pool

<http://ipip.ori.org/>

Web: John Johnson personality scales

<http://www.personal.psu.edu/j5j/IPIP/ipipneo120.htm>

Web: Personality trait systems compared

<http://www.personalityresearch.org/bigfive/goldberg.html>

Web: Sam Gosling website

<http://homepage.psy.utexas.edu/homepage/faculty/gosling/samgosling.htm>

Discussion Questions

1. Consider different combinations of the Big Five, such as O (Low), C (High), E (Low), A (High), and N (Low). What would this person be like? Do you know anyone who is like this? Can you select politicians, movie stars, and other famous people and rate them on the Big Five?

2. How do you think learning and inherited personality traits get combined in adult personality?
3. Can you think of instances where people do not act consistently—where their personality traits are not good predictors of their behavior?
4. Has your personality changed over time, and in what ways?
5. Can you think of a personality trait not mentioned in this module that describes how people differ from one another?
6. When do extremes in personality traits become harmful, and when are they unusual but productive of good outcomes?

Vocabulary

Agreeableness

A personality trait that reflects a person's tendency to be compassionate, cooperative, warm, and caring to others. People low in agreeableness tend to be rude, hostile, and to pursue their own interests over those of others.

Conscientiousness

A personality trait that reflects a person's tendency to be careful, organized, hardworking, and to follow rules.

Continuous distributions

Characteristics can go from low to high, with all different intermediate values possible. One does not simply have the trait or not have it, but can possess varying amounts of it.

Extraversion

A personality trait that reflects a person's tendency to be sociable, outgoing, active, and assertive.

Facets

Broad personality traits can be broken down into narrower facets or aspects of the trait. For example, extraversion has several facets, such as sociability, dominance, risk-taking and so forth.

Factor analysis

A statistical technique for grouping similar things together according to how highly they are associated.

Five-Factor Model

(also called the Big Five) The Five-Factor Model is a widely accepted model of personality traits. Advocates of the model believe that much of the variability in people's thoughts, feelings, and behaviors can be summarized with five broad traits. These five traits are Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism.

HEXACO model

The HEXACO model is an alternative to the Five-Factor Model. The HEXACO model includes six traits, five of which are variants of the traits included in the Big Five (Emotionality [E], Extraversion [X], Agreeableness [A], Conscientiousness [C], and Openness [O]). The sixth

factor, Honesty-Humility [H], is unique to this model.

Independent

Two characteristics or traits are separate from one another-- a person can be high on one and low on the other, or vice-versa. Some correlated traits are relatively independent in that although there is a tendency for a person high on one to also be high on the other, this is not always the case.

Lexical hypothesis

The lexical hypothesis is the idea that the most important differences between people will be encoded in the language that we use to describe people. Therefore, if we want to know which personality traits are most important, we can look to the language that people use to describe themselves and others.

Neuroticism

A personality trait that reflects the tendency to be interpersonally sensitive and the tendency to experience negative emotions like anxiety, fear, sadness, and anger.

Openness to Experience

A personality trait that reflects a person's tendency to seek out and to appreciate new things, including thoughts, feelings, values, and experiences.

Personality

Enduring predispositions that characterize a person, such as styles of thought, feelings and behavior.

Personality traits

Enduring dispositions in behavior that show differences across individuals, and which tend to characterize the person across varying types of situations.

Person-situation debate

The person-situation debate is a historical debate about the relative power of personality traits as compared to situational influences on behavior. The situationist critique, which started the person-situation debate, suggested that people overestimate the extent to which personality traits are consistent across situations.

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Self and Identity

Dan P. McAdams

For human beings, the self is what happens when “I” encounters “Me.” The central psychological question of selfhood, then, is this: How does a person apprehend and understand who he or she is? Over the past 100 years, psychologists have approached the study of self (and the related concept of identity) in many different ways, but three central metaphors for the self repeatedly emerge. First, the self may be seen as a social actor, who enacts roles and displays traits by performing behaviors in the presence of others. Second, the self is a motivated agent, who acts upon inner desires and formulates goals, values, and plans to guide behavior in the future. Third, the self eventually becomes an autobiographical author, too, who takes stock of life — past, present, and future — to create a story about who I am, how I came to be, and where my life may be going. This module briefly reviews central ideas and research findings on the self as an actor, an agent, and an author, with an emphasis on how these features of selfhood develop over the human life course.

Learning Objectives

- Explain the basic idea of reflexivity in human selfhood—how the “I” encounters and makes sense of itself (the “Me”).
- Describe fundamental distinctions between three different perspectives on the self: the self as actor, agent, and author.
- Describe how a sense of self as a social actor emerges around the age of 2 years and how it develops going forward.
- Describe the development of the self’s sense of motivated agency from the emergence of the child’s theory of mind to the articulation of life goals and values in adolescence and beyond.
- Define the term narrative identity, and explain what psychological and cultural functions

narrative identity serves.

Introduction

In the Temple of Apollo at Delphi, the ancient Greeks inscribed the words: “Know thyself.” For at least 2,500 years, and probably longer, human beings have pondered the meaning of the ancient aphorism. Over the past century, psychological scientists have joined the effort. They have formulated many theories and tested countless hypotheses that speak to the central question of human selfhood: *How does a person know who he or she is?*



We work on ourselves as we would any other interesting project. And when we do we generally focus on three psychological categories - The Social Actor, The Motivated Agent, and The Autobiographical Author. [Image: MakuKulden, <https://goo.gl/sMUsnj>], CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

The ancient Greeks seemed to realize that the self is inherently **reflexive**—it reflects back on itself. In the disarmingly simple idea made famous by the great psychologist William James (1892/1963), the self is what happens when “I” reflects back upon “Me.” The self is both the I and the Me—it is the knower, and it is what the knower knows when the knower reflects upon itself. When you look back at yourself, what do you see? When you look inside, what do you find? Moreover, when you try to *change* your self in some way, what is it that you are trying to change? The philosopher Charles Taylor (1989) describes the self as a reflexive *project*. In modern life, Taylor argues, we often try to manage, discipline, refine, improve, or develop the self. We *work on* our selves, as we might work on any other interesting project. But

what exactly is it that we work on?

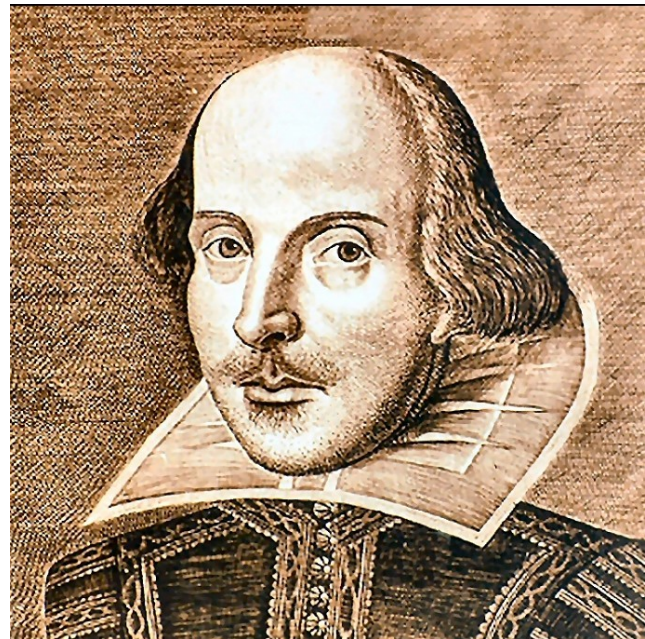
Imagine for a moment that you have decided to improve yourself. You might, say, go on a diet to improve your appearance. Or you might decide to be nicer to your mother, in order to improve that important social role. Or maybe the problem is at work—you need to find a

better job or go back to school to prepare for a different career. Perhaps you just need to work harder. Or get organized. Or recommit yourself to religion. Or maybe the key is to begin thinking about your whole life story in a completely different way, in a way that you hope will bring you more happiness, fulfillment, peace, or excitement.

Although there are many different ways you might reflect upon and try to improve the self, it turns out that many, if not most, of them fall roughly into three broad psychological categories (McAdams & Cox, 2010). The I may encounter the Me as (a) a social actor, (b) a motivated agent, or (c) an autobiographical author.

The Social Actor

Shakespeare tapped into a deep truth about human nature when he famously wrote, “All the world’s a stage, and all the men and women merely players.” He was wrong about the “merely,” however, for there is nothing more important for human adaptation than the manner in which we perform our roles as actors in the everyday theatre of social life. What Shakespeare may have sensed but could not have fully understood is that human beings evolved to live in social groups. Beginning with Darwin (1872/1965) and running through contemporary conceptions of human evolution, scientists have portrayed human nature as profoundly *social* (Wilson, 2012). For a few million years, *Homo sapiens* and their evolutionary forerunners have survived and flourished



In some ways people are just like actors on stage. We play roles and follow scripts every day. [Image: Brian, <https://goo.gl/z0VI3t>, CC BY-SA 2.0, <https://goo.gl/i4GXf5>]

by virtue of their ability to live and work together in complex social groups, cooperating with each other to solve problems and overcome threats and competing with each other in the face of limited resources. As social animals, human beings strive to *get along* and *get ahead* in the presence of each other (Hogan, 1982). Evolution has prepared us to care deeply about social acceptance and social status, for those unfortunate individuals who do not get along well in social groups or who fail to attain a requisite status among their peers have typically been severely compromised when it comes to survival and reproduction. It makes

consummate evolutionary sense, therefore, that the human "I" should apprehend the "Me" first and foremost as a *social actor*.

For human beings, the sense of the self as a social actor begins to emerge around the age of 18 months. Numerous studies have shown that by the time they reach their second birthday most toddlers recognize themselves in mirrors and other reflecting devices (Lewis & Brooks-Gunn, 1979; Rochat, 2003). What they see is an embodied actor who moves through space and time. Many children begin to use words such as "me" and "mine" in the second year of life, suggesting that the I now has linguistic labels that can be applied reflexively to itself: I call myself "me." Around the same time, children also begin to express social emotions such as embarrassment, shame, guilt, and pride (Tangney, Stuewig, & Mashek, 2007). These emotions tell the social actor how well he or she is performing in the group. When I do things that win the approval of others, I feel proud of myself. When I fail in the presence of others, I may feel embarrassment or shame. When I violate a social rule, I may experience guilt, which may motivate me to make amends.

Many of the classic psychological theories of human selfhood point to the second year of life as a key developmental period. For example, Freud (1923/1961) and his followers in the psychoanalytic tradition traced the emergence of an autonomous ego back to the second year. Freud used the term "ego" (in German *das Ich*, which also translates into "the I") to refer to an executive self in the personality. Erikson (1963) argued that experiences of trust and interpersonal attachment in the first year of life help to consolidate the autonomy of the ego in the second. Coming from a more sociological perspective, Mead (1934) suggested that the I comes to know the Me through reflection, which may begin quite literally with mirrors but later involves the reflected appraisals of others. I come to know who I am as a social actor, Mead argued, by noting how *other people* in my social world react to my performances. In the development of the self as a social actor, other people function like mirrors—they reflect who I am back to me.

Research has shown that when young children begin to make attributions about themselves, they start simple (Harter, 2006). At age 4, Jessica knows that she has dark hair, knows that she lives in a white house, and describes herself to others in terms of simple behavioral *traits*. She may say that she is "nice," or "helpful," or that she is "a good girl most of the time." By the time, she hits fifth grade (age 10), Jessica sees herself in more complex ways, attributing traits to the self such as "honest," "moody," "outgoing," "shy," "hard-working," "smart," "good at math but not gym class," or "nice except when I am around my annoying brother." By late childhood and early adolescence, the personality traits that people attribute to themselves, as well as those attributed to them by others, tend to correlate with each other in ways that conform to a well-established taxonomy of five broad trait domains, repeatedly derived in studies of

adult personality and often called the **Big Five**: (1) extraversion, (2) neuroticism, (3) agreeableness, (4) conscientiousness, and (5) openness to experience (Roberts, Wood, & Caspi, 2008). By late childhood, moreover, self-conceptions will likely also include important social roles: "I am a good student," "I am the oldest daughter," or "I am a good friend to Sarah."

Traits and roles, and variations on these notions, are the main currency of the **self as social actor** (McAdams & Cox, 2010). Trait terms capture perceived consistencies in social performance. They convey what I reflexively perceive to be my overall acting style, based in part on how I think others see me as an actor in many different social situations. Roles capture the quality, as I perceive it, of important structured relationships in my life. Taken together, traits and roles make up the main features of my **social reputation**, as I apprehend it in my own mind (Hogan, 1982).

If you have ever tried hard to change yourself, you may have taken aim at your social reputation, targeting your central traits or your social roles. Maybe you woke up one day and decided that you must become a more optimistic and emotionally upbeat person. Taking into consideration the reflected appraisals of others, you realized that even your friends seem to avoid you because you bring them down. In addition, it feels bad to feel so bad all the time: Wouldn't it be better to feel good, to have more energy and hope? In the language of traits, you have decided to "work on" your "neuroticism." Or maybe instead, your problem is the trait of "conscientiousness": You are undisciplined and don't work hard enough, so you resolve to make changes in that area. Self-improvement efforts such as these—aimed at changing one's traits to become a more effective social actor—are sometimes successful, but they are very hard—kind of like dieting. Research suggests that broad traits tend to be stubborn, resistant to change, even with the aid of psychotherapy. However, people often have more success working directly on their social roles. To become a more effective social actor, you may want to take aim at the important roles you play in life. What can I do to become a better son or daughter? How can I find new and meaningful roles to perform at work, or in my family, or among my friends, or in my church and community? By doing concrete things that enrich your performances in important social roles, you may begin to see yourself in a new light, and others will notice the change, too. Social actors hold the potential to transform their performances across the human life course. Each time you walk out on stage, you have a chance to start anew.

The Motivated Agent

Whether we are talking literally about the theatrical stage or more figuratively, as I do in this module, about the everyday social environment for human behavior, observers can never



When we observe others we only see how they act but are never able to access the entirety of their internal experience. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

fully know what is in the actor's head, no matter how closely they watch. We can see actors act, but we cannot know for sure what they *want* or what they *value*, unless they tell us straightaway. As a social actor, a person may come across as friendly and compassionate, or cynical and mean-spirited, but in neither case can we infer their motivations from their traits or their roles. What does the friendly person want? What is the cynical father trying to achieve? Many broad psychological theories of the self prioritize the motivational qualities of human behavior—the inner needs, wants, desires, goals, values, plans, programs, fears, and aversions that seem to give behavior its direction and purpose (Bandura, 1989; Deci & Ryan, 1991; Markus & Nurius, 1986). These kinds of theories explicitly conceive of the self as a *motivated agent*.

To be an agent is to act with direction and purpose, to move forward into the future in pursuit of self-chosen and valued goals. In a sense, human beings are agents even as infants, for babies can surely act in goal-directed ways. By age 1 year, moreover, infants show a strong preference for observing and imitating the goal-directed, intentional behavior of others, rather than random behaviors (Woodward, 2009). Still, it is one thing to act in goal-directed ways; it is quite another for the I to know itself (the Me) as an intentional and purposeful force who moves forward in life in pursuit of self-chosen goals, values, and other desired end states. In order to do so, the person must first realize that people indeed have desires and goals in their minds and that these inner desires and goals *motivate* (initiate, energize, put into motion) their behavior. According to a strong line of research in developmental psychology, attaining this kind of understanding means acquiring a theory of mind (Wellman, 1993), which occurs for most children by the age of 4. Once a child understands that other people's behavior is often motivated by inner desires and goals, it is a small step to apprehend the self in similar terms.

Building on theory of mind and other cognitive and social developments, children begin to construct the self as a motivated agent in the elementary school years, layered over their still-developing sense of themselves as social actors. Theory and research on what developmental psychologists call the age 5-to-7 shift converge to suggest that children become more planful,

intentional, and systematic in their pursuit of valued goals during this time (Sameroff & Haith, 1996). Schooling reinforces the shift in that teachers and curricula place increasing demands on students to work hard, adhere to schedules, focus on goals, and achieve success in particular, well-defined task domains. Their relative success in achieving their most cherished goals, furthermore, goes a long way in determining children's self-esteem (Robins, Tracy, & Trzesniewski, 2008). Motivated agents feel good about themselves to the extent they believe that they are making good progress in achieving their goals and advancing their most important values.

Goals and values become even more important for the self in adolescence, as teenagers begin to confront what Erikson (1963) famously termed the developmental challenge of identity. For adolescents and young adults, establishing a psychologically efficacious identity involves exploring different options with respect to life goals, values, vocations, and intimate relationships and eventually committing to a motivational and ideological agenda for adult life—an integrated and realistic sense of what I want and value in life and how I plan to achieve it (Kroger & Marcia, 2011). Committing oneself to an integrated suite of life goals and values is perhaps the greatest achievement for the self as motivated agent. Establishing an adult identity has implications, as well, for how a person moves through life as a social actor, entailing new role commitments and, perhaps, a changing understanding of one's basic dispositional traits. According to Erikson, however, identity achievement is always provisional, for adults continue to work on their identities as they move into midlife and beyond, often relinquishing old goals in favor of new ones, investing themselves in new projects and making new plans, exploring new relationships, and shifting their priorities in response to changing life circumstances (Freund & Riediger, 2006; Josselson, 1996).

There is a sense whereby *any* time you try to change yourself, you are assuming the role of a motivated agent. After all, to strive to change something is inherently what an agent does. However, what particular feature of selfhood you try to change may correspond to your self as actor, agent, or author, or some combination. When you try to change your traits or roles, you take aim at the social actor. By contrast, when you try to change your values or life goals, you are focusing on yourself as a motivated agent. Adolescence and young adulthood are periods in the human life course when many of us focus attention on our values and life goals. Perhaps you grew up as a traditional Catholic, but now in college you believe that the values inculcated in your childhood no longer function so well for you. You no longer believe in the central tenets of the Catholic Church, say, and are now working to replace your old values with new ones. Or maybe you still want to be Catholic, but you feel that your new take on faith requires a different kind of personal ideology. In the realm of the motivated agent, moreover, changing values can influence life goals. If your new value system prioritizes alleviating the suffering of others, you may decide to pursue a degree in social work, or to

become a public interest lawyer, or to live a simpler life that prioritizes people over material wealth. A great deal of the identity work we do in adolescence and young adulthood is about values and goals, as we strive to articulate a personal vision or dream for what we hope to accomplish in the future.

The Autobiographical Author

Even as the “I” continues to develop a sense of the “Me” as both a social actor and a motivated agent, a third standpoint for selfhood gradually emerges in the adolescent and early-adult years. The third perspective is a response to Erikson’s (1963) challenge of identity. According to Erikson, developing an identity involves more than the exploration of and commitment to life goals and values (the self as motivated agent), and more than committing to new roles and re-evaluating old traits (the self as social actor). It also involves achieving a sense of *temporal continuity* in life—a reflexive understanding of *how I have come to be the person I am becoming*, or put differently, how my past self has developed into my present self, and how my present self will, in turn, develop into an envisioned future self. In his analysis of identity formation in the life of the 15th-century Protestant reformer Martin Luther, Erikson (1958) describes the culmination of a young adult’s search for identity in this way:

"To be adult means among other things to see one’s own life in continuous perspective, both in retrospect and prospect. By accepting some definition of who he is, usually on the basis of a function in an economy, a place in the sequence of generations, and a status in the structure of society, the adult is able *to selectively reconstruct his past in such a way that, step for step, it seems to have planned him, or better, he seems to have planned it*. In this sense, psychologically we do choose our parents, our family history, and the history of our kings, heroes, and gods. By making them our own, we maneuver ourselves into the inner position of proprietors, of creators."

-- (Erikson, 1958, pp. 111–112; emphasis added).

In this rich passage, Erikson intimates that the development of a mature identity in young adulthood involves the I’s ability to construct a retrospective and prospective *story* about the Me (McAdams, 1985). In their efforts to find a meaningful identity for life, young men and women begin “to selectively reconstruct” their past, as Erikson wrote, and imagine their future to create an integrative life story, or what psychologists today often call a narrative identity. A narrative identity is an internalized and evolving story of the self that reconstructs the past and anticipates the future in such a way as to provide a person’s life with some degree of unity, meaning, and purpose over time (McAdams, 2008; McLean, Pasupathi, & Pals, 2007).

The self typically becomes an *autobiographical author* in the early-adult years, a way of being that is layered over the motivated agent, which is layered over the social actor. In order to provide life with the sense of temporal continuity and deep meaning that Erikson believed identity should confer, we must author a personalized life story that integrates our understanding of who we once were, who we are today, and who we may become in the future. The story helps to explain, for the author and for the author's world, why the social actor does what it does and why the motivated agent wants what it wants, and how the person as a whole has developed over time, from the past's reconstructed beginning to the future's imagined ending.

By the time they are 5 or 6 years of age, children can tell well-formed stories about personal events in their lives (Fivush, 2011). By the end of childhood, they usually have a good sense of what a typical biography contains and how it is sequenced, from birth to death (Thomsen & Bernsten, 2008). But it is not until adolescence, research shows, that human beings express advanced storytelling skills and what psychologists call **autobiographical reasoning** (Habermas & Bluck, 2000; McLean & Fournier, 2008). In autobiographical reasoning, a narrator is able to derive substantive conclusions about the self from analyzing his or her own personal experiences. Adolescents may develop the ability to string together events into causal chains and inductively derive general themes about life from a sequence of chapters and scenes (Habermas & de Silveira, 2008). For example, a 16-year-old may be able to explain to herself



Young people often "try on" many variations of identities to see which best fits their private sense of themselves. [Image: Sangudo, <https://goo.gl/Ay3UMR>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

and to others how childhood experiences in her family have shaped her vocation in life. Her parents were divorced when she was 5 years old, the teenager recalls, and this caused a great deal of stress in her family. Her mother often seemed anxious and depressed, but she (the now-teenager when she was a little girl—the story's protagonist) often tried to cheer her mother up, and her efforts seemed to work. In more recent years, the teenager notes that her friends often come to her with their boyfriend problems. She seems to be very adept at giving advice about love and relationships, which stems, the teenager now believes, from her early experiences with her mother. Carrying this causal narrative forward, the teenager now thinks that she would like to be a marriage

counselor when she grows up.

Unlike children, then, adolescents can tell a full and convincing story about an entire human life, or at least a prominent line of causation within a full life, explaining continuity and change in the story's protagonist over time. Once the cognitive skills are in place, young people seek interpersonal opportunities to share and refine their developing sense of themselves as storytellers (the I) who tell stories about themselves (the Me). Adolescents and young adults author a narrative sense of the self by telling stories about their experiences to other people, monitoring the feedback they receive from the tellings, editing their stories in light of the feedback, gaining new experiences and telling stories about those, and on and on, as selves create stories that, in turn, create new selves (McLean et al., 2007). Gradually, in fits and starts, through conversation and introspection, the I develops a convincing and coherent narrative about the Me.

Contemporary research on the self as autobiographical author emphasizes the strong effect of *culture* on narrative identity (Hammack, 2008). Culture provides a menu of favored plot lines, themes, and character types for the construction of self-defining life stories. Autobiographical authors sample selectively from the cultural menu, appropriating ideas that seem to resonate well with their own life experiences. As such, life stories reflect the culture, wherein they are situated as much as they reflect the authorial efforts of the autobiographical I.

As one example of the tight link between culture and narrative identity, McAdams (2013) and others (e.g., Kleinfeld, 2012) have highlighted the prominence of redemptive narratives in American culture. Epitomized in such iconic cultural ideals as the American dream, Horatio Alger stories, and narratives of Christian atonement, redemptive stories track the move from suffering to an enhanced status or state, while scripting the development of a chosen protagonist who journeys forth into a dangerous and unredeemed world (McAdams, 2013). Hollywood movies often celebrate redemptive quests. Americans are exposed to similar narrative messages in self-help books, 12-step programs, Sunday sermons, and in the rhetoric of political campaigns. Over the past two decades, the world's most influential spokesperson for the power of redemption in human lives may be Oprah Winfrey, who tells her own story of overcoming childhood adversity while encouraging others, through her media outlets and philanthropy, to tell similar kinds of stories for their own lives (McAdams, 2013). Research has demonstrated that American adults who enjoy high levels of mental health and civic engagement tend to construct their lives as narratives of redemption, tracking the move from sin to salvation, rags to riches, oppression to liberation, or sickness/abuse to health/recovery (McAdams, Diamond, de St. Aubin, & Mansfield, 1997; McAdams, Reynolds, Lewis, Patten, & Bowman, 2001; Walker & Frimer, 2007). In American society, these kinds of stories are often

seen to be inspirational.

At the same time, McAdams (2011, 2013) has pointed to shortcomings and limitations in the redemptive stories that many Americans tell, which mirror cultural biases and stereotypes in American culture and heritage. McAdams has argued that redemptive stories support happiness and societal engagement for some Americans, but the same stories can encourage moral righteousness and a naïve expectation that suffering will always be redeemed. For better and sometimes for worse, Americans seem to love stories of personal redemption and often aim to assimilate their autobiographical memories and aspirations to a redemptive form. Nonetheless, these same stories may not work so well in cultures that espouse different values and narrative ideals (Hammack, 2008). It is important to remember that every culture offers its own storehouse of favored narrative forms. It is also essential to know that no single narrative form captures all that is good (or bad) about a culture. In American society, the redemptive narrative is but one of many different kinds of stories that people commonly employ to make sense of their lives.

What is your story? What kind of a narrative are you working on? As you look to the past and imagine the future, what threads of continuity, change, and meaning do you discern? For many people, the most dramatic and fulfilling efforts to change the self happen when the I works hard, as an autobiographical author, to construct and, ultimately, to tell a new story about the Me. Storytelling may be the most powerful form of self-transformation that human beings have ever invented. Changing one's life story is at the heart of many forms of psychotherapy and counseling, as well as religious conversions, vocational epiphanies, and other dramatic transformations of the self that people often celebrate as turning points in their lives (Adler, 2012). Storytelling is often at the heart of the little changes, too, minor edits in the self that we make as we move through daily life, as we live and experience life, and as we later tell it to ourselves and to others.

Conclusion

For human beings, selves begin as social actors, but they eventually become motivated agents and autobiographical authors, too. The I first sees itself as an embodied actor in social space; with development, however, it comes to appreciate itself also as a forward-looking source of self-determined goals and values, and later yet, as a storyteller of personal experience, oriented to the reconstructed past and the imagined future. To "know thyself" in mature adulthood, then, is to do three things: (a) to apprehend and to perform with social approval my self-ascribed traits and roles, (b) to pursue with vigor and (ideally) success my most valued goals and plans, and (c) to construct a story about life that conveys, with vividness and cultural

resonance, how I became the person I am becoming, integrating my past as I remember it, my present as I am experiencing it, and my future as I hope it to be.

Outside Resources

Web: The website for the Foley Center for the Study of Lives, at Northwestern University. The site contains research materials, interview protocols, and coding manuals for conducting studies of narrative identity.

<http://www.sesp.northwestern.edu/foley/>

Discussion Questions

1. Back in the 1950s, Erik Erikson argued that many adolescents and young adults experience a tumultuous identity crisis. Do you think this is true today? What might an identity crisis look and feel like? And, how might it be resolved?
2. Many people believe that they have a true self buried inside of them. From this perspective, the development of self is about discovering a psychological truth deep inside. Do you believe this to be true? How does thinking about the self as an actor, agent, and author bear on this question?
3. Psychological research shows that when people are placed in front of mirrors they often behave in a more moral and conscientious manner, even though they sometimes experience this procedure as unpleasant. From the standpoint of the self as a social actor, how might we explain this phenomenon?
4. By the time they reach adulthood, does everybody have a narrative identity? Do some people simply never develop a story for their life?
5. What happens when the three perspectives on self—the self as actor, agent, and author—conflict with each other? Is it necessary for people's self-ascribed traits and roles to line up well with their goals and their stories?
6. William James wrote that the self includes all things that the person considers to be "mine." If we take James literally, a person's self might extend to include his or her material possessions, pets, and friends and family. Does this make sense?
7. To what extent can we control the self? Are some features of selfhood easier to control than others?
8. What cultural differences may be observed in the construction of the self? How might gender, ethnicity, and class impact the development of the self as actor, as agent, and as author?

Vocabulary

Autobiographical reasoning

The ability, typically developed in adolescence, to derive substantive conclusions about the self from analyzing one's own personal experiences.

Big Five

A broad taxonomy of personality trait domains repeatedly derived from studies of trait ratings in adulthood and encompassing the categories of (1) extraversion vs. introversion, (2) neuroticism vs. emotional stability, (3) agreeable vs. disagreeableness, (4) conscientiousness vs. nonconscientiousness, and (5) openness to experience vs. conventionality. By late childhood and early adolescence, people's self-attributions of personality traits, as well as the trait attributions made about them by others, show patterns of intercorrelations that confirm with the five-factor structure obtained in studies of adults.

Ego

Sigmund Freud's conception of an executive self in the personality. Akin to this module's notion of "the I," Freud imagined the ego as observing outside reality, engaging in rational thought, and coping with the competing demands of inner desires and moral standards.

Identity

Sometimes used synonymously with the term "self," identity means many different things in psychological science and in other fields (e.g., sociology). In this module, I adopt Erik Erikson's conception of identity as a developmental task for late adolescence and young adulthood. Forming an identity in adolescence and young adulthood involves exploring alternative roles, values, goals, and relationships and eventually committing to a realistic agenda for life that productively situates a person in the adult world of work and love. In addition, identity formation entails commitments to new social roles and reevaluation of old traits, and importantly, it brings with it a sense of temporal continuity in life, achieved through the construction of an integrative life story.

Narrative identity

An internalized and evolving story of the self designed to provide life with some measure of temporal unity and purpose. Beginning in late adolescence, people craft self-defining stories that reconstruct the past and imagine the future to explain how the person came to be the person that he or she is becoming.

Redemptive narratives

Life stories that affirm the transformation from suffering to an enhanced status or state. In American culture, redemptive life stories are highly prized as models for the good self, as in classic narratives of atonement, upward mobility, liberation, and recovery.

Reflexivity

The idea that the self reflects back upon itself; that the I (the knower, the subject) encounters the Me (the known, the object). Reflexivity is a fundamental property of human selfhood.

Self as autobiographical author

The sense of the self as a storyteller who reconstructs the past and imagines the future in order to articulate an integrative narrative that provides life with some measure of temporal continuity and purpose.

Self as motivated agent

The sense of the self as an intentional force that strives to achieve goals, plans, values, projects, and the like.

Self as social actor

The sense of the self as an embodied actor whose social performances may be construed in terms of more or less consistent self-ascribed traits and social roles.

Self-esteem

The extent to which a person feels that he or she is worthy and good. The success or failure that the motivated agent experiences in pursuit of valued goals is a strong determinant of self-esteem.

Social reputation

The traits and social roles that others attribute to an actor. Actors also have their own conceptions of what they imagine their respective social reputations indeed are in the eyes of others.

The Age 5-to-7 Shift

Cognitive and social changes that occur in the early elementary school years that result in the child's developing a more purposeful, planful, and goal-directed approach to life, setting the stage for the emergence of the self as a motivated agent.

The "I"

The self as knower, the sense of the self as a subject who encounters (knows, works on) itself (the Me).

The “Me”

The self as known, the sense of the self as the object or target of the I’s knowledge and work.

Theory of mind

Emerging around the age of 4, the child’s understanding that other people have minds in which are located desires and beliefs, and that desires and beliefs, thereby, motivate behavior.

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Gender

Christia Spears Brown, Jennifer A. Jewell & Michelle J. Tam

This module discusses gender and its related concepts, including sex, gender roles, gender identity, sexual orientation, and sexism. In addition, this module includes a discussion of differences that exist between males and females and how these real gender differences compare to the stereotypes society holds about gender differences. In fact, there are significantly fewer real gender differences than one would expect relative to the large number of stereotypes about gender differences. This module then discusses theories of how gender roles develop and how they contribute to strong expectations for gender differences. Finally, the module concludes with a discussion of some of the consequences of relying on and expecting gender differences, such as gender discrimination, sexual harassment, and ambivalent sexism.

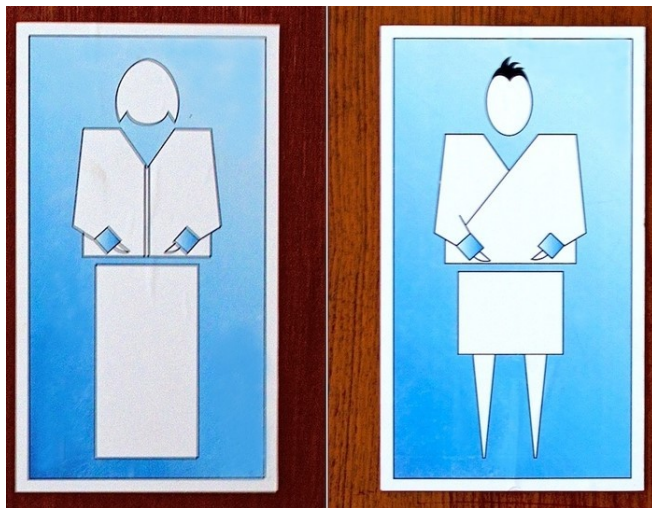
Learning Objectives

- Distinguish gender and sex, as well as gender identity and sexual orientation.
- Discuss gender differences that exist, as well as those that do not actually exist.
- Understand and explain different theories of how gender roles are formed.
- Discuss sexism and its impact on gender.

Introduction

Before we discuss gender in detail, it is important to understand what gender actually is. The terms sex and gender are frequently used interchangeably, though they have different

meanings. In this context, sex refers to the biological category of male or female, as defined by physical differences in genetic composition and in reproductive anatomy and function. On the other hand, gender refers to the cultural, social, and psychological meanings that are associated with masculinity and femininity (Wood & Eagly, 2002). You can think of “male” and “female” as distinct categories of sex (a person is typically born a male or a female), but “masculine” and “feminine” as continuums associated with gender (everyone has a certain degree of masculine and feminine traits and qualities).



Gender refers to the cultural, social, and psychological meanings that are associated with masculinity and femininity. [Photo: Michael Foley Photography, <https://goo.gl/B46jym>, CC BY-NC-ND 2.0, <https://goo.gl/aAX82f>]

Beyond sex and gender, there are a number of related terms that are also often misunderstood. Gender roles are the behaviors, attitudes, and personality traits that are designated as either masculine or feminine in a given culture. It is common to think of gender roles in terms of gender stereotypes, or the beliefs and expectations people hold about the typical characteristics, preferences, and behaviors of men and women. A person's gender identity refers to their psychological sense of being male or female. In contrast, a person's sexual orientation is the direction of their emotional and erotic attraction toward members of the opposite sex, the same

sex, or both sexes. These are important distinctions, and though we will not discuss each of these terms in detail, it is important to recognize that sex, gender, gender identity, and sexual orientation do not always correspond with one another. A person can be biologically male but have a female gender identity while being attracted to women, or any other combination of identities and orientations.

Defining Gender

Historically, the terms gender and sex have been used interchangeably. Because of this, gender is often viewed as a binary – a person is either male or female – and it is assumed that a person's gender matches their biological sex. This is not always the case, however, and more recent research has separated these two terms. While the majority of people do identify with the gender that matches their biological sex (cisgender), an estimated 0.6% of the population

identify with a gender that does not match their biological sex (**transgender**; Flores, Herman, Gates, & Brown, 2016). For example, an individual who is biologically male may identify as female, or vice versa.

In addition to separating gender and sex, recent research has also begun to conceptualize gender in ways beyond the gender binary. **Genderqueer or gender nonbinary** are umbrella terms used to describe a wide range of individuals who do not identify with and/or conform to the gender binary. These terms encompass a variety of more specific labels individuals may use to describe themselves. Some common labels are **genderfluid**, **agender**, and **bigender**. An individual who is genderfluid may identify as male, female, both, or neither at different times and in different circumstances. An individual who is agender may have no gender or describe themselves as having a neutral gender, while bigender individuals identify as two genders.

It is important to remember that sex and gender do not always match and that gender is not always binary; however, a large majority of prior research examining gender has not made these distinctions. As such, the following sections will discuss gender as a binary.

The (Trans)Gender-Bathroom Debate

In recent years, there has been much cultural and political debate over the right of transgender individuals to use the public bathroom of their choosing. This debate made major headlines in 2016 when North Carolina passed the Public Facilities Privacy & Security Act (commonly called House Bill 2 or HB2). This law required individuals to use the restroom that corresponded with their birth sex, meaning that transgender individuals could not use the bathroom that matched their gender identity. This law and the similar “bathroom bills” proposed by other states were met with widespread controversy, with opponents arguing that they were discriminatory and perpetuated inequality (Barnett, Nesbit, & Sorrentino, 2018). HB2 has since been repealed, but many states still do not protect the rights of transgender individuals to use their restrooms of choice.

Text Box 1

Gender Differences

Differences between males and females can be based on (a) actual gender differences (i.e., men and women are actually different in some abilities), (b) gender roles (i.e., differences in how men and women are supposed to act), or (c) gender stereotypes (i.e., differences in how we *think* men and women are). Sometimes gender stereotypes and gender roles reflect actual

gender differences, but sometimes they do not.

What are actual gender differences? In terms of language and language skills, girls develop language skills earlier and know more words than boys; this does not, however, translate into long-term differences. Girls are also more likely than boys to offer praise, to agree with the person they're talking to, and to elaborate on the other person's comments; boys, in contrast, are more likely than girls to assert their opinion and offer criticisms (Leaper & Smith, 2004). In terms of temperament, boys are slightly less able to suppress inappropriate responses and slightly more likely to blurt things out than girls (Else-Quest, Hyde, Goldsmith, & Van Hulle, 2006).

With respect to aggression, boys exhibit higher rates of unprovoked physical aggression than girls, but no difference in provoked aggression (Hyde, 2005). Some of the biggest differences involve the play styles of children. Boys frequently play organized rough-and-tumble games in large groups, while girls often play less physical activities in much smaller groups (Maccoby, 1998). There are also differences in the rates of depression, with girls much more likely than boys to be depressed after puberty. After puberty, girls are also more likely to be unhappy with their bodies than boys.

However, there is considerable variability between individual males and individual females. Also, even when there are mean level differences, the actual size of most of these differences is quite small. This means, knowing someone's gender does not help much in predicting his or her actual traits. For example, in terms of activity level, boys are considered more active than girls. However, 42% of girls are more active than the average boy (but so are 50% of boys; see Figure 1 for a depiction of this phenomenon in a comparison of male and female self-esteem). Furthermore, many gender differences do not reflect innate differences, but instead reflect differences in specific experiences and socialization. For example, one presumed gender difference is that boys show better spatial abilities than girls. However, Tzuriel and Egozi (2010) gave girls the chance to practice their spatial skills (by imagining a line drawing was different shapes) and discovered that, with practice, this gender difference



Boys exhibit higher rates of unprovoked physical aggression than girls and are more likely to play organized rough-and-tumble games. [Image: Aislinn Ritchie, <https://goo.gl/cVQ0Ab>, CC BY-SA 2.0, <https://goo.gl/jSSrcO>]

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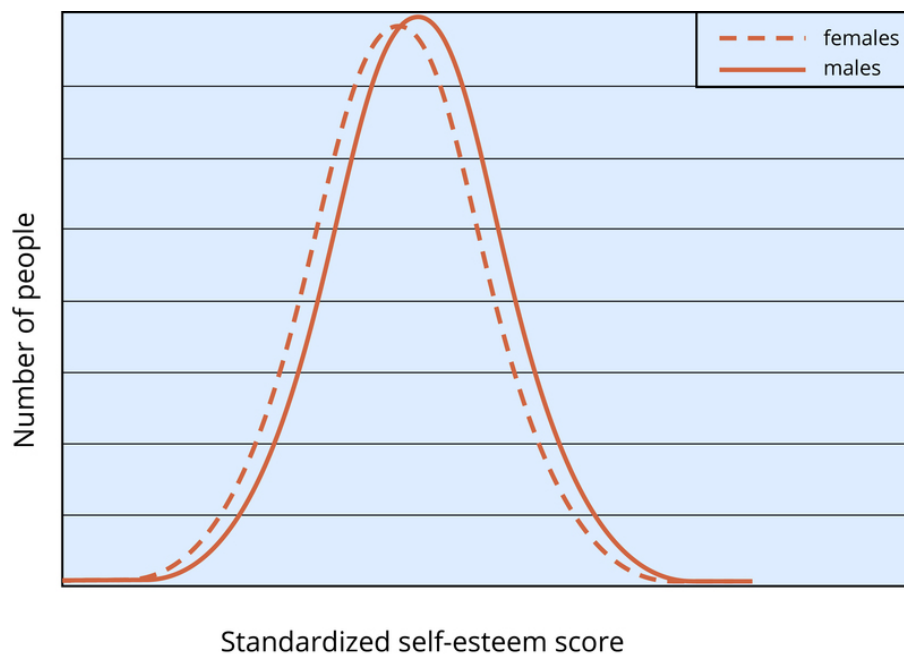


Figure 1. While our gender stereotypes paint males and females as drastically different from each other, even when a difference exists, there is considerable overlap in the presence of that trait between genders. This graph shows the average difference in self-esteem between boys and girls. Boys have a higher average self-esteem than girls, but the average scores are much more similar than different. Taken from Hyde (2005).

Many domains we assume differ across genders are really based on gender stereotypes and not actual differences. Based on large meta-analyses, the analyses of thousands of studies across more than one million people, research has shown: Girls are not more fearful, shy, or scared of new things than boys; boys are not more angry than girls and girls are not more emotional than boys; boys do not perform better at math than girls; and girls are not more talkative than boys (Hyde, 2005).

In the following sections, we'll investigate gender roles, the part they play in creating these stereotypes, and how they can affect the development of real gender differences.

Gender Roles

As mentioned earlier, gender roles are well-established social constructions that may change from culture to culture and over time. In American culture, we commonly think of gender roles in terms of gender stereotypes, or the beliefs and expectations people hold about the typical

characteristics, preferences, and behaviors of men and women.

By the time we are adults, our gender roles are a stable part of our personalities, and we usually hold many gender stereotypes. When do children start to learn about gender? Very early. By their first birthday, children can distinguish faces by gender. By their second birthday, they can label others' gender and even sort objects into gender-typed categories. By the third birthday, children can consistently identify their own gender (see Martin, Ruble, & Szkrybalo, 2002, for a review). At this age, children believe sex is determined by external attributes, not biological attributes. Between 3 and 6 years of age, children learn that gender is constant and can't change simply by changing external attributes, having developed gender constancy. During this period, children also develop strong and rigid gender stereotypes. Stereotypes can refer to play (e.g., boys play with trucks, and girls play with dolls), traits (e.g., boys are strong, and girls like to cry), and occupations (e.g., men are doctors and women are nurses). These stereotypes stay rigid until children reach about age 8 or 9. Then they develop cognitive abilities that allow them to be more flexible in their thinking about others.

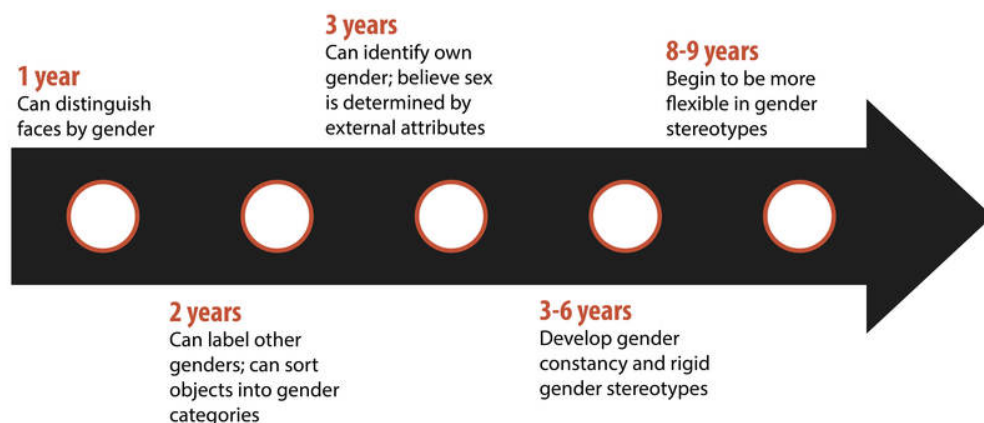


Figure 2: Children develop the ability to classify gender very early in life.

How do our gender roles and gender stereotypes develop and become so strong? Many of our gender stereotypes are so strong because we emphasize gender so much in culture (Bigler & Liben, 2007). For example, males and females are treated differently before they are even born. When someone learns of a new pregnancy, the first question asked is "Is it a boy or a girl?" Immediately upon hearing the answer, judgments are made about the child: Boys will be rough and like blue, while girls will be delicate and like pink. Developmental intergroup theory postulates that adults' heavy focus on gender leads children to pay attention to gender as a key source of information about themselves and others, to seek out any possible gender differences, and to form rigid stereotypes based on gender that are subsequently difficult to change.

There are also psychological theories that partially explain how children form their own gender roles after they learn to differentiate based on gender. The first of these theories is gender schema theory. Gender schema theory argues that children are active learners who essentially socialize themselves. In this case, children actively organize others' behavior, activities, and attributes into gender categories, which are known as schemas. These schemas then affect what children notice and remember later. People of all ages are more likely to remember schema-consistent behaviors and attributes than schema-inconsistent behaviors and attributes. So, people are more likely to remember men, and forget women, who are firefighters. They also misremember schema-inconsistent information. If research participants are shown pictures of someone standing at the stove, they are more likely to remember the person to be cooking if depicted as a woman, and the person to be repairing the stove if depicted as a man. By only remembering schema-consistent information, gender schemas strengthen more and more over time.



People are more likely to remember schema-consistent behaviors and attributes than schema-inconsistent behaviors and attributes. For example, people are more likely to remember men, and forget women, who are firefighters. [Photo: Billy V, <https://goo.gl/Kb2MuL>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

A second theory that attempts to explain the formation of gender roles in children is social learning theory. Social learning theory argues that gender roles are learned through reinforcement, punishment, and modeling. Children are rewarded and reinforced for behaving in concordance with gender roles and punished for breaking gender roles. In addition, social learning theory argues that children learn many of their gender roles by modeling the behavior of adults and older children and, in doing so, develop ideas about what behaviors are appropriate for each gender. Social learning theory has less support than gender schema theory—research shows that parents do reinforce gender-appropriate play, but for the most part treat their male and female children similarly (Lytton & Romney, 1991).

Gender Sexism and Socialization

Treating boys and girls, and men and women, differently is both a *consequence* of gender

differences and a *cause* of gender differences. Differential treatment on the basis of gender is also referred to gender discrimination and is an inevitable consequence of gender stereotypes. When it is based on unwanted treatment related to sexual behaviors or appearance, it is called sexual harassment. By the time boys and girls reach the end of high school, most have experienced some form of sexual harassment, most commonly in the form of unwanted touching or comments, being the target of jokes, having their body parts rated, or being called names related to sexual orientation.

Different treatment by gender begins with parents. A meta-analysis of research from the United States and Canada found that parents most frequently treated sons and daughters differently by encouraging gender-stereotypical activities (Lytton & Romney, 1991). Fathers, more than mothers, are particularly likely to encourage gender-stereotypical play, especially in sons. Parents also talk to their children differently based on stereotypes. For example, parents talk about numbers and counting twice as often with sons than daughters (Chang, Sandhofer, & Brown, 2011) and talk to sons in more detail about science than with daughters. Parents are also much more likely to discuss emotions with their daughters than their sons.

Children do a large degree of socializing themselves. By age 3, children play in gender-segregated play groups and expect a high degree of conformity. Children who are perceived as gender atypical (i.e., do not conform to gender stereotypes) are more likely to be bullied and rejected than their more gender-conforming peers.

Gender stereotypes typically maintain gender inequalities in society. The concept of ambivalent sexism recognizes the complex nature of gender attitudes, in which women are often associated with positive and negative qualities (Glick & Fiske, 2001). It has two components. First, hostile sexism refers to the negative attitudes of women as inferior and incompetent relative to men. Second, benevolent sexism refers to the perception that women need to be protected, supported, and adored by men. There has been considerable empirical support for benevolent sexism, possibly because it is seen as more socially acceptable than hostile sexism. Gender stereotypes are found not just in American culture. Across cultures, males tend to be associated with stronger and more active characteristics than females (Best, 2001).

In recent years, gender and related concepts have become a common focus of social change and social debate. Many societies, including American society, have seen a rapid change in perceptions of gender roles, media portrayals of gender, and legal trends relating to gender. For example, there has been an increase in children's toys attempting to cater to both genders (such as Legos marketed to girls), rather than catering to traditional stereotypes. Nationwide, the drastic surge in acceptance of homosexuality and gender questioning has resulted in a

rapid push for legal change to keep up with social change. Laws such as “Don’t Ask, Don’t Tell” and the Defense of Marriage Act (DOMA), both of which were enacted in the 1990s, have met severe resistance on the grounds of being discriminatory toward sexual minority groups and have been accused of unconstitutionality less than 20 years after their implementation. Change in perceptions of gender is also evident in social issues such as sexual harassment, a term that only entered the mainstream mindset in the 1991 Clarence Thomas/Anita Hill scandal. As society’s gender roles and gender restrictions continue to fluctuate, the legal system and the structure of American society will continue to change and adjust.

Important Gender-related Events in the United States

1920 -- *19th Amendment* (women's Suffrage Ratified)

1941-1945 -- World War II forces millions of women to enter the workforce

1948 -- Universal Declaration of Human Rights

1963 -- Congress passes *Equal Pay Act*

1964 -- Congress passes *Civil Rights Act*, which outlaws sex discrimination

1969 -- Stonewall riots in NYC, forcing gay rights into the American spotlight

1972 -- Congress passes *Equal Rights Amendment*; *Title IX* prohibits sex discrimination in schools and sports

1973 -- American Psychiatric Association removes homosexuality from the DSM

1981 -- First woman appointed to the US Supreme Court

1987 -- Average woman earned \$0.68 for every \$1.00 earned by a man

1992 -- World Health Organization no longer considers homosexuality an illness

1993 -- Supreme Court rules that sexual harassment in the workplace is illegal

2011 -- *Don't Ask Don't Tell* is repealed, allowing people who identify as gay serve openly in the US military

2012 -- President Barack Obama becomes the first American president to openly support LGBT rights and marriage equality

Outside Resources

PFLAG National Glossary of Terms

<https://pflag.org/glossary>

Resources for Transgender, Non-binary, and Gender Non-Conforming Individuals and their Allies

<https://sites.sph.harvard.edu/unity/resources-transgender/>

Video: Human Sexuality is Complicated

<http://www.youtube.com/watch?v=xXAoG8vAyzI>

Web: Big Think with Professor of Neuroscience Lise Eliot

<http://bigthink.com/users/liseeliot>

Web: Understanding Prejudice: Sexism

<http://www.understandingprejudice.org/links/sexism.htm>

Discussion Questions

1. What are the differences and associations among gender, sex, gender identity, and sexual orientation?
2. Are the gender differences that exist innate (biological) differences or are they caused by other variables?
3. Discuss the theories relating to the development of gender roles and gender stereotypes. Which theory do you support? Why?
4. Using what you've read in this module: a. Why do you think gender stereotypes are so inflated compared with actual gender differences? b. Why do you think people continue to believe in such strong gender differences despite evidence to the contrary?
5. Brainstorm additional forms of gender discrimination aside from sexual harassment. Have you seen or experienced gender discrimination personally?
6. How is benevolent sexism detrimental to women, despite appearing positive?

Vocabulary

Agender

An individual who may have no gender or may describe themselves as having a neutral gender.

Ambivalent sexism

A concept of gender attitudes that encompasses both positive and negative qualities.

Benevolent sexism

The “positive” element of ambivalent sexism, which recognizes that women are perceived as needing to be protected, supported, and adored by men.

Bigender

An individual who identifies as two genders.

Binary

The idea that gender has two separate and distinct categories (male and female) and that a person must be either one or the other.

Cisgender

A term used to describe individuals whose gender matches their biological sex.

Developmental intergroup theory

A theory that postulates that adults’ focus on gender leads children to pay attention to gender as a key source of information about themselves and others, to seek out possible gender differences, and to form rigid stereotypes based on gender.

Gender

The cultural, social, and psychological meanings that are associated with masculinity and femininity.

Gender constancy

The awareness that gender is constant and does not change simply by changing external attributes; develops between 3 and 6 years of age.

Gender discrimination

Differential treatment on the basis of gender.

Gender identity

A person's psychological sense of being male or female.

Gender roles

The behaviors, attitudes, and personality traits that are designated as either masculine or feminine in a given culture.

Gender schema theory

This theory of how children form their own gender roles argues that children actively organize others' behavior, activities, and attributes into gender categories or schemas.

Gender stereotypes

The beliefs and expectations people hold about the typical characteristics, preferences, and behaviors of men and women.

Genderfluid

An individual who may identify as male, female, both, or neither at different times and in different circumstances.

Genderqueer or gender nonbinary

An umbrella term used to describe a wide range of individuals who do not identify with and/or conform to the gender binary.

Hostile sexism

The negative element of ambivalent sexism, which includes the attitudes that women are inferior and incompetent relative to men.

Schemas

The gender categories into which, according to gender schema theory, children actively organize others' behavior, activities, and attributes.

Sex

Biological category of male or female as defined by physical differences in genetic composition and in reproductive anatomy and function.

Sexual harassment

A form of gender discrimination based on unwanted treatment related to sexual behaviors or appearance.

Sexual orientation

Refers to the direction of emotional and erotic attraction toward members of the opposite sex, the same sex, or both sexes.

Social learning theory

This theory of how children form their own gender roles argues that gender roles are learned through reinforcement, punishment, and modeling.

Transgender

A term used to describe individuals whose gender does not match their biological sex.

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Personality Stability and Change

M. Brent Donnellan

This module describes different ways to address questions about personality stability across the lifespan. Definitions of the major types of personality stability are provided, and evidence concerning the different kinds of stability and change are reviewed. The mechanisms thought to produce personality stability and personality change are identified and explained.

Learning Objectives

- Define heterotypic stability, homotypic stability, absolute stability, and differential stability.
- Describe evidence concerning the absolute and differential stability of personality attributes across the lifespan.
- Explain the maturity, cumulative continuity, and corresponive principles of personality development.
- Explain person-environment transactions, and distinguish between active, reactive, and evocative person-environment transactions.
- Identify the four processes that promote personality stability (attraction, selection, manipulation, and attrition). Provide examples of these processes.
- Describe the mechanisms behind the possibility of personality transformation.

Introduction

Personality psychology is about how individuals differ from each other in their characteristic



How much of your personality was determined as a child? How much of it developed as you aged? What aspects of yourself are you thankful to have “grown out of?” [Image: Ah Wei, <https://goo.gl/nD3y7q>, CC BY-SA 2.0, <https://goo.gl/rxiUsF>]

ways of thinking, feeling, and behaving. Some of the most interesting questions about personality attributes involve issues of stability and change. Are shy children destined to become shy adults? Are the typical personality attributes of adults different from the typical attributes of adolescents? Do people become more self-controlled and better able to manage their negative emotions as they become adults? What mechanisms explain personality stability and what mechanisms account for personality change?

Defining Different Kinds of Personality Stability

Something frustrating happens when you attempt to learn about personality stability[1]: As with many topics in psychology, there are a number of different ways to conceptualize and quantify personality stability (e.g., Caspi & Bem, 1990; Roberts, Wood, & Caspi, 2008). This means there are multiple ways to consider questions about personality stability. Thus, the simple (and obviously frustrating) way to respond to most blanket questions about personality stability is to simply answer that it depends on what one means by personality stability. To provide a more satisfying answer to questions about stability, I will first describe the different ways psychologists conceptualize and evaluate personality stability. I will make an important distinction between heterotypic and homotypic stability. I will then describe absolute and differential stability, two ways of considering homotypic stability. I will also draw your attention to the important concept of individual differences in personality development.

Heterotypic stability refers to the psychological coherence of an individual's thoughts, feelings, and behaviors across development. Questions about heterotypic stability concern the degree of consistency in underlying personality attributes. The tricky part of studying heterotypic stability is that the underlying psychological attribute can have different behavioral expressions at different ages. (You may already know that the prefix “hetero” means something like “different” in Greek.) Shyness is a good example of such an attribute because shyness is expressed differently by toddlers and young children than adults. The shy toddler might cling to a caregiver in a crowded setting and burst into tears when separated from this caregiver. The shy adult, on the other hand, may avoid making eye contact with strangers and seem

aloof and distant at social gatherings. It would be highly unusual to observe an adult burst into tears in a crowded setting. The observable behaviors typically associated with shyness “look” different at different ages. Researchers can study heterotypic continuity only once they have a theory that specifies the different behavioral manifestations of the psychological attribute at different points in the lifespan. As it stands, there is evidence that attributes such as shyness and aggression exhibit heterotypic stability across the lifespan (Caspi, Bem, & Elder, 1989). Individuals who act shy as children often act shy as adults, but the degree of correspondence is far from perfect because many things can intervene between childhood and adulthood to alter how an individual develops. Nonetheless, the important point is that the patterns of behavior observed in childhood sometimes foreshadow adult personality attributes.

Homotypic stability concerns the amount of similarity in the same observable personality characteristics across time. (The prefix “homo” means something like the “same” in Greek.) For example, researchers might ask whether **stress reaction** or the tendency to become easily distressed by the normal challenges of life exhibits homotypic stability from age 25 to age 45. The assumption is that this attribute has the same manifestations at these different ages. Researchers make further distinctions between absolute stability and differential stability when considering homotypic stability.

Absolute stability refers to the consistency of the level of the same personality attribute across time. If an individual received a score of 45 on a hypothetical measure of stress reaction at age 20 and at age 40, researchers would conclude there was evidence of absolute stability. Questions about absolute stability can be considered at the **group level** or the **individual level**. At the group level, it is common for personality researchers to compare average scores on personality measures for groups of different ages. For example, it is possible to investigate whether the average 40-year-old adult has a lower (or higher) level of stress reaction than the average 20-year-old. The answer to this question would tell researchers something about typical patterns of personality development.



When considering personality stability, researchers can think of it at the individual level (e.g., how is 18-year-old you different than 38-year-old you?) or at the group level (e.g., how are most 18-year-olds different than most 38-year-olds?). [Image: Ken Wytock, <https://goo.gl/G1qfcO>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

It is important to consider absolute stability from both the group and individual perspectives. The individual level is interesting because different people might have different patterns of absolute change over time. One person might report consistently low levels of stress reaction throughout adulthood, whereas another person may report dramatic increases in stress reaction during her 30s and 40s. These different individual patterns can be present even if the overall trend is for a decline in stress reaction with age. Personality psychology is about individual differences and whether an individual's attributes change or remain the same across time might be an important individual difference. Indeed, there are intriguing hints that the rate and direction of change in characteristics such as stress reaction (or neuroticism) predicts mortality (Mroczek & Spiro, 2007).

Differential stability refers to the consistency of a personality attribute in terms of an individual's rank-ordering. A typical question about differential stability might be whether a 20-year-old who is low in stress reaction relative to her same aged peers develops into a 40-year-old who is also low in stress reaction compared to her peers. Differential stability is often interesting because many psychological attributes show average changes across the lifespan. Regardless of average changes with age, however, it is common to assume that more trait-like attributes have a high degree of differential stability. Consider athletic performance as an attribute that may exhibit differential stability. The average 35-year-old is likely to run a 5K race faster than the average 55-year-old. Nonetheless, individuals who are fast relative to their peers in their 30s might also be fast relative to their peers in their 50s. Likewise, even if most people decline on a stress reaction as they age, it is still useful to investigate whether there is consistency over time in their relative standing on this attribute.

Basic Findings about Absolute and Differential Stability

Absolute Stability. There are two common ways to investigate average levels of personality attributes at different ages. The simplest approach is to conduct a **cross-sectional study** and compare different age groups on a given attribute assessed at the same time. For instance, researchers might collect data from a sample of individuals ranging in age from 18 to 99 years and compare stress reaction scores for groups of different ages. A more complicated design involves following the same group of individuals and assessing their personalities at multiple time points (often two). This is a **longitudinal study**, and it is a much better way to study personality stability than a cross-sectional study. If all of the individuals in the sample are roughly the same age at the start of the study, they would all be considered members of the same **birth cohort**. One of the chief drawbacks of a cross-sectional study is that individuals who are of different ages are *also* members of different birth cohorts. Thus, researchers have no way of knowing whether any personality differences observed in a cross-sectional study

are attributable to the influence of age per se or birth cohort. A longitudinal study is better able to isolate age effects (i.e., differences in personality related to maturation and development) from cohort effects (i.e., differences in personality related to being born at a particular point in history) than a cross-sectional study. Cohort is a constant (i.e., an unchanging value) in a longitudinal study when all participants start the study at roughly the same age.

A number of large-scale, cross-sectional studies have evaluated age differences in personality (Anusic, Lucas, & Donnellan, 2012; Lucas & Donnellan, 2009; McCrae & Costa, 2003; Soto, John, Gosling, & Potter, 2011; Srivastava, John, Gosling, & Potter, 2003) as have a number of longitudinal studies (Lucas & Donnellan, 2011; Specht, Egloff, & Schmukle, 2011; Terracciano, McCrae, Brant, & Costa, 2005; Wortman, Lucas, & Donnellan, in press). Fortunately, many of the general trends from these different designs converge on the same basic set of findings. Most notably, Roberts, Walton, and Viechtbauer (2006) combined the results of 92 longitudinal studies to provide an overview of absolute changes in personality across the lifespan. They used the Big Five taxonomy (e.g., John, Naumann, & Soto, 2008) to categorize the different personality attributes examined in the individual studies to make sense of the vast literature.

Big 5 Personality Traits (OCEAN)

- **Openness**
- **Conscientiousness**
- **Extraversion**
- **Agreeableness**
- **Neuroticism**

If you're curious where you rank on the Big Five Domains, head to "Psychology Today's" website to take the test for yourself:
<https://goo.gl/aQTqoD>

The Big Five domains include extraversion (attributes such as assertive, confident, independent, outgoing, and sociable), agreeableness (attributes such as cooperative, kind, modest, and trusting), conscientiousness (attributes such as hard working, dutiful, self-controlled, and goal-oriented), neuroticism (attributes such as anxious, tense, moody, and easily angered), and openness (attributes such as artistic, curious, inventive, and open-minded). The Big Five is one of the most common ways of organizing the vast range of personality attributes that seem to distinguish one person from the next. This organizing framework made it possible for Roberts et al. (2006) to draw broad conclusions from the literature.

In general, average levels of extraversion (especially the attributes linked to self-confidence and independence), agreeableness, and conscientiousness appear to increase with age

whereas neuroticism appears to decrease with age (Roberts et al., 2006). Openness also declines with age, especially after mid-life (Roberts et al., 2006). These changes are often viewed as positive trends given that higher levels of agreeableness and conscientiousness and lower levels of neuroticism are associated with seemingly desirable outcomes such as increased relationship stability and quality, greater success at work, better health, a reduced risk of criminality and mental health problems, and even decreased mortality (e.g., Kotov, Gamez, Schmidt, & Watson, 2010; Miller & Lynam 2001; Ozer & Benet-Martínez, 2006; Roberts, Kuncel, Shiner, Caspi, & Goldberg, 2007). This pattern of positive average changes in personality attributes is known as the **maturity principle of adult personality development** (Caspi, Roberts, & Shiner, 2005). The basic idea is that attributes associated with positive adaptation and attributes associated with the successful fulfillment of adult roles tend to increase during adulthood in terms of their average levels.

Beyond providing insights into the general outline of adult personality development, Roberts et al. (2006) found that young adulthood (the period between the ages of 18 and the late 20s) was the most active time in the lifespan for observing average changes, although average differences in personality attributes were observed across the lifespan. Such a result might be surprising in light of the intuition that adolescence is a time of personality change and maturation. However, young adulthood is typically a time in the lifespan that includes a number of life changes in terms of finishing school, starting a career, committing to romantic partnerships, and parenthood (Donnellan, Conger, & Burzette, 2007; Rindfuss, 1991). Finding that young adulthood is an active time for personality development provides circumstantial evidence that adult roles might generate pressures for certain patterns of personality development. Indeed, this is one potential explanation for the maturity principle of personality development.

It should be emphasized again that average trends are summaries that do not necessarily apply to all individuals. Some people do not conform to the maturity principle. The possibility of exceptions to general trends is the reason it is necessary to study individual patterns of personality development. The methods for this kind of



The maturity principle of adult personality development states that as we age we adjust our traits to fit with our new, adult responsibilities. [Image: Alex France, <https://goo.gl/h12CM3>, CC BY-SA 3.0, <https://goo.gl/eLCn2O>]

research are becoming increasingly popular (e.g., Vaidya, Gray, Haig, Mroczek, & Watson, 2008) and existing studies suggest that personality changes differ across people (Roberts & Mroczek, 2008). These new research methods work best when researchers collect more than two waves of longitudinal data covering longer spans of time. This kind of research design is still somewhat uncommon in psychological studies but it will likely characterize the future of research on personality stability.

Differential stability. The evaluation of differential stability requires a longitudinal study. The simplest strategy is to follow a large sample of participants of the same age and measure their personality attributes at two points separated by a meaningful span of time. The researcher then calculates the correlation between scores at the first assessment and scores at the second assessment (a coefficient sometimes called a test-retest correlation or even a stability coefficient). As you know, a correlation coefficient is a numerical summary of the linear association between two variables. Correlations around .1 or $-.1$ are often called “small” associations, whereas correlations around .50 and $-.50$ (or larger) are often called “large” associations (Cohen, 1988).

Roberts and DelVecchio (2000) summarized 3,217 test-retest correlations for a wide range of personality attributes reported in 152 longitudinal studies. They used statistical methods to equate the different test-retest correlations to a common interval of about seven years. This allowed them to compare results from studies of differing lengths of time because not all studies followed participants for the same interval of time. Roberts and DelVecchio found that differential stability increased with age. The correlations ranged from about .30 for samples involving young children to about .70 for samples involving older adults. Ferguson (2010) updated and replicated this basic pattern. This pattern of increasing stability with age is called the **cumulative continuity principle of personality development** (Caspi et al., 2005). This general pattern holds for both women and men and applies to a wide range of different personality attributes ranging from extraversion to openness and curiosity. It is important to emphasize, however, that the observed correlations are never perfect at any age (i.e., the correlations do not reach 1.0). This indicates that personality changes can occur at any time in the lifespan; it just seems that greater inconsistency is observed in childhood and adolescence than in adulthood.

Key Messages So Far

It is useful to summarize the key ideas of this module so far. The starting point was the realization that there are several different ways to define and measure personality stability. Heterotypic stability refers to the consistency of the underlying psychological attribute that

may have different behavioral manifestations at different ages. Homotypic stability, on the other hand, refers to the consistency of the same observable manifestations of a personality attribute. This type of stability is commonly studied in the current literature, and absolute and differential stability are a focus on many studies. A consideration of the broad literature on personality stability yields two major conclusions.

1. Average levels of personality attributes seem to change in predictable ways across the lifespan in line with maturity principle of personality development. Traits that are correlated with positive outcomes (such as conscientiousness) seem to increase from adolescence to adulthood. This perspective on personality stability is gained from considering absolute stability in the form of average levels of personality attributes at different ages.
2. Personality attributes are relatively enduring attributes that become increasingly consistent during adulthood in line with the cumulative continuity principle. This perspective on stability is gained from considering differential stability in the form of test-retest correlations from longitudinal studies.



It's quite easy to imagine, "Once I'm 30, married, and with a family, I will be that person for the rest of my life." But the research shows that while some traits are stable, others continue to develop and adjust to our new environments. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

In general, the picture that emerges from the literature is that personality traits are relatively enduring attributes that become more stable from childhood to adulthood. Nonetheless, the stability of personality attributes is not perfect at any period in the lifespan. This is an important conclusion because it challenges two extreme perspectives that have been influential in psychological research. More than 100 years ago, the famous psychologist William James remarked that character (personality) was "set like plaster" for most people by age 30. This perspective implies near perfect stability of personality in adulthood. In contrast, other psychologists have sometimes denied there was any stability to personality at all. Their perspective is that individual thoughts and feelings are

simply responses to transitory situational influences that are unlikely to show much consistency across the lifespan. As discussed so far, current research does not support either

of these extreme perspectives. Nonetheless, the existence of some degree of stability raises important questions about the exact processes and mechanisms that produce personality stability (and personality change).

The How and Why of Personality Stability and Change: Different Kinds of Interplay Between Individuals and Their Environments

Personality stability is the result of the interplay between the individual and her/his environment. Psychologists use the term **person-environment transactions** (e.g., Roberts et al., 2008) to capture the mutually transforming interplay between individuals and their contextual circumstances. Several different types of these transactions have been described by psychological researchers. **Active person-environment transactions** occur when individuals seek out certain kinds of environments and experiences that are consistent with their personality characteristics. Risk-taking individuals may spend their leisure time very differently than more cautious individuals. Some prefer extreme sports whereas others prefer less intense experiences. **Reactive person-environment transactions** occur when individuals react differently to the same objective situation because of their personalities. A large social gathering represents a psychologically different context to the highly extraverted person compared with the highly introverted person. **Evocative person-environment transactions** occur whenever individuals draw out or evoke certain kinds of responses from their social environments because of their personality attributes. A warm and secure individual invites different kinds of responses from peers than a cold and aloof individual.

Current researchers make distinctions between the mechanisms likely to produce personality stability and the mechanisms likely to produce changes (Roberts, 2006; Roberts et al., 2008). Brent Roberts coined the helpful acronym ASTMA to aid in remembering many of these mechanisms: Attraction (A), selection (S), manipulation (M), and attrition (A) tend to produce personality stability, whereas transformation (T) explains personality change.

Individuals sometimes select careers,



Think to your own preference for hobbies and jobs. How do these activities reflect core attributes of your own personality?
[Image: Dave Scriven, <https://goo.gl/4g9tCz>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

friends, social clubs, and lifestyles because of their personality attributes. This is the active process of **attraction**—individuals are attracted to environments because of their personality attributes. Situations that match with our personalities seem to feel “right” (e.g., Cesario, Grant, & Higgins, 2004). On the flipside of this process, gatekeepers, such as employers, admissions officers, and even potential relationship partners, often **select** individuals because of their personalities. Extraverted and outgoing individuals are likely to make better salespeople than quiet individuals who are uncomfortable with social interactions. All in all, certain individuals are “admitted” by gatekeepers into particular kinds of environments because of their personalities. Likewise, individuals with characteristics that are a bad fit with a particular environment may leave such settings or be asked to leave by gatekeepers. A lazy employee will not last long at a demanding job. These examples capture the process of **attrition** (dropping out). The processes of selection and attrition reflect evocative person–environment transactions. Last, individuals can actively **manipulate** their environments to match their personalities. An outgoing person will find ways to introduce more social interactions into the workday, whereas a shy individual may shun the proverbial water cooler to avoid having contact with others.

These four processes of attraction, selection, attrition, and manipulation explain how a kind of matching occurs between personality attributes and environmental conditions for many individuals. This positive matching typically produces personality consistency because the “press” of the situation reinforces the attributes of the person. This observation is at the core of the **corresponsive principle of personality development** (Caspi et al., 2005; Roberts, Caspi, & Moffitt, 2003). Preexisting personality attributes and environmental contexts work in concert to promote personality continuity. The idea is that environments often reinforce those personality attributes that were partially responsible for the initial environmental conditions in the first place. For example, ambitious and confident individuals might be attracted to and selected for more demanding jobs (Roberts et al., 2003). These kinds of jobs often require drive, dedication, and achievement striving thereby accentuating dispositional tendencies toward ambition and confidence.

Additional considerations related to person–environment transactions may help to further explain personality stability. Individuals gain more autonomy to select their own environment as they transition from childhood to adulthood (Scarr & McCartney, 1983). This might help explain why the differential stability of personality attributes increases from adolescence into adulthood. Reactive and evocative person–environment transactions also facilitate personality stability. The overarching idea is that personality attributes shape how individuals respond to situations and shape the kinds of responses individuals elicit from their environments. These responses and reactions can generate self-fulfilling cycles. For example, aggressive individuals seem to interpret ambiguous social cues as threatening (something

called a hostile attribution bias or a hostile attribution of intent; see Crick & Dodge, 1996; Orobio de Castro, Veerman, Koops, Bosch, & Monshouwer, 2002). If a stranger runs into you and you spill your hot coffee all over a clean shirt, how do you interpret the situation? Do you believe the other person was being aggressive, or were you just unlucky? A rude, caustic, or violent response might invite a similar response from the individual who ran into you. The basic point is that personality attributes help shape reactions to and responses from the social world, and these processes often (but not always) end up reinforcing dispositional tendencies.

Although a number of mechanisms account for personality continuity by generating a match between the individual's characteristics and the environment, personality change or transformation is nonetheless possible. Recall that differential stability is not perfect. The simplest mechanism for producing change is a cornerstone of behaviorism: Patterns of behavior that produce positive consequences (pleasure) are repeated, whereas patterns of behavior that produce negative consequences (pain) will diminish (Thorndike, 1933). Social settings may have the power to transform personality if the individual is exposed to different rewards and punishments and the setting places limitations on how a person can reasonably behave (Caspi & Moffitt, 1993). For example, environmental contexts that limit agency and have very clear reward structures such as the military might be particularly powerful contexts for producing lasting personality changes (e.g., Jackson, Thoemmes, Jonkmann, Lüdke, & Trautwein, 2012).

It is also possible that individuals might change their personality attributes by actively striving to change their behaviors and emotional reactions with help from outsiders. This idea lies at the heart of psychotherapy. As it stands, the conditions that produce lasting personality changes are an active area of research. Personality researchers have historically sought to demonstrate the existence of personality stability, and they are now turning their full attention to the conditions that facilitate personality change. There are currently a few examples of interventions that end up producing short-term personality changes (Jackson, Hill, Payne, Roberts, & Stine-Morrow, 2012), and this is an exciting area for future research (Edmonds, Jackson, Fayard, & Roberts, 2008). Insights about personality change are important for creating effective interventions designed to foster positive human development. Finding ways to promote self-control, emotional stability, creativity, and an agreeable disposition would likely lead to improvements for both individuals and society as a whole because these attributes predict a range of consequential life outcomes (Ozer & Benet-Martínez, 2006; Roberts et al., 2007)

Conclusion



Our personality is the result of the self interacting with the environment. Who you are helps to determine what you do and like, and what you do and like helps to determine who you are. [Image: Nick Sheerbart, CC0 Public Domain, <https://goo.gl/m25gce>]

There are multiple ways to evaluate personality stability. The existing evidence suggests that personality attributes are relatively enduring attributes that show predictable average-level changes across the lifespan. Personality stability is produced by a complicated interplay between individuals and their social settings. Many personality attributes are linked to life experiences in a mutually reinforcing cycle: Personality attributes seem to shape environmental contexts, and those contexts often then accentuate and reinforce those very personality attributes. Even so, personality change or transformation is possible because individuals respond to their environments. Individuals may also want to change their personalities. Personality researchers are now beginning to address important

questions about the possibility of lasting personality changes through intervention efforts.

[1] Throughout most of this module I will use the term stability to refer to continuity, stability/change, and consistency/inconsistency.

Discussion Questions

1. Why is it difficult to give a simple answer to the question of whether personality is stable across the lifespan?
2. What happens during young adulthood that might explain findings about average changes in personality attributes?
3. Why does differential stability increase during adulthood?
4. What are some concrete examples of the ASTMA processes?
5. Can you explain the corresponsive principle of personality development? Provide several clear examples.
6. Do you think dramatic personality changes are likely to happen in adulthood? Why or why not?
7. What kinds of environments might be particularly powerful for changing personality? What specific features of these environments seem to make them powerful for producing change?
8. Is it easy to change your personality in adulthood? What steps do you think are needed to produce noticeable and lasting changes in your personality? What steps are needed to change the personalities of others?
9. Do you find the evidence that personality attributes are relatively enduring attributes reflects a largely positive aspect of adult development or a more unpleasant aspect? Why?

Vocabulary

Absolute stability

Consistency in the level or amount of a personality attribute over time.

Active person–environment transactions

The interplay between individuals and their contextual circumstances that occurs whenever individuals play a key role in seeking out, selecting, or otherwise manipulating aspects of their environment.

Age effects

Differences in personality between groups of different ages that are related to maturation and development instead of birth cohort differences.

Attraction

A connection between personality attributes and aspects of the environment that occurs because individuals with particular traits are drawn to certain environments.

Attrition

A connection between personality attributes and aspects of the environment that occurs because individuals with particular traits drop out from certain environments.

Birth cohort

Individuals born in a particular year or span of time.

Cohort effects

Differences in personality that are related to historical and social factors unique to individuals born in a particular year.

Corresponsive principle

The idea that personality traits often become matched with environmental conditions such that an individual's social context acts to accentuate and reinforce their personality attributes.

Cross-sectional study/design

A research design that uses a group of individuals with different ages (and birth cohorts) assessed at a single point in time.

Cumulative continuity principle

The generalization that personality attributes show increasing stability with age and experience.

Differential stability

Consistency in the rank-ordering of personality across two or more measurement occasions.

Evocative person–environment transactions

The interplay between individuals and their contextual circumstances that occurs whenever attributes of the individual draw out particular responses from others in their environment.

Group level

A focus on summary statistics that apply to aggregates of individuals when studying personality development. An example is considering whether the average score of a group of 50 year olds is higher than the average score of a group of 21 year olds when considering a trait like conscientiousness.

Heterotypic stability

Consistency in the underlying psychological attribute across development regardless of any changes in how the attribute is expressed at different ages.

Homotypic stability

Consistency of the exact same thoughts, feelings, and behaviors across development.

Hostile attribution bias

The tendency of some individuals to interpret ambiguous social cues and interactions as examples of aggressiveness, disrespect, or antagonism.

Individual level

A focus on individual level statistics that reflect whether individuals show stability or change when studying personality development. An example is evaluating how many individuals increased in conscientiousness versus how many decreased in conscientiousness when considering the transition from adolescence to adulthood.

Longitudinal study/design

A research design that follows the same group of individuals at multiple time points.

Manipulation

A connection between personality attributes and aspects of the environment that occurs whenever individuals with particular traits actively shape their environments.

Maturity principle

The generalization that personality attributes associated with the successful fulfillment of adult roles increase with age and experience.

Person–environment transactions

The interplay between individuals and their contextual circumstances that ends up shaping both personality and the environment.

Reactive person–environment transactions

The interplay between individuals and their contextual circumstances that occurs whenever attributes of the individual shape how a person perceives and responds to their environment.

Selection

A connection between personality attributes and aspects of the environment that occurs whenever individuals with particular attributes choose particular kinds of environments.

Stress reaction

The tendency to become easily distressed by the normal challenges of life.

Transformation

The term for personality changes associated with experience and life events.

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Personality Assessment

David Watson

This module provides a basic overview to the assessment of personality. It discusses objective personality tests (based on both self-report and informant ratings), projective and implicit tests, and behavioral/performance measures. It describes the basic features of each method, as well as reviewing the strengths, weaknesses, and overall validity of each approach.

Learning Objectives

- Appreciate the diversity of methods that are used to measure personality characteristics.
- Understand the logic, strengths and weaknesses of each approach.
- Gain a better sense of the overall validity and range of applications of personality tests.

Introduction

Personality is the field within psychology that studies the thoughts, feelings, behaviors, goals, and interests of normal individuals. It therefore covers a very wide range of important psychological characteristics. Moreover, different theoretical models have generated very different strategies for measuring these characteristics. For example, humanistically oriented models argue that people have clear, well-defined goals and are actively striving to achieve them (McGregor, McAdams, & Little, 2006). It, therefore, makes sense to ask them directly about themselves and their goals. In contrast, psychodynamically oriented theories propose that people lack insight into their feelings and motives, such that their behavior is influenced by processes that operate outside of their awareness (e.g., McClelland, Koestner, &

Weinberger, 1989; Meyer & Kurtz, 2006). Given that people are unaware of these processes, it does not make sense to ask directly about them. One, therefore, needs to adopt an entirely different approach to identify these nonconscious factors. Not surprisingly, researchers have adopted a wide range of approaches to measure important personality characteristics. The most widely used strategies will be summarized in the following sections.



Do people possess the necessary awareness to see themselves as they are and provide accurate insights into their own personalities?
[Image: fotEK10, <https://goo.gl/GCBDJL>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

Objective Tests

Definition

Objective tests (Loevinger, 1957; Meyer & Kurtz, 2006) represent the most familiar and widely used approach to assessing personality. Objective tests involve administering a standard set of items, each of which is answered using a limited set of response options (e.g., true or false; strongly disagree, slightly disagree, slightly agree, strongly agree). Responses to these items then are scored in a standardized, predetermined way. For example, self-ratings on items assessing talkativeness, assertiveness, sociability, adventurousness, and energy can be

summed up to create an overall score on the personality trait of extraversion.

It must be emphasized that the term “objective” refers to the method that is used to score a person’s responses, rather than to the responses themselves. As noted by Meyer and Kurtz (2006, p. 233), “What is *objective* about such a procedure is that the psychologist administering the test does not need to rely on judgment to classify or interpret the test-taker’s response; the intended response is clearly indicated and scored according to a pre-existing key.” In fact, as we will see, a person’s test responses may be highly subjective and can be influenced by a number of different rating biases.

Basic Types of Objective Tests

Self-report measures

Objective personality tests can be further subdivided into two basic types. The first type—which easily is the most widely used in modern personality research—asks people to describe themselves. This approach offers two key advantages. First, self-raters have access to an unparalleled wealth of information: After all, who knows more about you than you yourself? In particular, self-raters have direct access to their own thoughts, feelings, and motives, which may not be readily available to others (Oh, Wang, & Mount, 2011; Watson, Hubbard, & Weise, 2000). Second, asking people to describe themselves is the simplest, easiest, and most cost-effective approach to assessing personality. Countless studies, for instance, have involved administering self-report measures to college students, who are provided some relatively simple incentive (e.g., extra course credit) to participate.

The items included in self-report measures may consist of single words (e.g., *assertive*), short phrases (e.g., *am full of energy*), or complete sentences (e.g., *I like to spend time with others*). Table 1 presents a sample self-report measure assessing the general traits comprising the influential five-factor model (FFM) of personality: neuroticism, extraversion, openness, agreeableness, and conscientiousness (John & Srivastava, 1999; McCrae, Costa, & Martin, 2005). The sentences shown in Table 1 are modified versions of items included in the International Personality Item Pool (IPIP) (Goldberg et al., 2006), which is a rich source of personality-related content in the public domain (for more information about IPIP, go to: <http://ipip.ori.org/>).

Self-report personality tests show impressive validity in relation to a wide range of important

Please read each statement carefully and then mark the appropriate response below. Use the following scale to record your responses:

1	2	3	4	5
strongly disagree	slightly disagree	neutral or cannot decide	slightly agree	strongly agree

-----	1. I get upset easily.
-----	2. I enjoy being part of a group.
-----	3. I like to solve complex problems.
-----	4. I believe that others have good intentions.
-----	5. I am always prepared.
-----	6. I have a low opinion of myself.
-----	7. I have a natural talent for influencing people.
-----	8. I enjoy the beauty of nature.
-----	9. I try to anticipate the needs of others.
-----	10. I can be trusted to keep my promises.
-----	11. I get irritated easily.
-----	12. I have a lot of fun.
-----	13. I like to visit new places.
-----	14. I love to help others.
-----	15. I set high standards for myself and others.

Sum up the following items to see how you score on five general personality traits. The numbers below indicate which questions correspond to each trait. A high score indicates a stronger level of that trait:

1	6	11	Neuroticism
2	7	12	Extraversion
3	8	13	Openness/Intellect
4	9	14	Agreeableness
5	10	15	Conscientiousness

Table1: Sample Self-Report Personality Measure

outcomes. For example, self-ratings of conscientiousness are significant predictors of both overall academic performance (e.g., cumulative grade point average; Poropat, 2009) and job performance (Oh, Wang, and Mount, 2011). Roberts, Kuncel, Shiner, Caspi, and Goldberg (2007) reported that self-rated personality predicted occupational attainment, divorce, and mortality. Similarly, Friedman, Kern, and Reynolds (2010) showed that personality ratings collected early in life were related to happiness/well-being, physical health, and mortality risk

assessed several decades later. Finally, self-reported personality has important and pervasive links to psychopathology. Most notably, self-ratings of neuroticism are associated with a wide array of clinical syndromes, including anxiety disorders, depressive disorders, substance use disorders, somatoform disorders, eating disorders, personality and conduct disorders, and schizophrenia/schizotypy (Kotov, Gamez, Schmidt, & Watson, 2010; Mineka, Watson, & Clark, 1998).

At the same time, however, it is clear that this method is limited in a number of ways. First, raters may be motivated to present themselves in an overly favorable, socially desirable way (Paunonen & LeBel, 2012). This is a particular concern in **“high-stakes testing,”** that is, situations in which test scores are used to make important decisions about individuals (e.g., when applying for a job). Second, personality ratings reflect a **self-enhancement bias** (Vazire & Carlson, 2011); in other words, people are motivated to ignore (or at least downplay) some of their less desirable characteristics and to focus instead on their more positive attributes. Third, self-ratings are subject to the **reference group effect** (Heine, Buchtel, & Norenzayan, 2008); that is, we base our self-perceptions, in part, on how we compare to others in our sociocultural reference group. For instance, if you tend to work harder than most of your friends, you will see yourself as someone who is relatively conscientious, even if you are not particularly conscientious in any absolute sense.

Informant ratings

Another approach is to ask someone who knows a person well to describe his or her personality characteristics. In the case of children or adolescents, the informant is most likely to be a parent or teacher. In studies of older participants, informants may be friends, roommates, dating partners, spouses, children, or bosses (Oh et al., 2011; Vazire & Carlson, 2011; Watson et al., 2000).

Generally speaking, informant ratings are similar in format to self-ratings. As was the case with self-report, items may consist of single words, short phrases, or complete sentences. Indeed, many popular instruments include parallel self- and informant-rating versions, and it often is relatively easy to convert a self-report measure so that it can be used to obtain informant ratings. Table 2 illustrates how the self-report instrument shown in Table 1 can be converted to obtain spouse-ratings (in this case, having a husband describe the personality characteristics of his wife).

Informant ratings are particularly valuable when self-ratings are impossible to collect (e.g., when studying young children or cognitively impaired adults) or when their validity is suspect

Please read each statement carefully and then mark the appropriate response below. For each item, select the answer that best represents the personality characteristics of *your wife*. Use the following scale to record your responses:

1	2	3	4	5
strongly disagree	slightly disagree	neutral or cannot decide	slightly agree	strongly agree

Table 2: Sample Spouse-Report Personality Measure

(e.g., as noted earlier, people may not be entirely honest in high-stakes testing situations). They also may be combined with self-ratings of the same characteristics to produce more reliable and valid measures of these attributes (McCrae, 1994).

Informant ratings offer several advantages in comparison to other approaches to assessing personality. A well-acquainted informant presumably has had the opportunity to observe large samples of behavior in the person he or she is rating. Moreover, these judgments presumably are not subject to the types of defensiveness that potentially can distort self-ratings (Vazire & Carlson, 2011). Indeed, informants typically have strong incentives for being accurate in their judgments. As Funder and Dobroth (1987, p. 409), put it, “Evaluations of the people in our social environment are central to our decisions about who to befriend and avoid, trust and distrust, hire and fire, and so on.”

Informant personality ratings have demonstrated a level of validity in relation to important

life outcomes that is comparable to that discussed earlier for self-ratings. Indeed, they outperform self-ratings in certain circumstances, particularly when the assessed traits are highly evaluative in nature (e.g., intelligence, charm, creativity; see Vazire & Carlson, 2011). For example, Oh et al. (2011) found that informant ratings were more strongly related to job performance than were self-ratings. Similarly, Oltmanns and Turkheimer (2009) summarized evidence indicating that informant ratings of Air Force cadets predicted early, involuntary discharge from the military better than self-ratings.

Nevertheless, informant ratings also are subject to certain problems and limitations. One general issue is the level of relevant information that is available to the rater (Funder, 2012). For instance, even under the best of circumstances, informants lack full access to the thoughts, feelings, and motives of the person they are rating. This problem is magnified when the informant does not know the person particularly well and/or only sees him or her in a limited range of situations (Funder, 2012; Beer & Watson, 2010).



Informant personality ratings are generally a reliable and valid assessment instrument, however in certain cases the informant may have some significant biases that make the rating less reliable. Newly married individuals for example are likely to rate their partners in an unrealistically positive way. [Image: Sociales El Heraldo de Saltillo, <https://goo.gl/3g3Qhh>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

Informant ratings also are subject to some of the same response biases noted earlier for self-ratings. For instance, they are not immune to the reference group effect. Indeed, it is well-established that parent ratings often are subject to a **sibling contrast effect**, such that parents exaggerate the true magnitude of differences between their children (Pinto, Rijdsdijk, Frazier-Wood, Asherson, & Kuntsi, 2012). Furthermore, in many studies, individuals are allowed to nominate (or even recruit) the informants who will rate them. Because of this, it most often is the case that informants (who, as noted earlier, may be friends, relatives, or romantic partners) like the people they are rating. This, in turn, means that informants may produce overly favorable personality ratings. Indeed, their ratings actually can be more favorable than the corresponding self-ratings (Watson & Humrichouse,

2006). This tendency for informants to produce unrealistically positive ratings has been termed the **letter of recommendation effect** (Leising, Erbs, & Fritz, 2010) and the **honeymoon effect**

when applied to newlyweds (Watson & Humrichouse, 2006).

Other Ways of Classifying Objective Tests

Comprehensiveness

In addition to the source of the scores, there are at least two other important dimensions on which personality tests differ. The first such dimension concerns the extent to which an instrument seeks to assess personality in a reasonably comprehensive manner. At one extreme, many widely used measures are designed to assess a single core attribute. Examples of these types of measures include the Toronto Alexithymia Scale (Bagby, Parker, & Taylor, 1994), the Rosenberg Self-Esteem Scale (Rosenberg, 1965), and the Multidimensional Experiential Avoidance Questionnaire (Gamez, Chmielewski, Kotov, Ruggero, & Watson, 2011). At the other extreme, a number of omnibus inventories contain a large number of specific scales and purport to measure personality in a reasonably comprehensive manner. These instruments include the California Psychological Inventory (Gough, 1987), the Revised HEXACO Personality Inventory (HEXACO-PI-R) (Lee & Ashton, 2006), the Multidimensional Personality Questionnaire (Patrick, Curtin, & Tellegen, 2002), the NEO Personality Inventory-3 (NEO-PI-3) (McCrae et al., 2005), the Personality Research Form (Jackson, 1984), and the Sixteen Personality Factor Questionnaire (Cattell, Eber, & Tatsuoka, 1980).

Breadth of the target characteristics

Second, personality characteristics can be classified at different levels of breadth or generality. For example, many models emphasize broad, “big” traits such as neuroticism and extraversion. These general dimensions can be divided up into several distinct yet empirically correlated component traits. For example, the broad dimension of extraversion contains such specific component traits as dominance (extraverts are assertive, persuasive, and exhibitionistic), sociability (extraverts seek out and enjoy the company of others), positive emotionality (extraverts are active, energetic, cheerful, and enthusiastic), and adventurousness (extraverts enjoy intense, exciting experiences).

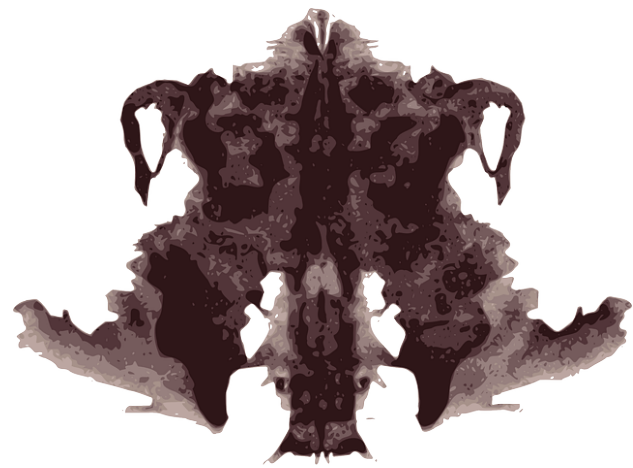
Some popular personality instruments are designed to assess only the broad, general traits. For example, similar to the sample instrument displayed in Table 1, the **Big Five** Inventory (John & Srivastava, 1999) contains brief scales assessing the broad traits of neuroticism, extraversion, openness, agreeableness, and conscientiousness. In contrast, many instruments—including several of the omnibus inventories mentioned earlier—were

designed primarily to assess a large number of more specific characteristics. Finally, some inventories—including the HEXACO-PI-R and the NEO-PI-3—were explicitly designed to provide coverage of both general and specific trait characteristics. For instance, the NEO-PI-3 contains six specific facet scales (e.g., Gregariousness, Assertiveness, Positive Emotions, Excitement Seeking) that then can be combined to assess the broad trait of extraversion.

Projective and Implicit Tests

Projective Tests

As noted earlier, some approaches to personality assessment are based on the belief that important thoughts, feelings, and motives operate outside of conscious awareness. Projective tests represent influential early examples of this approach. Projective tests originally were based on the **projective hypothesis** (Frank, 1939; Lilienfeld, Wood, & Garb, 2000): If a person is asked to describe or interpret ambiguous stimuli—that is, things that can be understood in a number of different ways—their responses will be influenced by nonconscious needs, feelings, and experiences (note, however, that the theoretical rationale underlying these measures has evolved over time) (see, for example, Spangler, 1992). Two prominent examples of projective tests are the Rorschach Inkblot Test (Rorschach, 1921) and the Thematic Apperception Test (TAT) (Morgan & Murray, 1935). The former asks respondents to interpret symmetrical blots of ink, whereas the latter asks them to generate stories about a series of pictures.



Projective tests, such as the famous Rorschach inkblot test require a person to give spontaneous answers that "project" their unique personality onto an ambiguous stimulus. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

For instance, one TAT picture depicts an elderly woman with her back turned to a young man; the latter looks downward with a somewhat perplexed expression. Another picture displays a man clutched from behind by three mysterious hands. What stories could you generate in response to these pictures?

In comparison to objective tests, projective tests tend to be somewhat cumbersome and labor intensive to administer. The biggest challenge, however, has been to develop a reliable and

valid scheme to score the extensive set of responses generated by each respondent. The most widely used Rorschach scoring scheme is the Comprehensive System developed by Exner (2003). The most influential TAT scoring system was developed by McClelland, Atkinson and colleagues between 1947 and 1953 (McClelland et al., 1989; see also Winter, 1998), which can be used to assess motives such as the need for achievement.

The validity of the Rorschach has been a matter of considerable controversy (Lilienfeld et al., 2000; Mihura, Meyer, Dumitrascu, & Bombel, 2012; Society for Personality Assessment, 2005). Most reviews acknowledge that Rorschach scores do show some ability to predict important outcomes. Its critics, however, argue that it fails to provide important incremental information beyond other, more easily acquired information, such as that obtained from standard self-report measures (Lilienfeld et al., 2000).

Validity evidence is more impressive for the TAT. In particular, reviews have concluded that TAT-based measures of the need for achievement (a) show significant validity to predict important criteria and (b) provide important information beyond that obtained from objective measures of this motive (McClelland et al., 1989; Spangler, 1992). Furthermore, given the relatively weak associations between objective and projective measures of motives, McClelland et al. (1989) argue that they tap somewhat different processes, with the latter assessing implicit motives (Schultheiss, 2008).

Implicit Tests

In recent years, researchers have begun to use implicit measures of personality (Back, Schmuckle, & Egloff, 2009; Vazire & Carlson, 2011). These tests are based on the assumption that people form automatic or implicit associations between certain concepts based on their previous experience and behavior. If two concepts (e.g., *me* and *assertive*) are strongly associated with each other, then they should be sorted together more quickly and easily than two concepts (e.g., *me* and *shy*) that are less strongly associated. Although validity evidence for these measures still is relatively sparse, the results to date are encouraging: Back et al. (2009), for example, showed that implicit measures of the FFM personality traits predicted behavior even after controlling for scores on objective measures of these same characteristics.

Behavioral and Performance Measures

A final approach is to infer important personality characteristics from direct samples of behavior. For example, Funder and Colvin (1988) brought opposite-sex pairs of participants into the laboratory and had them engage in a five-minute “getting acquainted” conversation;



Observing real world behavior is one way to assess personality. Tendencies such as messiness and neatness are clues to personality. [Image: Crumley Roberts, <https://goo.gl/6Ahn8q>, CC BY 2.0, <https://goo.gl/BRvSA7>]

raters watched videotapes of these interactions and then scored the participants on various personality characteristics. Mehl, Gosling, and Pennebaker (2006) used the electronically activated recorder (EAR) to obtain samples of ambient sounds in participants' natural environments over a period of two days; EAR-based scores then were related to self- and observer-rated measures of personality. For instance, more frequent talking over this two-day period was significantly related to both self- and observer-ratings of extraversion. As a final example, Gosling, Ko, Mannarelli, and Morris (2002) sent observers into college students' bedrooms and then had them rate the students' personality characteristics on the Big Five traits. The averaged observer ratings correlated significantly

with participants' self-ratings on all five traits. Follow-up analyses indicated that conscientious students had neater rooms, whereas those who were high in openness to experience had a wider variety of books and magazines.

Behavioral measures offer several advantages over other approaches to assessing personality. First, because behavior is sampled directly, this approach is not subject to the types of response biases (e.g., self-enhancement bias, reference group effect) that can distort scores on objective tests. Second, as is illustrated by the Mehl et al. (2006) and Gosling et al. (2002) studies, this approach allows people to be studied in their daily lives and in their natural environments, thereby avoiding the artificiality of other methods (Mehl et al., 2006). Finally, this is the only approach that actually assesses what people *do*, as opposed to what they think or feel (see Baumeister, Vohs, & Funder, 2007).

At the same time, however, this approach also has some disadvantages. This assessment strategy clearly is much more cumbersome and labor intensive than using objective tests, particularly self-report. Moreover, similar to projective tests, behavioral measures generate a rich set of data that then need to be scored in a reliable and valid way. Finally, even the most ambitious study only obtains relatively small samples of behavior that may provide a somewhat distorted view of a person's true characteristics. For example, your behavior during a "getting acquainted" conversation on a single given day inevitably will reflect a number of

transient influences (e.g., level of stress, quality of sleep the previous night) that are idiosyncratic to that day.

Conclusion

No single method of assessing personality is perfect or infallible; each of the major methods has both strengths and limitations. By using a diversity of approaches, researchers can overcome the limitations of any single method and develop a more complete and integrative view of personality.

Discussion Questions

1. Under what conditions would you expect self-ratings to be most similar to informant ratings? When would you expect these two sets of ratings to be most different from each other?
2. The findings of Gosling, et al. (2002) demonstrate that we can obtain important clues about students' personalities from their dorm rooms. What other aspects of people's lives might give us important information about their personalities?
3. Suppose that you were planning to conduct a study examining the personality trait of honesty. What method or methods might you use to measure it?

Vocabulary

Big Five

Five, broad general traits that are included in many prominent models of personality. The five traits are neuroticism (those high on this trait are prone to feeling sad, worried, anxious, and dissatisfied with themselves), extraversion (high scorers are friendly, assertive, outgoing, cheerful, and energetic), openness to experience (those high on this trait are tolerant, intellectually curious, imaginative, and artistic), agreeableness (high scorers are polite, considerate, cooperative, honest, and trusting), and conscientiousness (those high on this trait are responsible, cautious, organized, disciplined, and achievement-oriented).

High-stakes testing

Settings in which test scores are used to make important decisions about individuals. For example, test scores may be used to determine which individuals are admitted into a college or graduate school, or who should be hired for a job. Tests also are used in forensic settings to help determine whether a person is competent to stand trial or fits the legal definition of sanity.

Honeymoon effect

The tendency for newly married individuals to rate their spouses in an unrealistically positive manner. This represents a specific manifestation of the letter of recommendation effect when applied to ratings made by current romantic partners. Moreover, it illustrates the very important role played by relationship satisfaction in ratings made by romantic partners: As marital satisfaction declines (i.e., when the “honeymoon is over”), this effect disappears.

Implicit motives

These are goals that are important to a person, but that he/she cannot consciously express. Because the individual cannot verbalize these goals directly, they cannot be easily assessed via self-report. However, they can be measured using projective devices such as the Thematic Apperception Test (TAT).

Letter of recommendation effect

The general tendency for informants in personality studies to rate others in an unrealistically positive manner. This tendency is due a pervasive bias in personality assessment: In the large majority of published studies, informants are individuals who like the person they are rating (e.g., they often are friends or family members) and, therefore, are motivated to depict them in a socially desirable way. The term reflects a similar tendency for academic letters of recommendation to be overly positive and to present the referent in an unrealistically

desirable manner.

Projective hypothesis

The theory that when people are confronted with ambiguous stimuli (that is, stimuli that can be interpreted in more than one way), their responses will be influenced by their unconscious thoughts, needs, wishes, and impulses. This, in turn, is based on the Freudian notion of projection, which is the idea that people attribute their own undesirable/unacceptable characteristics to other people or objects.

Reference group effect

The tendency of people to base their self-concept on comparisons with others. For example, if your friends tend to be very smart and successful, you may come to see yourself as less intelligent and successful than you actually are. Informants also are prone to these types of effects. For instance, the sibling contrast effect refers to the tendency of parents to exaggerate the true extent of differences between their children.

Reliability

The consistency of test scores across repeated assessments. For example, test-retest reliability examines the extent to which scores change over time.

Self-enhancement bias

The tendency for people to see and/or present themselves in an overly favorable way. This tendency can take two basic forms: defensiveness (when individuals actually believe they are better than they really are) and impression management (when people intentionally distort their responses to try to convince others that they are better than they really are). Informants also can show enhancement biases. The general form of this bias has been called the letter-of-recommendation effect, which is the tendency of informants who like the person they are rating (e.g., friends, relatives, romantic partners) to describe them in an overly favorable way. In the case of newlyweds, this tendency has been termed the honeymoon effect.

Sibling contrast effect

The tendency of parents to use their perceptions of all of their children as a frame of reference for rating the characteristics of each of them. For example, suppose that a mother has three children; two of these children are very sociable and outgoing, whereas the third is relatively average in sociability. Because of operation of this effect, the mother will rate this third child as less sociable and outgoing than he/she actually is. More generally, this effect causes parents to exaggerate the true extent of differences between their children. This effect represents a specific manifestation of the more general reference group effect when applied to ratings made by parents.

Validity

Evidence related to the interpretation and use of test scores. A particularly important type of evidence is criterion validity, which involves the ability of a test to predict theoretically relevant outcomes. For example, a presumed measure of conscientiousness should be related to academic achievement (such as overall grade point average).

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Self-Regulation and Conscientiousness

Roy F. Baumeister

Self-regulation means changing oneself based on standards, that is, ideas of how one should or should not be. It is a centrally important capacity that contributes to socially desirable behavior, including moral behavior. Effective self-regulation requires knowledge of standards for proper behavior, careful monitoring of one's actions and feelings, and the ability to make desired changes.

Learning Objectives

- Understand what self-regulation means and how it works.
- Understand the requirements and benefits of effective self-regulation.
- Understand differences in state (ego depletion) and trait (conscientiousness).

Introduction

Self-regulation is the capacity to alter one's responses. It is broadly related to the term "self-control". The term "regulate" means to change something—but not just any change, rather change to bring it into agreement with some idea, such as a rule, a goal, a plan, or a moral principle. To illustrate, when the government regulates how houses are built, that means the government inspects the buildings to check that everything is done "up to code" or according to the rules about good building. In a similar fashion, when you regulate yourself, you watch and change yourself to bring your responses into line with some ideas about how they should be.



When you find that quiet spot in the library and keep yourself focused on your study tasks for a few hours you're demonstrating self-regulation. Certainly you're controlling your thinking, but you may also be controlling your impulses to do other things. [Image: Clemson University Library, <https://goo.gl/RtZrqu>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

People regulate four broad categories of responses. They control their thinking, such as in trying to concentrate or to shut some annoying earworm tune out of their mind. They control their emotions, as in trying to cheer themselves up or to calm down when angry (or to stay angry, if that's helpful). They control their impulses, as in trying not to eat fattening food, trying to hold one's tongue, or trying to quit smoking. Last, they try to control their task performances, such as in pushing themselves to keep working when tired and discouraged, or deciding whether to speed up (to get more done) or slow down (to make sure to get it right).

Early Work on Delay of Gratification

Research on self-regulation was greatly stimulated by early experiments conducted by Walter Mischel and his colleagues (e.g., Mischel, 1974) on the capacity to delay gratification, which means being able to refuse current temptations and pleasures to work toward future benefits. In a typical study with what later came to be called the "marshmallow test," a 4-year-old child would be seated in a room, and a favorite treat such as a cookie or marshmallow was placed on the table. The experimenter would tell the child, "I have to leave for a few minutes and then I'll be back. You can have this treat any time, but if you can wait until I come back, you can have two of them." Two treats are better than one, but to get the double treat, the child had to wait. Self-regulation was required to resist that urge to gobble down the marshmallow on the table so as to reap the larger reward.

Many situations in life demand similar delays for best results. Going to college to get an education often means living in poverty and debt rather than getting a job to earn money right away. But in the long run, the college degree increases your lifetime income by hundreds of thousands of dollars. Very few nonhuman animals can bring themselves to resist immediate temptations so as to pursue future rewards, but this trait is an important key to success in human life.

Benefits of Self-Control

People who are good at self-regulation do better than others in life. Follow-up studies with Mischel's samples found that the children who resisted temptation and delayed gratification effectively grew into adults who were better than others in school and work, more popular with other people, and who were rated as nicer, better people by teachers and others (Mischel, Shoda, & Peake, 1988; Shoda, Mischel, & Peake, 1990). College students with high self-control get better grades, have better close relationships, manage their emotions better, have fewer problems with drugs and alcohol, are less prone to eating disorders, are better adjusted, have higher self-esteem, and get along better with other people, as compared to people with low self-control (Tangney, Baumeister, & Boone, 2004). They are happier and have less stress and conflict (Hofmann, Vohs, Fisher, Luhmann, & Baumeister, 2013). Longitudinal studies have found that children with good self-control go through life with fewer problems, are more successful, are less likely to be arrested or have a child out of wedlock, and enjoy other benefits (Moffitt et al., 2011). Criminologists have concluded that low self-control is a—if not the—key trait for understanding the criminal personality (Gottfredson & Hirschi, 1990; Pratt & Cullen, 2000).



If you have never seen a 4-year-old try to resist eating a marshmallow, you may not realize how difficult (and funny) a task like this is. See the “Outside Resources” of this module for a great video demonstration. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Some researchers have searched for evidence that too much self-control can be bad (Tangney et al., 2004)—but without success. There is such a thing as being highly inhibited or clinically “over-controlled,” which can impair initiative and reduce happiness, but that does not appear to be an excess of self-regulation. Rather, it may stem from having been punished excessively as a child and, therefore, adopting a fearful, inhibited approach to life. In general, self-control resembles intelligence in that the more one has, the better off one is, and the benefits are found through a broad range of life activities.

Three Ingredients of Effective Self-Regulation

For self-regulation to be effective, three parts or ingredients are involved. The first is standards, which are ideas about how things should (or should not) be. The second is monitoring, which means keeping track of the target behavior that is to be regulated. The third is the capacity to change.

Standards are an indispensable foundation for self-regulation. We already saw that self-regulation means change in relation to some idea; without such guiding ideas, change would largely be random and lacking direction. Standards include goals, laws, moral principles, personal rules, other people's expectations, and social norms. Dieters, for example, typically have a goal in terms of how much weight they wish to lose. They help their self-regulation further by developing standards for how much or how little to eat and what kinds of foods they will eat.

The second ingredient is monitoring. It is hard to regulate something without being aware of it. For example, dieters count their calories. That is, they keep track of how much they eat and how fattening it is. In fact, some evidence suggests that dieters stop keeping track of how much they eat when they break their diet or go on an eating binge, and the failure of monitoring contributes to eating more (Polivy, 1976). Alcohol has been found to impair all sorts of self-regulation, partly because intoxicated persons fail to keep track of their behavior and compare it to their standards.

The combination of standards and monitoring was featured in an influential theory about self-regulation by Carver and Scheier (1981, 1982, 1998). Those researchers started their careers studying self-awareness, which is a key human trait.

The study of self-awareness recognized early on that people do not simply notice themselves the way they might notice a tree or car. Rather, self-awareness always seemed to involve comparing oneself to a standard. For example, when a man looks in a mirror, he does not just think, "Oh, there I am," but more likely thinks, "Is my hair a mess? Do my clothes look good?" Carver and Scheier proposed that the reason for this comparison to standards is that



With some self-regulation goals (like increasing your distance in preparation for a 10K race), it is easier to monitor your actual progress. With other goals, however, if there isn't a helpful standard to compare oneself to it may be harder to know if you are progressing. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

it enables people to regulate themselves, such as by changing things that do not measure up to their standards. In the mirror example, the man might comb his hair to bring it into line with his standards for personal appearance. Good students keep track of their grades, credits, and progress toward their degree and other goals. Athletes keep track of their times, scores, and achievements, as a way to monitor improvement.

The process of monitoring oneself can be compared to how a thermostat operates. The thermostat checks the temperature in the room, compares it to a standard (the setting for desired temperature), and if those do not match, it turns on the heat or air conditioner to change the temperature. It checks again and again, and when the room temperature matches the desired setting, the thermostat turns off the climate control. In the same way, people compare themselves to their personal standards, make changes as needed, and stop working on change once they have met their goals. People feel good not just when they reach their goals but even when they deem they are making good progress (Carver & Scheier, 1990). They feel bad when they are not making sufficient progress.

That brings up the third ingredient, which is the capacity to change oneself. In effective self-regulation, people operate on themselves to bring about these changes. The popular term for this is “willpower,” which suggests some kind of energy is expended in the process. Psychologists hesitate to adopt terms associated with folk wisdom, because there are many potential implications. Here, the term is used to refer specifically to some energy that is involved in the capacity to change oneself.

Consistent with the popular notion of willpower, people do seem to expend some energy during self-regulation. Many studies have found that after people exert self-regulation to change some response, they perform worse on the next unrelated task if it too requires self-regulation (Hagger, Wood, Stiff, & Chatzisarantis, 2010). That pattern suggests that some energy such as willpower was used up during the first task, leaving less available for the second task. The term for this state of reduced energy available for self-regulation is **ego depletion** (Baumeister, Bratslavsky, Muraven, & Tice, 1998). Current research provides mixed results on ego depletion, and we need further study to better understand when and how it occurs. It may be that as people go about their daily lives, they gradually become ego depleted because they are exerting self-control and resisting temptations. Some research suggests that during the state of ego depletion people become less helpful and more aggressive, prone to overeat, misbehave sexually, and express more prejudice (Hofmann, Vohs, & Baumeister, 2012).

Thus, a person’s capacity for self-regulation is not constant, but rather it fluctuates. To be sure, some people are generally better than others at controlling themselves (Tangney et al., 2004). But even someone with excellent self-control may occasionally find that control breaks down



People can get worn down from exercising self-control. And when they do they're more likely to make the kinds of decisions that are not in their own best interests. [Image: Tim Caynes, <https://goo.gl/vaoc3q>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

under ego depletion. In general, self-regulation can be improved by getting enough sleep and healthy food, and by minimizing other demands on one's willpower.

There is some evidence that regular exercise of self-control can build up one's willpower, like strengthening a muscle (Baumeister & Tierney, 2011; Oaten & Cheng, 2006). Even in early adulthood, one's self-control can be strengthened. Furthermore, research has shown that disadvantaged, minority children who take part in preschool programs such as Head Start (often based on the Perry program) end up doing better in life even as adults. This was thought for a while to be due to increases

in intelligence quotient (IQ), but changes in IQ from such programs are at best temporary. Instead, recent work indicates that improvement in self-control and related traits may be what produce the benefits (Heckman, Pinto, & Savelyev, in press). It's not doing math problems or learning to spell at age 3 that increases subsequent adult success—but rather the benefit comes from having some early practice at planning, getting organized, and following rules.

Conscientiousness

Conscientiousness is a stable dimension of personality, which means that some people are typically higher on it than others. Being a personality trait does not mean that it is unchangeable. Most people do show some changes over time, particularly becoming higher on conscientiousness as they grow older. Some psychologists look specifically at the trait of self-control, which is understood (and measured) in personality psychology in a very specific, narrowly focused, well-defined sense. Conscientiousness, in contrast, is one of five super-traits that supposedly account for all the other traits, in various combinations. The trait self-control is one big part of conscientiousness, but there are other parts.

Two aspects of conscientiousness that have been well documented are being orderly and being industrious (Roberts, Lejuez, Krueger, Richards, & Hill, 2012). Orderliness includes being clean and neat, making and following plans, and being punctual (which is helpful with following

plans!). Low conscientious means the opposite: being disorganized, messy, late, or erratic. Being industrious not only means working hard but also persevering in the face of failures and difficulties, as well as aspiring to excellence. Most of these reflect good self-control.

Conscientious people are careful, disciplined, responsible, and thorough, and they tend to plan and think things through before acting. People who are low in conscientiousness tend to be more impulsive and spontaneous, even reckless. They are easygoing and may often be late or sloppy, partly because they are not strongly focused on future goals for success and not highly concerned to obey all rules and stay on schedule. Psychologists prefer not to make a value judgment about whether it is better to be high or low in any personality trait. But when it comes specifically to self-control, it is difficult to resist the conclusion that high self-control is better, both for the person and for society at large.

Some aspects of conscientiousness have less apparent connection to self-control, however. People high in conscientiousness tend to be decisive. They are often formal, in the sense of following social norms and rules, such as dressing properly, waiting one's turn, or holding doors for others. They tend to respect traditions and traditional values.

Conscientious people behave differently from people who score low on that trait. People scoring low on conscientiousness are more likely than others to report driving without wearing seatbelts, daydreaming, swearing, telling dirty jokes, and picking up hitchhikers (Hirsh, DeYoung, & Peterson, 2009). In terms of more substantial life outcomes, people low on conscientiousness are more likely than others to get divorced, presumably because they make bad choices and misbehave during the marriage such as by saying hurtful things, getting into arguments and fights, and behaving irresponsibly (Roberts, Jackson, Fayard, Edmonds, & Meints, 2009). People low on conscientiousness are more likely than others to lose their jobs, to become homeless, to do time in prison, to have money problems, and to have drug problems.

Conscientious people make better spouses. They are less likely than others to get divorced,



One of the most important characteristics of people high in conscientiousness is orderliness. If someone could take a look at your desk or your room right now, how conscientious would they judge you to be? [Image: William Iven, CC0 Public Domain, <https://goo.gl/m25gce>]

partly because they avoid many behaviors that undermine intimacy, such as abusing their partners, drinking excessively, or having extramarital affairs (Roberts et al., 2009).

Encompassing self-control, conscientiousness is the personality trait with the strongest effect on life or death: People high on that trait live longer than others (Deary, Weiss, & Batty, 2010). Why? Among other things, they avoid many behavior patterns associated with early death, including alcohol abuse, obesity and other eating problems, drug abuse, smoking, failure to exercise, risky sex, suicide, violence, and unsafe driving (Bogg & Roberts, in press). They also visit physicians more regularly and take their prescribed medicines more reliably than people low in conscientiousness. Their good habits help avoid many life-threatening diseases.

Outside Resources

Book: For more advanced and in-depth coverage, consult *The Handbook of Self-Regulation* (2nd Edition), edited by Kathleen Vohs and Roy Baumeister. This book contains different chapters by different experts in the field, covering large amounts of research findings.

Book: To read more, the easiest and most fun source would be *The New York Times* bestseller *Willpower: Rediscovering the Greatest Human Strength*, by Roy Baumeister and John Tierney, published by Penguin. This is intended not as a purely scientific work but as an entertaining summary for the general public.

Video: For an enjoyable and brief re-enactment of Mischel's "marshmallow" studies on delay of gratification, try the following video. Watching those children struggle to resist temptation is sure to bring a smile.

<http://vimeo.com/5239013>

Discussion Questions

1. Why do you think criminals are often poor at self-regulation?
2. On average, children growing up without both parents present do worse at many things, from math achievement in school to the likelihood of being arrested for crimes. Might self-control be part of the explanation? Why?
3. Many people make New Year's resolutions to change themselves in various ways, but often they fail at these. Why?
4. Is good self-control something one is born with or something that is learned?
5. How would a parent teach his or her children to have good self-control?
6. Why are people with good self-control happier than other people?

Vocabulary

Conscientiousness

A personality trait consisting of self-control, orderliness, industriousness, and traditionalism.

Ego depletion

The state of diminished willpower or low energy associated with having exerted self-regulation.

Monitoring

Keeping track of a target behavior that is to be regulated.

Self-regulation

The process of altering one's responses, including thoughts, feelings, impulses, actions, and task performance.

Standards

Ideas about how things should (or should not) be.

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Creativity

Dean Keith Simonton

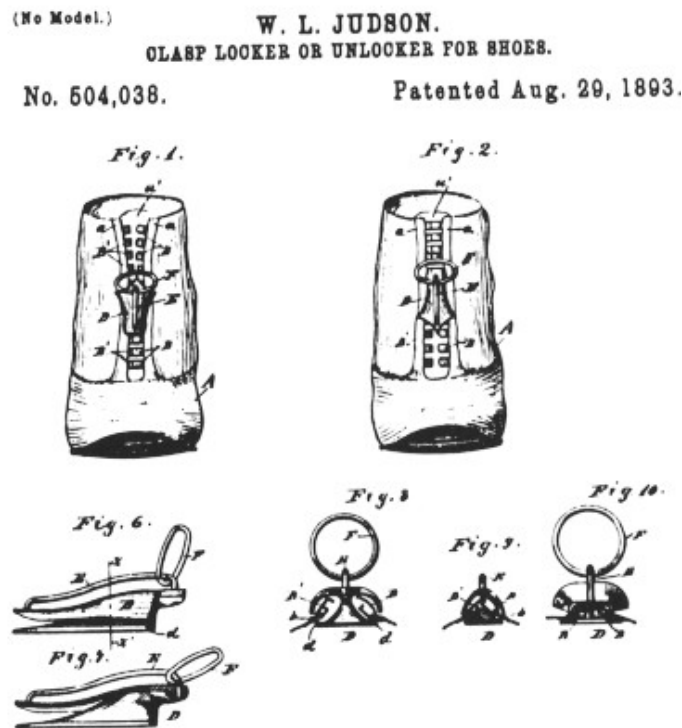
An idea or solution is considered creative if it is original, useful, and surprising. However, depending on who actually judges these three criteria, we must distinguish personal “little-c creativity” from consensual “Big-C Creativity.” In any case, psychologists who investigate creativity most often adopt one of three perspectives. First, they can ask how creators think, and thus focus on the cognitive processes behind creativity. Second, they can ask who is creative, and hence investigate the personal characteristics of highly creative people. Third, they can ask about the social context, and, thereby, examine the environments that influence creativity. Although psychologists have made major advances in the study of creativity, many exciting and important questions remain to be answered.

Learning Objectives

- Comprehend the three criteria that have to be satisfied to conclude that an idea is creative.
- Appreciate some of the cognitive processes that provide the basis for creativity.
- Know some of the personal characteristics of highly creative people.
- Understand how certain social environments influence creativity.

What do the following have in common: the drug penicillin, the Eiffel Tower, the film *Lord of the Rings*, the General Theory of Relativity, the hymn *Amazing Grace*, the iPhone, the novel *Don Quixote*, the painting *The Mona Lisa*, a recipe for chocolate fudge, the soft drink Coca-Cola, the video game *Wii Sports*, the West Coast offense in football, and the zipper? You guessed right! All of the named items were products of the creative mind. Not one of them existed until

somebody came up with the idea. Creativity is not something that you just pick like apples from a tree. Because creative ideas are so special, creators who come up with the best ideas are often highly rewarded with fame, fortune, or both. Nobel Prizes, Oscars, Pulitzers, and other honors bring fame, and big sales and box office bring fortune. Yet what is creativity in the first place?



The original patent drawings for the "clasp locker" - what we today call a zipper. This ubiquitous device was the product of the creative mind of American inventor Whitcomb Judson, who also invented a machine to make zippers quickly and cheaply. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Creativity: What Is It?

Creativity happens when someone comes up with a creative idea. An example would be a creative solution to a difficult problem. But what makes an idea or solution creative? Although psychologists have offered several definitions (Plucker, Beghetto, & Dow, 2004; Runco & Jaeger, 2012), probably the best definition is the one recently adapted from the three criteria that the U.S. Patent Office uses to decide whether an invention can receive patent protection (Simonton, 2012).



Knowing that magnifying glasses were already widely known, how would you judge the creativity of the invention of eyeglasses using the criteria of originality, usefulness, and surprise? [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

is a straightforward derivation from acquired expertise cannot be considered surprising even if it were original. Einstein's relativity theory was not a step-by-step deduction from classical physics but rather the theory was built upon a new foundation that challenged the very basis of traditional physics.

When applying these three criteria, it is critical to recognize that originality, usefulness, and surprise are all quantitative rather than qualitative attributes of an idea. Specifically, we really have to speak of degree to which an idea satisfies each of the three criteria. In addition, the three attributes should have a zero point, that is, it should be possible to speak of an idea lacking any originality, usefulness, or surprise whatsoever. Finally, we have to assume that if an idea scores zero on any one criterion then it must have zero creativity as well. For example, someone who reinvents the wheel is definitely producing a useful idea, but the idea has zero originality and hence no creativity whatsoever. Similarly, someone who invented a parachute made entirely out of steel reinforced concrete would get lots of credit for originality—and surprise!—but none for usefulness.

Yet, certainly, we have to ask: Who makes these judgments? The person who generated the idea or other people who the person expects to appreciate the idea? If the former, we can

The first criterion is originality. The idea must have a low probability. Indeed, it often should be unique. Albert Einstein's special theory of relativity certainly satisfied this criterion. No other scientist came up with the idea.

The second criterion is *usefulness*. The idea should be valuable or work. For example, a solution must, in fact, solve the problem. An original recipe that produces a dish that tastes too terrible to eat cannot be creative. In the case of Einstein's theory, his relativity principle provided explanations for what otherwise would be inexplicable empirical results.

The third and last criterion is *surprise*. The idea should be surprising, or at least nonobvious (to use the term used by the Patent Office). For instance, a solution that

speak of subjective or personal “little-c creativity,” and if the later, we have objective or consensual “Big-C Creativity” (Simonton, in press). This distinction is important because sometimes personal and consensual assessments do not have to agree. Such disagreements are especially conspicuous in “neglected geniuses,” such as the poet Emily Dickinson, the painter Vincent Van Gogh, and the scientist Gregor Mendel—all producing ideas that received only posthumous recognition for their creativity.

Creativity is a very complex phenomenon (Hennessey & Amabile, 2010; Runco, 2004). As a result, psychologists who study creativity can do so from many different perspectives. Nevertheless, the three most common perspectives are cognitive processes, personal characteristics, and social contexts.

Cognitive Processes: How Do Creators Think?

Cognitive scientists have long been interested in the thinking processes that lead to creative ideas (Simonton & Damian, 2013). Indeed, many so-called “creativity tests” are actually measures of the thought processes believed to underlie the creative act (Simonton, 2003b). The following two measures are among the best known.



What do you see? An ordinary piece of building material? Those who score higher on the Unusual Uses Task are able to imagine many possibilities for the humble brick. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

The first is the Remote Associates Test, or RAT, that was introduced by Mednick (1962). Mednick believed that the creative process requires the ability to associate ideas that are considered very far apart conceptually. The RAT consists of items that require the respondent to identify a word that can be associated to three rather distinct stimulus words. For example, what word can be associated with the words “widow, bite, monkey”? The answer is spider (black widow spider, spider bite, spider monkey). This particular question is relatively easy, others are much more difficult, but it gives you the basic idea.

The second measure is the Unusual Uses Task (Guilford, 1967; Torrance, 1974). Here, the participant is asked to generate

alternative uses for a common object, such as a brick. The responses can be scored on four dimensions: (a) *fluency*, the total number of appropriate uses generated; (b) *originality*, the statistical rarity of the uses given; (c) *flexibility*, the number of distinct conceptual categories implied by the various uses; and (d) *elaboration*, the amount of detail given for the generated uses. For example, using a brick as a paperweight represents a different conceptual category than using its volume to conserve water in a toilet tank. The capacity to produce **unusual uses** is but one example of the general cognitive ability to engage in **divergent thinking** (Guilford, 1967). Unlike **convergent thinking**, which converges on the single best answer or solution, divergent thinking comes up with multiple possibilities that might vary greatly in usefulness.

Unfortunately, many different cognitive processes have been linked to creativity (Simonton & Damian, 2013). That is why we cannot use the singular; there is no such thing as *the* “creative process.” Nonetheless, the various processes do share one feature: All enable the person to “think outside the box” imposed by routine thinking—to venture into territory that would otherwise be ignored (Simonton, 2011). Creativity requires that you go where you don’t know where you’re going.

Personal Characteristics: Who Is Creative?

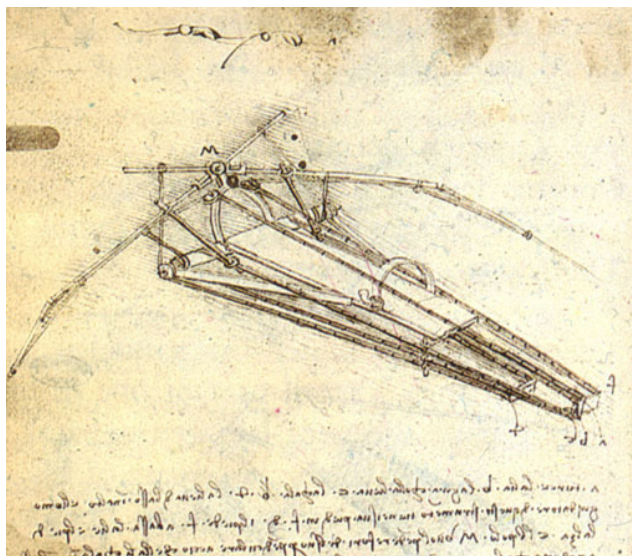
Add one point if you check		Subtract one point if you check	
capable	inventive	artificial	submissive
clever	original	cautious	suspicious
confident	reflective	commonplace	well-mannered
egotistical	resourceful	conservative	
humorous	self-confident	conventional	
individualistic	sexy	dissatisfied	
informal	snobbish	honest	
insightful	unconventional	narrow interests	
intelligent	wide interests	sincere	

Table 1. The Creative Personality Scale of the Gough (1979) Adjective Check List. Note. The Adjective Check List actually contains 300 adjectives arranged in alphabetical order from “absent-minded” to “zany,” but the above adjectives are the only ones scored.

Can anybody be creative? Or is creativity subject to individual differences, such as intelligence? Might creativity even be normally distributed just like scores on IQ tests? The answer is complex. Unlike general intelligence, which represents a more or less cohesive cognitive ability, creativity is just as much a personal attribute as an intellectual capacity. This feature is evident in the fact that some “creativity tests” are actually measures of personality, interests, and values (Simonton, 2003b). An example is the Creative Personality Scale of the Gough Adjective Check List (Gough, 1979; see also Carson, Peterson, & Higgins, 2005). In this measure, a person is asked to check off whatever adjectives are viewed as especially self-descriptive. The relevant adjectives are shown in Table 1. How would you describe *yourself*? Would you use more adjectives in the left column or the right column?

Another reason to speak of the “creative personality” is that creativity correlates with scores on standard personality measures (Feist, 1998). Most notably, the creative person is more likely to score on the openness-to-experience factor of the Big Five Factor Model (Carson, Peterson, & Higgins, 2005; Harris, 2004; McCrae, 1987). This factor concerns whether a person has a strong intellectual curiosity, preference for variety, and an active imagination and is aesthetically sensitive, attentive to inner feelings, as well as receptive to new ideas and values. It would seem obvious that persons high on this factor would *behave* differently than those scoring low. For instance, we would expect such persons to be less conventional, to have a wider range of leisure activities, and to be more versatile. Yet, it is equally important to note that people high in openness also *think* differently. Besides scoring higher in divergent thinking

(Carson, Peterson, & Higgins, 2005), openness is also associated with the diminished capacity to filter out extraneous information, a tendency often called cognitive disinhibition or reduced **latent inhibition** (Peterson & Carson, 2000). This “defocused attention” enables the creative person to make observations that others would overlook—such as what happens in serendipitous discovery. A classic example was when Alexander Flemming noticed that a bacteria culture was being killed by a certain mold, a discovery that directly led to penicillin.



Leonardo da Vinci's design for a flying machine. Leonardo spent years as a young man working as an artist's apprentice and later studied subjects such as engineering, anatomy, and mathematics. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Now you may wonder, isn't cognitive disinhibition a *bad* way of thinking? Isn't it a good thing to be able to ignore irrelevant

stimuli? The answer is yes. In fact, reduced latent inhibition is also connected with mental illness (Carson, 2011; Eysenck, 1995). Thus arises a link between creativity and psychopathology (Simonton, 2010). Even so, creative individuals are seldom outright mentally ill. Instead, creators possess other personal traits and capacities that convert a potential cognitive disability into an exceptional ability (Carson, 2011). Among the most important of these characteristics is high general intelligence (Carson, Peterson, & Higgins, 2005; Kéri, 2011). The creator then has the capacity not just to generate original and surprising ideas but also to test and develop them for usefulness. Mental illness arises when the person always skips the last step—the reality check.

At this point, we must add an important qualification: For Big-C Creativity, you have to have more going for you than boasting a creative personality that can engage in creative thought. You also must acquire appropriate expertise in the domain in which you hope to make creative contributions. Einstein had to learn physics and mathematics; Leonardo da Vinci had to learn how to draw and paint. In fact, it typically requires about a decade of extensive training and practice before a person can become a Big-C Creator (Ericsson, 1996). Even so, just because you become an expert in your field it does not mean that you'll be creative, too.

Social Contexts: What Environments Affect Creativity?

Although creativity is often viewed as an entirely psychological phenomenon, research by social psychologists shows that certain social environments have a part to play as well. These contextual influences can assume many forms. Sometimes these effects are relatively short term or transient. Other times the effects can be more long lasting.

To illustrate the former possibility, creativity is often enhanced when persons are exposed to incongruous or novel stimuli. For example, one recent experiment used virtual reality to create three conditions (Ritter et al., 2012). In one condition, participants walked around in a room in which the normal laws of physics were violated. Objects fell up rather than fell down, and the objects got smaller as you approached them rather than getting bigger. In a second condition, the participants were in the same virtual reality situation, but everything behaved as it would in normal reality. In the third and last condition, the participants merely saw a film clip of what the participants in the first condition experienced—a passive rather than active exposure to an otherworldly environment. Only those who directly experienced the strange environment showed an increase in cognitive flexibility, an important component of creativity, as noted earlier. In a second experiment, the participants were again subjected to three conditions, but this time the manipulation concerned cultural scripts—in this case, the customary way to make a popular breakfast meal. Only those participants who directly

experienced the violation of the norms showed an increase in cognitive flexibility. Those who made breakfast the normal way or who vicariously watched somebody else make breakfast an unusual way showed no effect.

The above effect is most likely transient. It is doubtful that those participants exposed to such incongruous experiences would exhibit any long-term change in their creativity. But what would happen if the exposure was much longer, years rather than minutes? Then the benefit might endure a lifetime. An example is the long-term benefits that accrue to persons who have acquired multicultural experiences, such as living in a foreign country for a significant amount of time (Leung, Maddux, Galinsky, & Chiu, 2008). Daily life abroad exposes a person to different ways of doing everyday activities. Moreover, because the visitor quickly learns that “when in Rome do as the Romans do,” the exposure becomes direct rather than vicarious (Maddux, Adam, & Galinsky, 2010). To be sure, not everybody’s creativity benefits from multicultural environments. The person also has to score high on openness to experience (Leung & Chiu, 2008). Otherwise, they will close themselves off from the potential stimulation, and then just gripe about the “peculiar customs of the natives” rather than actively practice those customs—such as making a totally different breakfast!



Multicultural experiences are advantageous for creative thinking. [Image: Ryan Maple, <https://goo.gl/Cu5qrn>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Finally, both little-c and Big-C creativity—but especially the latter—are more likely to appear in specific sociocultural systems (Simonton, 2003a). Some political, social, cultural, and economic environments are supportive of exceptional creativity, whereas others tend to suppress if not destroy creativity. For this reason, the history of any civilization or nation tends to have “Dark Ages” as well as “Golden Ages.” Early medieval Europe illustrates the former, while Renaissance Italy exemplifies the latter. It would take us too far beyond introductory

psychology to discuss all of the relevant factors. Yet, one factor fits nicely with what was discussed in the previous paragraph. Highly creative societies are far more likely to be multicultural, with abundant influences from other civilizations. For instance, Japanese civilization tended to undergo a revival of creativity after the infusion of new ideas from other civilizations, including Korean, Chinese, Indian, and European (Simonton, 1997). This influx involved not just Japanese living abroad but also non-Japanese immigrating to Japan.

Conclusion

Creativity certainly must be considered a crucial human behavior. Indeed, like language, creativity sets *Homo sapiens* well apart from even our closest evolutionary relatives. It is virtually impossible to imagine a world in which all of the products of the creative mind were removed. I couldn't even type this very sentence at this instant. Even the alphabet was invented. Creativity permeates every aspect of modern life: technology, science, literature, the visual arts, music, cooking, sports, politics, war, business, advertising ... well, I could go on and on. Fortunately, psychologists have made major strides in understanding the phenomenon. In fact, some of the best studies of creativity are also excellent examples of scientific creativity. At the same time, it remains clear that we still have a long ways to go before we know everything we need to know about the psychology of creativity. Hence, creativity research has a bright future.

Outside Resources

Video: Amy Tan: Where does creativity hide?

http://www.ted.com/talks/amy_tan_on_creativity.html

Video: Creativity science

<http://www.youtube.com/watch?v=EL4bVMeuxlM&feature=youtu.be>

Video: How to be creative

<https://www.youtube.com/watch?v=weIQlthC3Ks>

Web: American Creativity Association

<http://www.aca.cloverpad.org/>

Web: Be More Creative

<http://www.bemorecreative.com/>

Web: Creating Minds

<http://creatingminds.org/>

Web: Creative Quotations

<http://www.creativequotations.com/>

Web: Creativity at Work

<http://www.creativityatwork.com/>

Discussion Questions

1. To be creative an idea must be useful. Although it is easy to see how a new invention can be useful, what does it mean for a scientific discovery or artistic composition to be useful? When, in 1865, Mendel discovered that the traits of peas were inherited according to genetic laws, what possible use could that finding have at the time? What conceivable utility could there be for a painting by Van Gogh or a poem by Dickinson? Should some other word be used, such as valuable or appropriate? Or, should we acknowledge that a theory, painting, or poem is useful in a different way than an invention? Can a new idea be creative just because it satisfies our intellectual curiosity or aesthetic appreciation?

2. Computers can do some amazing things—such as beat humans at chess and Jeopardy! But, do you think that they can ever display genuine creativity? Will a computer one day make a scientific discovery or write a poem that earns it a Nobel Prize? If not, why not? If so, who should get the award money, the computer or the computer's programmer?
3. All of the personal characteristics of very creative people are also highly inheritable. For instance, intelligence, openness to experience, and cognitive disinhibition all have a partial genetic basis. Does that mean that creators are born and not made?
4. Highly creative people believe that they possess certain personality traits. If you make yourself have the same traits, will that make you more creative? For example, will you become more creative if you become more egotistical, individualistic, informal, reflective, self-confident, sexy, and unconventional? Or, how about widening your interests and becoming more open to experience? Which comes first, the personality or the capacity?

Vocabulary

Big-C Creativity

Creative ideas that have an impact well beyond the everyday life of home or work. At the highest level, this kind of creativity is that of the creative genius.

Convergent thinking

The opposite of divergent thinking, the capacity to narrow in on the single “correct” answer or solution to a given question or problem (e.g., giving the right response on an intelligence tests).

Divergent thinking

The opposite of convergent thinking, the capacity for exploring multiple potential answers or solutions to a given question or problem (e.g., coming up with many different uses for a common object).

Latent inhibition

The ability to filter out extraneous stimuli, concentrating only on the information that is deemed relevant. Reduced latent inhibition is associated with higher creativity.

Little-c creativity

Creative ideas that appear at the personal level, whether the home or the workplace. Such creativity needs not have a larger impact to be considered creative.

Multicultural experiences

Individual exposure to two or more cultures, such as obtained by living abroad, emigrating to another country, or working or going to school in a culturally diverse setting.

Openness to experience

One of the factors of the Big Five Model of personality, the factor assesses the degree that a person is open to different or new values, interests, and activities.

Originality

When an idea or solution has a low probability of occurrence.

Remote associations

Associations between words or concepts that are semantically distant and thus relatively unusual or original.

Unusual uses

A test of divergent thinking that asks participants to find many uses for commonplace objects, such as a brick or paperclip.

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Self-Efficacy

James E Maddux & Evan Kleiman

The term “self-efficacy” refers to your beliefs about your ability to effectively perform the tasks needed to attain a valued goal. Self-efficacy does not refer to your abilities but to how strongly you believe you can use your abilities to work toward goals. Self-efficacy is not a unitary construct or trait; rather, people have self-efficacy beliefs in different domains, such as academic self-efficacy, problem-solving self-efficacy, and self-regulatory self-efficacy. Stronger self-efficacy beliefs are associated with positive outcomes, such as better grades, greater athletic performance, happier romantic relationships, and a healthier lifestyle.

Learning Objectives

- Define self-efficacy.
- List the major factors that influence self-efficacy.
- Explain how self-efficacy develops.
- Understand the influence of self-efficacy on psychological and physical health and well-being as well as academic and vocational success.
- Define collective efficacy and explain why it is important.

Introduction: What Is Self-Efficacy?

Imagine two students, Sally and Lucy, who are about to take the same math test. Sally and Lucy have the same exact ability to do well in math, the same level of intelligence, and the same motivation to do well on the test. They also studied together. They even have the same



It makes intuitive sense that if you believe you're going to do well on a math test, you'll do better than if you believe you won't do well. But why should this be the case? [Image: Steve S., <https://goo.gl/4JOZdG>, CC BY 2.0, <https://goo.gl/BRvSA7>]

brand of shoes on. The only difference between the two is that Sally is very confident in her mathematical and her test-taking abilities, while Lucy is not. So, who is likely to do better on the test? Sally, of course, because she has the confidence to use her mathematical and test-taking abilities to deal with challenging math problems and to accomplish goals that are important to her—in this case, doing well on the test. This difference between Sally and Lucy—the student who got the A and the student who got the B-, respectively—is *self-efficacy*. As you will read later, self-efficacy influences behavior and emotions in particular ways that help people better manage challenges and achieve valued goals.

A concept that was first introduced by Albert Bandura in 1977, *self-efficacy* refers to a person's beliefs that he or she is able to effectively perform the tasks needed to attain a valued goal (Bandura, 1977). Since then, self-efficacy has become one of the most thoroughly researched concepts in psychology. Just about every important domain of human behavior has been investigated using self-efficacy theory (Bandura, 1997; Maddux, 1995; Maddux & Gosselin, 2011, 2012). Self-efficacy does not refer to your abilities but rather to your beliefs about what you can do with your abilities. Also, self-efficacy is not a trait—there are not certain types of people with high self-efficacies and others with low self-efficacies (Stajkovic & Luthans, 1998). Rather, people have self-efficacy beliefs about specific goals and life domains. For example, if you believe that you have the skills necessary to do well in school and believe you can use those skills to excel, then you have high *academic self-efficacy*.

Self-efficacy may sound similar to a concept you may be familiar with already—*self-esteem*—but these are very different notions. Self-esteem refers to how much you like or “esteem” yourself—to what extent you believe you are a good and worthwhile person. Self-efficacy, however, refers to your self-confidence to perform well and to achieve in specific areas of life such as school, work, and relationships. Self-efficacy does influence self-esteem because how you feel about yourself overall is greatly influenced by your confidence in your ability to perform well in areas that are important to you and to achieve valued goals. For example, if performing well in athletics is very important to you, then your self-efficacy for athletics will

greatly influence your self-esteem; however, if performing well in athletics is not at all important to you, then your self-efficacy for athletics will probably have little impact on your self-esteem.

How Do We Measure Self-Efficacy?

Like many other concepts in psychology, self-efficacy is not necessarily measured in a straightforward manner and requires much thought to be measured accurately. Self-efficacy is unlike weight, which is simple to objectively measure by using a scale, or height, which is simple to objectively measure by using a tape measure. Rather, self-efficacy is an abstract concept you can't touch or see. To measure an abstract concept like self-efficacy, we use something called a **self-report measure**. A self-report measure is a type of questionnaire, like a survey, where people answer questions usually with answers that correspond to numerical values that can be added to create an overall index of some construct. For example, a well-known self-report measure is the Perceived Stress Scale (Cohen, Kamarck, & Mermelstein, 1983). It asks questions like, "In the last month, how often have you been upset because of something that happened unexpectedly?" and "In the last month, how often have you been angered because of things that were outside of your control?" Participants answer the questions on a 1 through 5 scale, where 1 means "not often" and 5 means "very often." Then all of the answers are summed together to create a total "stress" score, with higher scores equating to higher levels of stress. It is very important to develop tools to measure self-efficacy that take people's subjective beliefs about their self-efficacy and turn them into the most objective possible measure. This means that one person's score of 6 out of 10 on a measure of self-efficacy will be similar to another person's score of 6 out of 10 on the same measure.

We will discuss two broad types of self-report measures for self-efficacy. The first category includes measures of general self-efficacy (e.g., Schwarzer & Jerusalem, 1995; Sherer et al., 1982). These scales ask people



How confident would you be about doing daily exercise if you had small children to take care of? Task specific measures of self efficacy would take this sort of circumstance into account. [Image: Ed Yourdon, <https://goo.gl/DH582s>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

to rate themselves on general items, such as “It is easy for me to stick to my aims and accomplish my goals” and “I can usually handle whatever comes my way.” If you remember from earlier in this module, however, self-efficacy is not a global trait, so there are problems with lumping all types of self-efficacy together in one measure. Thus, the second category of self-efficacy measures includes **task-specific measures of self-efficacy**. Rather than gauge self-efficacy in general, these measures ask about a person’s self-efficacy beliefs about a particular task. There can be an unlimited number of these types of measures. Task-specific measures of self-efficacy describe several situations relating to a behavior and then ask the participant to write down how confidently he or she feels about doing that behavior. For example, a measure of dieting self-efficacy would list a variety of situations where it can be hard to stick to a diet—such as during vacations, when bored, or when going out to eat with others who are not on a diet. A measure of exercise self-efficacy would list a variety of situations where it can be hard to exercise—such as when feeling depressed, when feeling tired, and when you are with other people who do not want to exercise. Finally, a measure of children’s or teens’ self-regulatory self-efficacy would include a variety of situations where it can be hard to resist impulses—such as controlling temper, resisting peer pressure to smoke cigarettes, and defying pressure to have unprotected sex. Most studies agree that the task-specific measures of self-efficacy are better predictors of behavior than the general measures of self-efficacy (Bandura, 2006).

Influence	Definition
Performance Experiences	When you do well and succeed at a particular task to attain a valued goal, you usually believe that you will succeed again at this task. When you fail you often expect that you will fail again in the future if you try that task.
Vicarious Performances	If someone who seems similar to you succeeds, then you may come to believe that you will succeed as well.
Verbal Persuasion	This involves people telling you what they believe you are and are not capable of doing. Not all people will be equally persuasive.
Imaginal Performances	What you imagine yourself doing and how well or poorly you imagine yourself doing it.
Affective States & Physical Sensations	When you associate negative moods and negative physical sensations with failure, and positive physical sensations with success.

Table 1: Major Influences on Self-efficacy

What Are the Major Influences on Self-Efficacy?

Self-efficacy beliefs are influenced in five different ways (Bandura, 1997), which are summarized in Table 1 .

These five types of self-efficacy influence can take many real-world forms that almost everyone has experienced. You may have had previous performance experiences affect your academic self-efficacy when you did well on a test and believed that you would do well on the next test. A vicarious performance may have affected your athletic self-efficacy when you saw your best friend skateboard for the first time and thought that you could skateboard well, too. **Verbal persuasion** could have affected your academic self-efficacy when a teacher that you respect told you that you could get into the college of your choice if you studied hard for the SATs. It's important to know that not all people are equally likely to influence your self-efficacy though verbal persuasion. People who appear trustworthy or attractive, or who seem to be experts, are more likely to influence your self-efficacy than are people who do not possess these qualities (Petty & Brinol, 2010). That's why a teacher you respect is more likely to influence your self-efficacy than a teacher you do not respect. **Imaginal performances** are an effective way to increase your self-efficacy. For example, imagining yourself doing well on a job interview actually leads to more effective interviewing (Knudstrup, Segrest, & Hurley, 2003). Affective



Our assessments of our own self-efficacy aren't created in a vacuum. Those around us (e.g., our parents, our friends) greatly influence our perceptions of self-efficacy. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

states and physical sensations abound when you think about the times you have given presentations in class. For example, you may have felt your heart racing while giving a presentation. If you believed your heart was racing because you had just had a lot of caffeine, it likely would not affect your performance. If you believed your heart was racing because you were doing a poor job, you might believe that you cannot give the presentation well. This is because you associate the feeling of anxiety with failure and expect to fail when you are feeling anxious.

When and How Does Self-Efficacy Develop?

Self-efficacy begins to develop in very young children. Once self-efficacy is developed, it does not remain constant—it can change and grow as an individual has different experiences throughout his or her lifetime. When children are very young, their parents' self-efficacies are important (Jones & Prinz, 2005). Children of parents who have high parental self-efficacies perceive their parents as more responsive to their needs (Gondoli & Silverberg, 1997). Around the ages of 12 through 16, adolescents' friends also become an important source of self-efficacy beliefs. Adolescents who associate with peer groups that are not academically motivated tend to experience a decline in academic self-efficacy (Wentzel, Barry, & Caldwell, 2004). Adolescents who watch their peers succeed, however, experience a rise in academic self-efficacy (Schunk & Miller, 2002). This is an example of gaining self-efficacy through **vicarious performances**, as discussed above. The effects of self-efficacy that develop in adolescence are long lasting. One study found that greater social and academic self-efficacy measured in people ages 14 to 18 predicted greater life satisfaction five years later (Vecchio, Gerbino, Pastorelli, Del Bove, & Caprara, 2007).

What Are the Benefits of High Self-Efficacy?

Academic Achievement

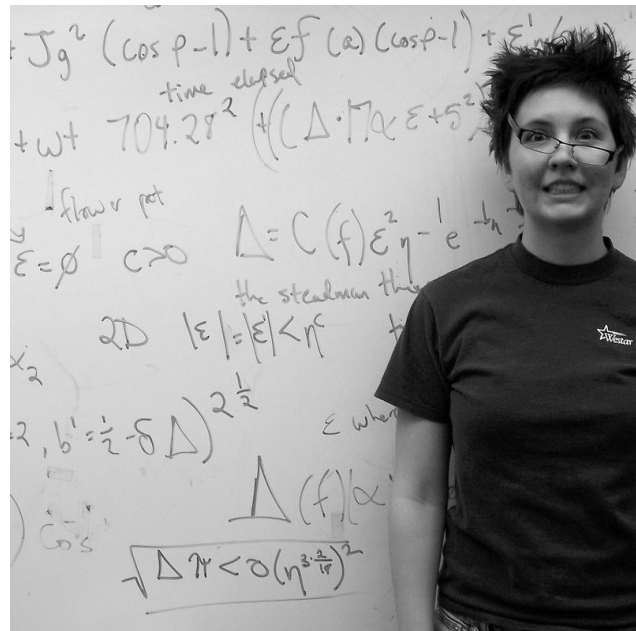
Consider academic self-efficacy in your own life and recall the earlier example of Sally and Lucy. Are you more like Sally, who has high academic self-efficacy and believes that she can use her abilities to do well in school, or are you more like Lucy, who does not believe that she can effectively use her academic abilities to excel in school? Do you think your own self-efficacy has ever affected your academic ability? Do you think you have ever studied more or less intensely because you did or did not believe in your abilities to do well? Many researchers have considered how self-efficacy works in academic settings, and the short answer is that academic self-efficacy affects every possible area of academic achievement (Pajares, 1996).

Students who believe in their ability to do well academically tend to be more motivated in school (Schunk, 1991). When self-efficacious students attain their goals, they continue to set even more challenging goals (Schunk, 1990). This can all lead to better performance in school in terms of higher grades and taking more challenging classes (Multon, Brown, & Lent, 1991). For example, students with high academic self-efficacies might study harder because they believe that they are able to use their abilities to study effectively. Because they studied hard, they receive an A on their next test. Teachers' self-efficacies also can affect how well a student performs in school. Self-efficacious teachers encourage parents to take a more active role in their children's learning, leading to better academic performance (Hoover-Dempsey, Bassler, & Brissie, 1987).

Although there is a lot of research about how self-efficacy is beneficial to school-aged children, college students can also benefit from self-efficacy. Freshmen with higher self-efficacies about their ability to do well in college tend to adapt to their first year in college better than those with lower self-efficacies (Chemers, Hu, & Garcia, 2001). The benefits of self-efficacy continue beyond the school years: people with strong self-efficacy beliefs toward performing well in school tend to perceive a wider range of career options (Lent, Brown, & Larkin, 1986). In addition, people who have stronger beliefs of self-efficacy toward their professional work tend to have more successful careers (Stajkovic & Luthans, 1998).

One question you might have about self-efficacy and academic performance is how a student's actual academic ability interacts with self-efficacy to influence academic performance. The answer is that a student's actual ability does play a role, but it is also influenced by self-efficacy. Students with greater ability perform better than those with lesser ability. But, among a group of students with the same exact level of academic ability, those with stronger academic self-efficacies outperform those with weaker self-efficacies. One study (Collins, 1984) compared performance on difficult math problems among groups of students with different levels of math ability and different levels of math self-efficacy. Among a group of students with average levels of math ability, the students with weak math self-efficacies got about 25% of the math problems correct. The

students with average levels of math ability and strong math self-efficacies got about 45% of the questions correct. This means that by just having stronger math self-efficacy, a student of average math ability will perform 20% better than a student with similar math ability but weaker math self-efficacy. You might also wonder if self-efficacy makes a difference only for people with average or below-average abilities. Self-efficacy is important even for above-average students. In this study, those with above-average math abilities and low math self-efficacies answered only about 65% of the questions correctly; those with above-average math abilities and high math self-efficacies answered about 75% of the questions correctly.



Consider getting a math problem you automatically believe you can't solve. Will you even attempt to answer it? Our perception of self-efficacy affects our motivation to engage with challenges in the first place. [Image: Andrea Allen, <https://goo.gl/EdsSwo>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Healthy Behaviors

Think about a time when you tried to improve your health, whether through dieting, exercising, sleeping more, or any other way. Would you be more likely to follow through on these plans if you believed that you could effectively use your skills to accomplish your health goals? Many researchers agree that people with stronger self-efficacies for doing healthy things (e.g., exercise self-efficacy, dieting self-efficacy) engage in more behaviors that prevent health problems and improve overall health (Strecher, DeVellis, Becker, & Rosenstock, 1986). People who have strong self-efficacy beliefs about quitting smoking are able to quit smoking more easily (DiClemente, Prochaska, & Gibertini, 1985). People who have strong self-efficacy beliefs about being able to reduce their alcohol consumption are more successful when treated for drinking problems (Maisto, Connors, & Zywiak, 2000). People who have stronger self-efficacy beliefs about their ability to recover from heart attacks do so more quickly than those who do not have such beliefs (Ewart, Taylor, Reese, & DeBusk, 1983).

One group of researchers (Roach Yadrick, Johnson, Boudreaux, Forsythe, & Billon, 2003) conducted an experiment with people trying to lose weight. All people in the study participated in a weight loss program that was designed for the U.S. Air Force. This program had already been found to be very effective, but the researchers wanted to know if increasing people's self-efficacies could make the program even more effective. So, they divided the participants into two groups: one group received an intervention that was designed to increase weight loss self-efficacy along with the diet program, and the other group received only the diet program. The researchers tried several different ways to increase self-efficacy, such as having participants read a copy of *Oh, The Places You'll Go!* by Dr. Seuss (1990), and having them talk to someone who had successfully lost weight. The people who received the diet program and an intervention to increase self-efficacy lost an average of 8.2 pounds over the 12 weeks of the study; those participants who had only the diet program lost only 5.8 pounds. Thus, just by increasing weight loss self-efficacy, participants were able to lose over 50% more weight.

Studies have found that increasing a person's nutritional self-efficacy can lead them to eat more fruits and vegetables (Luszczynska, Tryburcy, & Schwarzer, 2006). Self-efficacy plays a large role in successful physical exercise (Maddux & Dawson, 2014). People with stronger self-efficacies for exercising are more likely to plan on beginning an exercise program, actually beginning that program (DuCharme & Brawley, 1995), and continuing it (Marcus, Selby, Niaura, & Rossi, 1992). Self-efficacy is especially important when it comes to safe sex. People with greater self-efficacies about condom usage are more likely to engage in safe sex (Kaneko, 2007), making them more likely to avoid sexually transmitted diseases, such as HIV (Forsyth & Carey, 1998).

Athletic Performance



Self-efficacy can work in an upward, feedback loop. That is, when you do well in a sport, you have greater self-efficacy, and by having greater self-efficacy, you subsequently do better in the sport! [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

If you are an athlete, self-efficacy is especially important in your life. Professional and amateur athletes with stronger self-efficacy beliefs about their athletic abilities perform better than athletes with weaker levels of self-efficacy (Wurtele, 1986). This holds true for athletes in all types of sports, including track and field (Gernigon & Delloye, 2003), tennis (Sheldon & Eccles, 2005), and golf (Bruton, Mellalieu, Shearer, Roderique-Davies, & Hall, 2013). One group of researchers found that basketball players with strong athletic self-efficacy beliefs hit more foul shots than did basketball players with weak self-efficacy beliefs (Haney & Long, 1995). These researchers also found that the players who hit more foul shots had greater increases in self-efficacy after they hit the foul shots compared to those who hit

fewer foul shots and did not experience increases in self-efficacy. This is an example of how we gain self-efficacy through performance experiences.

Self-Regulation

One of the major reasons that higher self-efficacy usually leads to better performance and greater success is that self-efficacy is an important component of self-regulation. Self-regulation is the complex process through which you control your thoughts, emotions, and actions (Gross, 1998). It is crucial to success and well-being in almost every area of your life. Every day, you are exposed to situations where you might want to act or feel a certain way that would be socially inappropriate or that might be unhealthy for you in the long run. For example, when sitting in a boring class, you might want to take out your phone and text your friends, take off your shoes and take a nap, or perhaps scream because you are so bored. Self-regulation is the process that you use to avoid such behaviors and instead sit quietly through class. Self-regulation takes a lot of effort, and it is often compared to a muscle that can be exhausted (Baumeister, Bratslavsky, Muraven, & Tice, 1998). For example, a child might

be able to resist eating a pile of delicious cookies if he or she is in the room with the cookies for only a few minutes, but if that child were forced to spend hours with the cookies, his or her ability to regulate the desire to eat the cookies would wear down. Eventually, his or her self-regulatory abilities would be exhausted, and the child would eat the cookies. A person with strong self-efficacy beliefs might become less distressed in the face of failure than might someone with weak self-efficacy. Because self-efficacious people are less likely to become distressed, they draw less on their self-regulation reserves; thus, self-efficacious people persist longer in the face of a challenge.

Self-efficacy influences self-regulation in many ways to produce better performance and greater success (Maddux & Volkmann, 2010). First, people with stronger self-efficacies have greater *motivation to perform* in the area for which they have stronger self-efficacies (Bandura & Locke, 2003). This means that people are motivated to work harder in those areas where they believe they can effectively perform. Second, people with stronger self-efficacies are more likely to *persevere through challenges* in attaining goals (Vancouver, More, & Yoder, 2008). For example, people with high academic self-efficacies are better able to motivate themselves to persevere through such challenges as taking a difficult class and completing their degrees because they believe that their efforts will pay off. Third, self-efficacious people believe that *they have more control over a situation*. Having more control over a situation means that self-efficacious people might be more likely to engage in the behaviors that will allow them to achieve their desired goal. Finally, self-efficacious people *have more confidence* in their problem-solving abilities and, thus, are able to better use their cognitive resources and make better decisions, especially in the face of challenges and setbacks (Cervone, Jiwani, & Wood, 1991).



Self-efficacy is all about your belief of control over your environment. But when cookies that look this good are nearby, you may feel like you have no control to resist eating one (or more). [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Collective Efficacy

Collective efficacy is a concept related to self-efficacy. Collective efficacy refers to the shared

beliefs among members of a group about the group's ability to effectively perform the tasks needed to attain a valued goal (Bandura, 1997). Groups and teams that have higher collective efficacies perform better than groups and teams with lower collective efficacies (Marks, 1999). Collective efficacy is especially important during tasks that require a lot of teamwork (Katz-Navon & Erez, 2005). For example, when you have to do a group project that involves each group member contributing a portion of the final project, your group's performance will be much better if all members share the belief that your group can perform the necessary tasks together. Collective efficacy plays a role in romantic relationships. Married couples who strongly believe in their ability to accomplish shared goals are happier than couples with weaker efficacy beliefs (Kaplan & Maddux, 2002). Although collective efficacy is an important part of how well a team or group performs, self-efficacy also plays a role in team situations. For example, better decision-making self-efficacy predicts better performance in team sports, such as baseball (Hepler & Feltz, 2012).

Conclusion

Self-efficacy refers to your beliefs about your ability to effectively perform the tasks needed to attain a valued goal and it affects your daily life in many ways. Self-efficacious adolescents perform better at school and self-efficacious adults perform better at work. These individuals have happier romantic relationships and work better in teams. People with strong self-efficacies have better health than those with weak self-efficacies; they are more likely to engage in behaviors that prevent health problems and actually increase their health. They are more likely to begin and continue exercise, have safer sex, and eat better foods. Higher self-efficacy is also useful for getting out of bad habits. People with strong self-efficacies are able to lose weight, quit smoking, and cut down on alcohol consumption more successfully than can people with low self-efficacies. As illustrated by the well-known children's book *The Little Engine That Could* (Piper, 1930), telling yourself "I think I can" can be a powerful motivator and can increase your chances for success.

Our own final words on self-efficacy also draw from children's literature. Many people receive a copy of *Oh, The Places You'll Go!* when they reach a major milestone, such as graduating high school to go on to college or graduating college to enter the workforce. Whether or not you or whoever gave you the book knew it, *Oh, The Places You'll Go!* is all about self-efficacy. This book speaks directly to readers by talking about all of the challenges they might face on their journeys. Throughout the book, the narrator continues to assure readers that they will be able to use their abilities to effectively handle these challenges. So, we leave you with Dr. Seuss' wise words: "You're on your own. And you know what you know. And you are the guy who'll decide where to go.... And will you succeed? Yes! You will, indeed! 98 and 3/4 percent

guaranteed."

Outside Resources

Video: Association for Psychological Science presents an interview with Albert Bandura

http://www.youtube.com/watch?v=-_U-pSZwHy8

Video: Self-efficacy's role and sources

<http://www.youtube.com/watch?v=wrzzbaomLmc>

Web: Professor Frank Pajares' self-efficacy site.

<http://www.uky.edu/~eushe2/Pajares/self-efficacy.html>

Discussion Questions

1. Can you think of ways your own self-efficacy beliefs play a role in your daily life? In which areas do you have strong self-efficacy? In which areas would you like your self-efficacy to be a bit stronger? How could you increase your self-efficacy in those areas?
2. Can you think of a time when a teacher, coach, or parent did something to encourage your self-efficacy? What did he or she do and say? How did it enhance your self-efficacy?
3. What are some ways that you can help strengthen the self-efficacies of the people in your life?
4. Can you think of a time when collective efficacy played a role in your team or group activities? What did you notice about being on a team or in a group that had high collective efficacy? What about a team or group with low collective efficacy?

Vocabulary

Collective efficacy

The shared beliefs among members of a group about the group's ability to effectively perform the tasks needed to attain a valued goal.

Imaginal performances

When imagining yourself doing well increases self-efficacy.

Performance experiences

When past successes or failures lead to changes in self-efficacy.

Self-efficacy

The belief that you are able to effectively perform the tasks needed to attain a valued goal.

Self-regulation

The complex process through which people control their thoughts, emotions, and actions.

Self-report measure

A type of questionnaire in which participants answer questions whose answers correspond to numerical values that can be added to create an overall index of some construct.

Task-specific measures of self-efficacy

Measures that ask about self-efficacy beliefs for a particular task (e.g., athletic self-efficacy, academic self-efficacy).

Verbal persuasion

When trusted people (friends, family, experts) influence your self-efficacy for better or worse by either encouraging or discouraging you about your ability to succeed.

Vicarious performances

When seeing other people succeed or fail leads to changes in self-efficacy.

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Intellectual Abilities, Interests, and Mastery

David Lubinski

Psychologists interested in the study of human individuality have found that accomplishments in education, the world of work, and creativity are a joint function of talent, passion, and commitment — or how much effort and time one is willing to invest in personal development when the opportunity is provided. This module reviews models and measures that psychologists have designed to assess intellect, interests, and energy for personal development. The module begins with a model for organizing these three psychological domains, which is useful for understanding talent development. This model is not only helpful for understanding the many different ways that positive development may unfold among people, but it is also useful for conceptualizing personal development and ways of selecting opportunities in learning and work settings that are more personally meaningful. Data supporting this model are reviewed.

Learning Objectives

- Compare and contrast satisfaction and satisfactoriness.
- Discuss why the model of talent development offered in this module places equal emphasis on assessing the person and assessing the environment.
- Articulate the relationship between ability and learning and performance.
- Understand the issue of an "ability threshold" beyond which more ability may or may not matter.
- List personal attributes other than interests and abilities that are important to individual accomplishment.

An amalgam of intelligence, interests, and mastery are appropriate topics for an essay on the cross-cutting themes running through these vast domains of psychological diversity. For effective performance and valued accomplishments, these three classes of determinants are needed for comprehensive treatments of psychological phenomena supporting learning, occupational performance, and for advancing knowledge through innovative solutions. Historically, these personal attributes go back to at least Plato's triarchic view of the human psyche, described in *Phaedra*, wherein he depicts the *intellect* as a charioteer, and *affect*(interests) and *will*(to master) as horses that draw the chariot. Ever since that time, cognitive, affective, and conative factors have all been found in comprehensive models of human development, or "The Trilogy of Mind" (Hilgard, 1980). To predict the magnitude, nature, and sophistication of intellectual development toward learning, working, and creating, all three classes are indispensable and deficits on any one can markedly hobble the effectiveness of the others in meeting standards for typical as well as extraordinary performance. These three aspects of human individuality all operate in parallel confluences of behaviors, perceptions, and stimuli to engender stream of consciousness experiences as well as effective functioning. Hilgard (1980) was indeed justified to criticize formulations in cognitive psychology, which neglect affection and conation; technically, such truncated frameworks of human psychological phenomena are known as **under-determined or misspecified causal models** (Lubinski, 2000; Lubinski & Humphreys, 1997).



Although Plato's view of human intelligence may be quite dated, modern-day analyses describe a similar model for understanding humans' intellectual application and development. [Image: A Health Blog, <https://goo.gl/O8FoHH>, CC BY-SA 2.0, <https://goo.gl/rxiUsF>]

A Framework for Understanding Talent Development

Figure 1 is an adaptation of the Theory of Work Adjustment (TWA; Dawis & Lofquist, 1984; Lubinski & Benbow, 2000). It provides a useful organizational scheme for this treatment by outlining critical dimensions of human individuality for performance in learning and work settings (and in transitioning between such settings). Here, the dominant models of intellectual abilities and educational-occupational interests are assembled. Because this review will be

restricted to measures of individual differences that harbor real-world significance, these two models are linked to corresponding features of learning and work environments, ability requirements and incentive or reward structures, which set standards for meeting expectations (performance) and rewarding valued performance (compensation). Correspondence between abilities and ability requirements constitutes satisfactoriness ("competence"), whereas correspondence between an interests and reward structures constitutes **satisfaction** ("fulfillment"). To the extent that **satisfactoriness** and satisfaction co-occur, the individual is motivated to maintain contact with the environment and the environment is motivated to retain the individual; if one of these dimensions is discordant, the individual is motivated to leave the environment or the environment is motivated to dismiss.

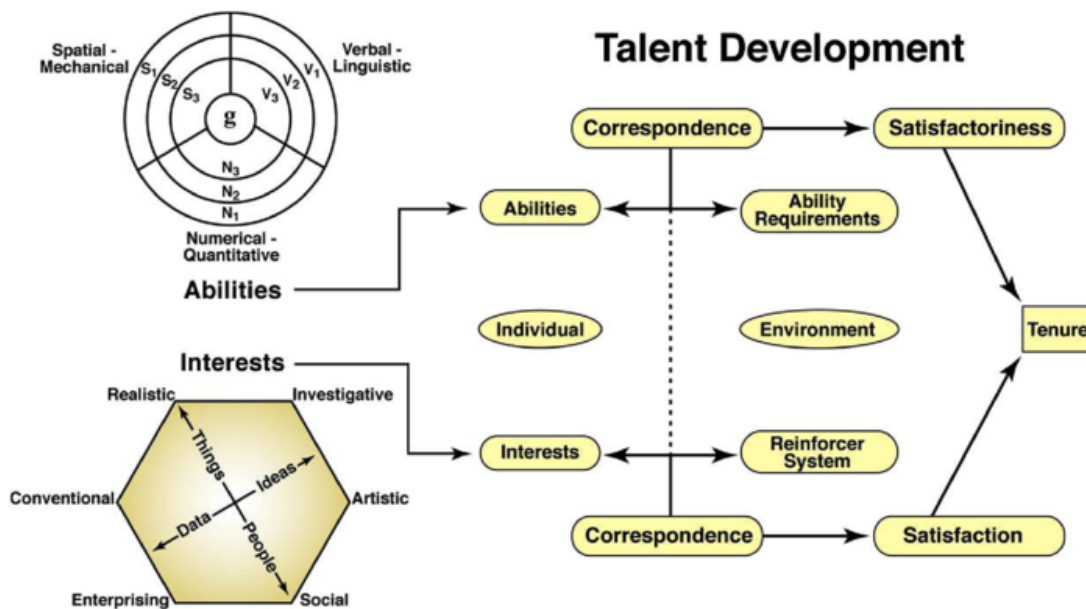


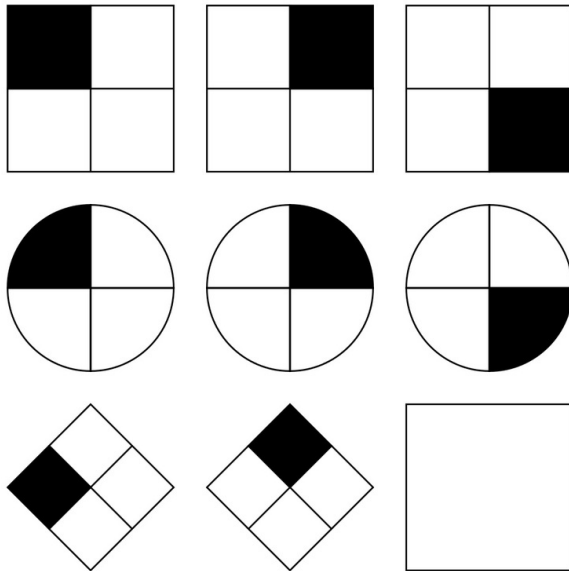
Figure 1. The above is an adaptation of the Theory of Work Adjustment (Dawis & Lofquist, 1984), following Lubinski and Benbow (2000) to highlight its general role in talent development over the life span. The radex scaling of cognitive abilities (upper left) and the RIASEC hexagon of interests (lower left) outline personal attributes relevant to learning and work. The letters within the cognitive ability arrangement denote different regions of concentration, whereas their accompanying numbers increase as a function of complexity. Contained within the RIASEC is a simplification of this hexagon. Following Prediger (1982), it amounts to a two-dimensional structure of independent dimensions: people/things and data/ideas, which underlie RIASEC. The dotted line running down the individual and environment sectors underscores the idea that equal emphasis is placed on assessing personal attributes (abilities and interests) and assessing the environment (abilities requirements and reward structure). Correspondence between abilities and ability requirements constitutes satisfactoriness whereas correspondence between interests and reinforce systems constitutes satisfaction. Jointly, these two dimensions predict tenure or a longstanding relationship between the individual and the environment.

This model of talent development places equal emphasis on assessing the individual (abilities and interests) and the environment (response requirements and reward structures). Comprehensive reviews of outcomes within education (Lubinski, 1996; Lubinski & Benbow, 2000), counseling (Dawis, 1992; Gottfredson, 2003; Rounds & Tracey, 1990), and industrial/organizational psychology all emphasize this person/environment tandem (Dawis, 1991; Katzell, 1994; Lubinski & Dawis, 1992; Strong, 1943): aligning competency/motivational proclivities to performance standards and reward structures for learning and work (Bouchard, 1997; Scarr, 1996; Scarr & McCartney, 1983). And indeed, educational, counseling, and industrial psychology can be contiguously sequenced by this framework. They all share a common feature: the scientific study of implementing interventions or opportunities, based on individual differences, for maximizing positive psychological growth across different stages of life span development (Lubinski, 1996). For making individual decisions about personal development, or institutional decisions about organizational development, it is frequently useful to go beyond a minimum requisite approach of “do you like it” (satisfaction) and “can you do it” (satisfactoriness), and instead consider what individuals like the most and can do the best (Lubinski & Benbow, 2000, 2001). This framework is useful for identifying “optimal promise” for personal as well as organizational development. For now, however, cognitive abilities and interests will be reviewed and, ultimately, linked to conative determinants that mobilize, and in part account for, individual differences in how capabilities and motives are expressed.

Cognitive Abilities

Over the past several decades—the past 20 years in particular—a remarkable consensus has emerged that cognitive abilities are organized hierarchically (Carroll, 1993). A general outline of this hierarchy is represented graphically by a radex (Guttman, 1954), depicted in the upper left region of Figure 1. This illustrates the reliable finding that cognitive ability assessments covary as a function of their content or complexity (Corno, Cronbach et al., 2002; Lubinski & Dawis, 1992; Snow & Lohman, 1989). Cognitive ability tests can be scaled in this space based on how highly they covary with one another. The more that two tests share complexity and content, the more they covary and the closer they are to one another as points within the radex. Test *complexity* is scaled from the center of the radex (“g”) out, and, along lines emanating from the origin, complexity decreases but test content remains the same. Test *content* is scaled around the circular bands with equal distance from the center of the radex and, progressing around these bands, the relative density of test content changes from spatial/mechanical to verbal/linguistic to quantitative/numerical, but test complexity remains constant. Therefore, test content varies *within* each band (but complexity remains constant), whereas test complexity varies *between* bands (but on lines from the origin to the periphery, content remains

constant). Because the extent to which tests covary is represented by how close together they are within this space (Lubinski & Dawis, 1992; Snow & Lohman, 1989; Wai, Lubinski, & Benbow, 2009), this model is helpful in organizing the many different kinds of specific ability tests.



As you can tell already, trying to document one's intelligence in problem-solving tests is very complicated. So although online "IQ tests" can be fun, rarely are they actually accurate. [Image: Life of Riley, <https://goo.gl/siweyC>, CC BY-SA 3.0, <https://goo.gl/eLcN2OJ>]

As Piaget astutely pointed out, "Intelligence is what you use when you don't know what to do," and this model affords an excellent overview of the content and sophistication of thought applied to familiar and novel problem-solving tasks. Mathematical, spatial, and verbal reasoning constitute the chief specific abilities with implications for different choices and performance after those choices in learning and work settings (Corno et al., 2002; Dawis, 1992; Gottfredson, 2003; Lubinski, 2004; Wai et al., 2009). The content of measures or tests of these specific abilities index individual differences in different modalities of thought: reasoning with numbers, words, and figures or shapes. Yet, despite this disparate content and focus, contrasting specific ability tests are all positively correlated, because they all index an underlying general property of intellectual

thought.

This general (common) dimension, identified over 100 years ago (Spearman, 1904) and corroborated by a massive quantity of subsequent research (Carroll, 1993; Jensen, 1998), is *general mental ability*, the *general factor*, or simply *g* (Gottfredson, 1997). **General mental ability** represents the complexity/sophistication of a person's intellectual repertoire (Jensen, 1998; Lubinski & Dawis, 1992). The more complex a test is, regardless of its content, the better a measure of *g* it is. Further, because *g* underlies all cognitive reasoning processes, any test that assesses a specific ability is also, to some extent, a measure of *g* (Lubinski, 2004). In school, work, and a variety of everyday life circumstances, assessments of this general dimension covary more broadly and deeper than any other measure of human individuality (Hunt, 2011; Jensen, 1998; Lubinski, 2000, 2004).

Measures of *g* manifest their life importance by going beyond educational settings (where

they covary with educational achievement assessments in the .70–.80 range), by playing a role in shaping phenomena within Freud's two important life domains, *arbeiten* and *lieben*, working and loving (or, resource acquisition and mating). Measures of *g* covary .20–.60 with work performance as a function of job complexity, .30–.40 with income, and –.20 with criminal behavior, .40 with SES of origin, and .50–.70 with achieved SES; assortative mating correlations on *g* are around .50 (Jensen, 1998; Lubinski, 2004; Schmidt & Hunter, 1998). Furthermore, Malcolm Gladwell (2008) notwithstanding, there does *not* appear to be an ability threshold; that is, the idea that after a certain point more ability does not matter. More ability does matter.

Although other determinants are certainly needed (interests, persistence, opportunity), more ability does make a difference in learning, working, and creating, even among the top 1% of ability, or IQ equivalents ranging from approximately 137 to over 200 (see Figure 2). When appropriate assessment and criterion measures are utilized to capture the breadth of ability and accomplishment differences among the profoundly talented, individual differences within the top 1% of ability are shown to matter a great deal. In the past this has been difficult to demonstrate, because intellectual assessments and criterion measures lacked sufficient scope in gifted or intellectually talented populations, which resulted in no variation in assessments among the able and exceptionally able (ceiling effects). Without variation there cannot be co-variation, but modern methods have now corrected for this (Kell, Lubinski, & Benbow, 2013a; Lubinski, 2009; Park, Lubinski, & Benbow, 2007, 2008). Yet, even when *g* is measured in its full scope, and validated with large samples and appropriate low-base-rate-criteria over protracted longitudinal intervals, there is much more to intellectual functioning than measures of *g* or general ability.

To reveal how general and specific abilities operate over the course of development, Figure 3 contains data from over 400,000 high schools students assessed between grades 9 through 12, and tracked for 11 years. Specifically, Figure 3 graphs the general and specific ability profiles of students earning terminal degrees in nine disciplines (Wai et al., 2009). Given that highly congruent findings were observed for all four cohorts (grades 9 through 12), the cohorts were combined. High general intelligence and an intellectual orientation dominated by high mathematical and spatial abilities, relative to verbal ability, were salient characteristics of individuals who pursued advanced education credentials in science, technology, engineering, and mathematics (STEM). These participants occupy a region in the intellectual space defined by the dimensions of ability level and ability pattern different from participants who earn undergraduate and graduate degrees in other domains.

Two major differences distinguish the STEM from the non-STEM educational groups. First, students who ultimately secure educational credentials in STEM domains are more capable than those earning degrees in other areas, especially in nonverbal intellectual abilities. Within

Accomplishments Across Individual Differences within the Top 1% of General Cognitive Ability: 25+ Years After Identified at Age 13

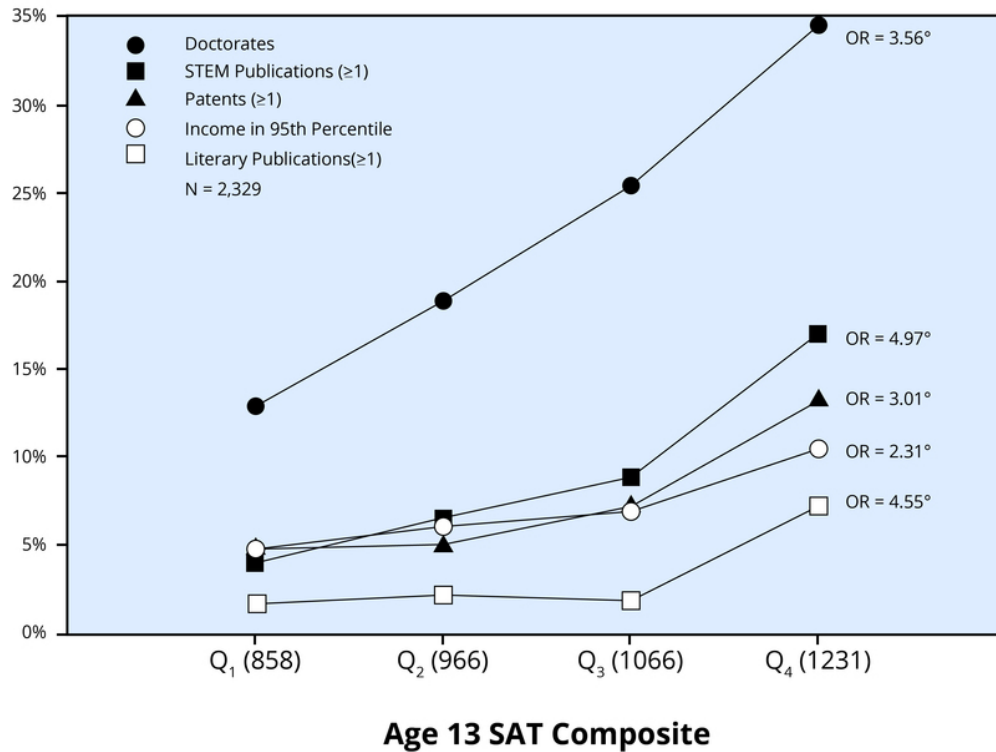


Figure 2. Frey and Detterman (2004) have shown that the SAT composite is an excellent measure of general intelligence for high ability samples; here, intellectually precocious youth were assessed on this composite at age 13 and separated into quartiles (Lubinski, 2009). The mean age 13 SAT composite scores for each quartile are displayed in parentheses along the x-axis. Odds ratios comparing the likelihood of each outcome in the top (Q4) and bottom (Q1) SAT quartiles are displayed at the end of every respective criterion line. An asterisk indicates that the 95% confidence interval for the odds ratio did not include 1.0, meaning that the likelihood of the outcome in Q4 was significantly greater than in Q1. These SAT assessments by age 13 were conducted before the re-centering of the SAT in the mid-1990s (i.e., during the 1970s and early 1980s); at that time, cutting scores for the top 1 in 200 were SAT-M $\geq$ 500, SAT-V $\geq$ 430; for the top 1 in 10,000, cutting scores were SAT-M $\geq$ 700, SAT-V $\geq$ 630 by age 13.

all educational domains, more advanced degrees are associated with more general *and* specific abilities. Second, for all three STEM educational groupings (and the advanced degrees within these groupings), spatial ability > verbal ability—whereas for all others, ranging from education to biology, spatial ability < verbal ability (with business being an exception). Young adolescents who subsequently secured advanced educational credentials in STEM manifested a spatial-verbal ability pattern opposite that of those who ultimately earned educational

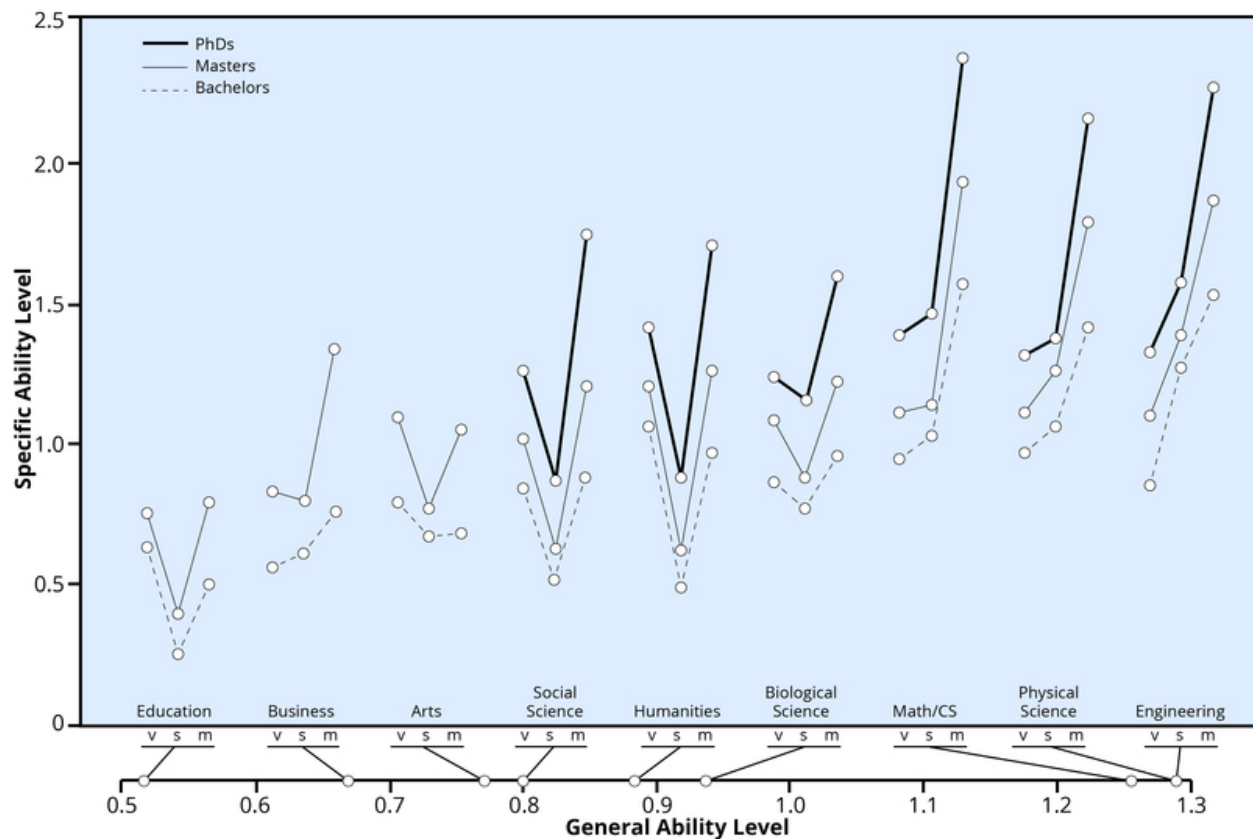


Figure 3. Average z scores of participants on verbal, spatial, and mathematical ability for terminal bachelor's degrees, terminal master's degrees, and doctoral degrees are plotted by field. The groups are plotted in rank order of their normative standing on g (verbal [V] + spatial [S] + mathematical [M]) along the x-axis, and the line with the arrows from each field pointing to it indicates on the continuous scale where they are in general mental ability in z-score units. This figure is standardized in relation to all participants with complete ability data at the time of initial testing. Respective N s for each group (men + women) were as follows for bachelor's, master's, and doctorates, respectively: engineering (1,143, 339, 71), physical science (633, 182, 202), math/computer science (877, 266, 57), biological science (740, 182, 79), humanities (3,226, 695, 82), social science (2,609, 484, 158), arts (615, 171 [master's only]), business (2,386, 191 [master's + doctorate]), and education (3,403, 1,505 [master's + doctorate]). For education and business, master's degrees and doctorates were combined because the doctorate samples for these groups were too small to obtain stability ($N = 30$). For the specific N for each degree by sex that composed the major groupings, see Appendix A in Wai et al. (2009).

credentials in other areas. These same patterns play out in occupational arenas in predictable ways (Kell, Lubinski, Benbow, & Steiger, 2013b). In the past decade, individual differences within the top 1% of ability have revealed that these patterns portend important outcomes for technical innovation and creativity, with respect to both ability level (Lubinski, 2009; Park et al., 2008) and pattern (Kell et al. 2013a, Kell et al., 2013b; Park et al., 2007). Level of general ability has predictive validity for the *magnitude* of accomplishment (how extraordinary they are), whereas ability pattern has predictive validity for the nature of accomplishments (the domains they occur in).

Interests

Just because people can do something well doesn't mean they like doing it. Psychological information on motivational differences (personal passions) is needed to understand attractions and aversions, different ways to create a meaningful life, and how differential development unfolds. Even people with the same intellectual equipment vary widely in their motivational proclivities. Paraphrasing Plato, different horses drive intellectual development down different life paths. The lower left region of Figure 1 provides the dominant model of vocational interests, one developed from decades of large-scale longitudinal and cross-cultural research. It shows a hexagonal structure consisting of six general themes: Realistic (R) = working with gadgets and things, the outdoors, need for structure; Investigative (I) = scientific pursuits, especially mathematics and the physical science, an interest in theory; Artistic (A) = creative expression in art and writing, little need for structure; Social (S) = people interests, the helping professions, teaching, nursing, counseling; Enterprising (E) = likes leadership roles directed toward economic objectives; and Conventional (C) = liking of well-structured environments and clear chains of command, such as office practices.



It's pretty easy to think of things you're skilled at but don't enjoy doing. However, can you think of something you're not very skilled at but still enjoy doing? [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

These six themes covary inversely with the distance between them, hence, the hexagonal structure circling around R-I-A-S-E-C. John Holland (1959, 1996) justifiably receives most of the credit for this model (Day & Rounds, 1998), although Guilford et al. (1954) uncovered a similar framework based on military data and labeled them Mechanical, Scientific, Aesthetic Expression, Social Welfare, Business, and Clerical. Although each theme contains multiple subcomponents, Holland's hexagon, like the radex of cognitive abilities, captures the general outlines of the educational/occupational interest domain, but there are molecular strands of intellectual and interest dimensions that add nuance to these general outlines (for abilities, see Carroll, 1993; for interests, see Dawis, 1991; Savickas & Spokane, 1999). There are also super-ordinal themes such as *people*

versus things (Su, Rounds, & Armstrong, 2009), which manifest arguably the largest sex-difference on a psychological dimension of human individuality.

At superordinate levels of *people versus things* or *data versus ideas* (Prediger, 1982), or at the RIASEC level of analysis, interest dimensions covary in different ways with mathematical, spatial, and verbal abilities (Ackerman, 1996; Ackerman & Heggestad, 1997; Schmidt, Lubinski, & Benbow, 1998); and intense selection, when exclusively restricted to a specific ability, will eventuate in distinctive interest profiles across the three abilities with implications for differential development (Humphreys, Lubinski, & Yao, 1993; Webb, Lubinski, & Benbow, 2007). Although correlations between abilities and interests are “only” in the .20–.30 range, when selection is extreme, distinct profiles emerge and reflect different “types” (Lubinski & Benbow, 2000, 2006). For basic science, this shows how ostensibly different kinds of intelligence at the extreme do not stem from different qualities, but rather from endpoint extremes within a multivariate space of systematic sources of individual differences, which “pull” with them constellations of nonintellectual personal attributes. For applied practice, skilled educational-vocational counselors routinely combine information on abilities and interests to distill learning and work environments that individuals are likely to thrive in competence and experience fulfillment (Dawis, 1992; Rounds & Tracy, 1990). For further insights, a final class of important psychological determinants is needed, however.

Mastery

As all parents of more than one child know, there are huge individual differences in the extent to which individuals embrace opportunities for positive development. Seasoned faculty at top institutions for graduate training have observed the same phenomenon—among highly select graduate students, task commitment varies tremendously. Even among the intellectual elite, individual differences in accomplishments stem from more than abilities, interests, and opportunity; conative determinants are critical catalysts. Galton (1869) called it “zeal,” Hull (1928) called it “industriousness,” and Webb (1915) called it “will.” Such labels as “grit” or “strivers” are sometimes used



According to some researchers, it takes about 10,000 hours of practicing a skill to become an “expert” in it. Of course, one’s natural ability would vary this amount. However, either way, get to practicing! [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

to define resources that people call upon to mobilize their abilities and interests over protracted intervals. Conative factors are distinct from abilities and preferences, having more to do with individual differences in energy or psychological tempo rather than the content of what people can do or how rapidly they learn. Indeed, characteristic across scientific studies of expertise and world-class accomplishment are attributes specifically indicative of indefatigable capacities for study and work. This is an underappreciated class of individual differences, although Ackerman (1996) has discussed *typical intellectual engagement* (TIE) and Dawis and Lofquist (1984) have discussed *pace* and *endurance*. This class of attributes simply has not received the attention it deserves.

Nevertheless, in the field of talent development and identification, the greatest consensus appears to be found on the topic of conation, rather than cognition or affect. Exceptional performers are deeply committed to what they do, and they devote a great deal of time to doing it. Regardless of the theorist, Howard Gardner, Dean Simonton, Arthur Jensen, Anders Erikson, and Harriet Zuckerman all agree that this is a uniform characteristic of world class performers at the top of their game. In the words of Dean Simonton and E. O. Wilson, respectively:

[M]aking it big [becoming a star] is a career. People who wish to do so must organize their whole lives around a single enterprise. They must be monomaniacs, even megalomaniacs, about their pursuits. They must start early, labor continuously, and never give up the cause. Success is not for the lazy, procrastinating, or mercurial. (Simonton, 1994, p. 181)

I have been presumptuous enough to counsel new Ph.D.'s in biology as follows: If you choose an academic career you will need forty hours a week to perform teaching and administrative duties, another twenty hours on top of that to conduct respectable research, and still another twenty hours to accomplish really important research. This formula is not boot-camp rhetoric. (Wilson, 1998, pp. 55–56)

Figure 4 contains data from two extraordinary populations of individuals (Lubinski, Benbow, Webb, Bleske-Rechek, 2006). One group consists of a sample of profoundly gifted adolescents identified at age 12 as in the top 1 in 10,000 in mathematical or verbal reasoning ability; they were subsequently tracked for 20 years. Members of the second group were identified in their early twenties, as first- or second-year STEM graduate students enrolled in a top-15 U.S. university; they were subsequently tracked for 10 years. Now in their mid-thirties, subjects were asked how much they would be *willing to work* in their “ideal job” and, second, how much they actually *do work*. The data are clear. There are huge individual differences associated with how much time people are willing to invest in their career development and work. The STEM graduate students are particularly interesting inasmuch as in their mid-twenties they

were assessed on abilities, interests, and personality, and both sexes were found to be highly similar on these psychological dimensions (Lubinski, Benbow, Shea, Eftekhari-Sanjani, & Halvorson, 2001). But subsequently, over the life span, they markedly diverged in time allocation and life priorities (Ceci & Williams, 2011; Ferriman, Lubinski, & Benbow, 2009).

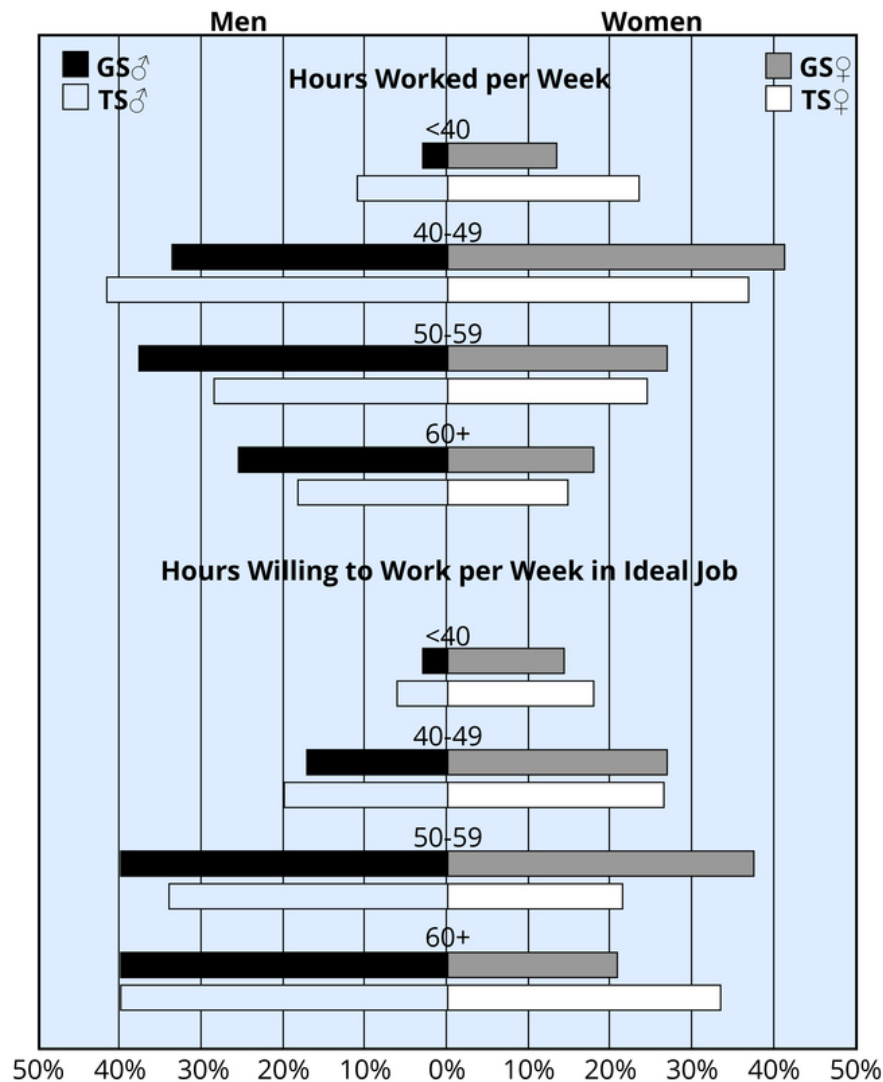


Figure 4: Hours worked per week (top) and hours willing to work per week in ideal job (bottom) for top STEM Graduate Students (GS) and Profoundly Gifted Talent Search (TS) participants now in their mid-thirties (from Lubinski et al., 2006).

These figures reveal huge *noncognitive* individual differences among individuals with exceptional intellectual talent. One only needs to imagine the ticking of a tenure clock and the differences likely to accrue over a 5-year interval between two faculty working 45- versus 65-hour weeks (other things being equal). Making partner in a prestigious law firm is no different, nor is achieving genuine excellence in most intellectually demanding areas.

Conclusion

Since Spearman (1904) advanced the idea of general intelligence, a steady stream of systematic scientific knowledge has accrued in the psychological study of human individuality. We have learned that the intellect is organized hierarchically, that interests are multidimensional and only covary slightly with abilities, and that individual differences are huge in terms of investing in personal development. When these aspects of human psychological diversity are combined with commensurate attention devoted to opportunities for learning, work, and personal growth, a framework for understanding human development begins to take shape. Because frameworks may be found that emphasize only one set of these determinants, this essay closes with the recommendation—based on the empirical evidence—to stress all three.

Outside Resources

Book: Human Cognitive Abilities, by John Carroll constitutes a definitive treatment of the nature and hierarchical organization of cognitive abilities, based on a conceptual and empirical analysis of the past century's factor analytic research.

http://www.amazon.com/Human-Cognitive-Abilities-Factor-Analytic-Studies/dp/0521382750-/ref=sr_1_1?s=books&ie=UTF8&qid=1390944516&sr=1-1&keywords=human+cognitive+abilities

Book: Human Intelligence, by Earl Hunt, provides a superb overview of empirical research on cognitive abilities. Collectively, these three sources capture the psychological significance of what this important domain of human psychological diversity affords.

<http://www.amazon.com/Human-Intelligence-Earl-Hunt/dp/0521707811>

Book: The g Factor, by Arthur Jensen, explicates the depth and breadth of the central dimension running through all cognitive abilities, the summit of Carroll's (1993) hierarchical organization: general intelligence (or "g"). Revealed here is the practical and scientific significance for coming to terms with a rich array of critical human outcomes found in schools, work, and everyday life.

<http://www.amazon.com/The-Factor-Evolution-Behavior-Intelligence/dp/0275961036>

Book: For additional reading on the history of intellectual assessment, read Century of Ability Testing, by Robert Thorndike and David F. Lohman

<http://www.amazon.com/Century-Ability-Testing-Robert-Thorndike/dp/0829251561>

Discussion Questions

1. Why are abilities and interests insufficient for conceptualizing educational and occupational development?
2. Why does the model of talent development discussed in this module place equal emphasis on assessing the individual and assessing the environment.
3. What is the most widely agreed on empirical finding, among investigators who study the development of truly outstanding careers?
4. Besides what you can do and what you like, what other factors are important to consider when making choices about your personal development in learning and work settings?

Vocabulary

g or general mental ability

The general factor common to all cognitive ability measures, “a very general mental capacity that, among other things, involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience. It is not merely book learning, a narrow academic skill, or test-taking smarts. Rather, it reflects a broader and deeper capability for comprehending our surroundings—‘catching on,’ ‘making sense of things,’ or ‘figuring out’ what to do” (Gottfredson, 1997, p. 13).

Satisfaction

Correspondence between an individual's needs or preferences and the rewards offered by the environment.

Satisfactoriness

Correspondence between an individual's abilities and the ability requirements of the environment.

Specific abilities

Cognitive abilities that contain an appreciable component of g or general ability, but also contain a large component of a more content-focused talent such as mathematical, spatial, or verbal ability; patterns of specific abilities channel development down different paths as a function of an individual's relative strengths and weaknesses.

Under-determined or misspecified causal models

Psychological frameworks that miss or neglect to include one or more of the critical determinants of the phenomenon under analysis.

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